The Microstructure of AI Diffusion: Evidence from Firms, Business Functions, and Worker Tasks

by

Kathryn Bonney
U.S. Census Bureau

Cory Breaux
U.S. Census Bureau

Emin Dinlersoz
U.S. Census Bureau

Lucia Foster
U.S. Census Bureau (retired)

John Haltiwanger
University of Maryland and U.S. Census Bureau

Aditya Pande
University of Maryland and U.S. Census Bureau

This paper is a revised version of "The Microstructure of AI Diffusion: Evidence from Firms, Business Functions, and Worker Tasks" (CES 26-25) from April 2026. A copy of the original paper is available upon request.

CES 26-25R October 2026

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

Papers in the series are posted on www.census.gov/library/working-papers/series/ces-wp.html. For information about the series, contact Danielle H. Sandler, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov.

Abstract

Using novel, nationally representative data from the 2026 AI supplement to the U.S. Census Bureau's Business Trends and Outlook Survey (BTOS), we characterize AI diffusion across three interconnected layers: overall firm use, deployment across business functions, and worker-task use. This multi-layered approach provides a nuanced picture of business AI adoption. During the supplement reference period (Nov 2025-Jan 2026), 18% of firms used AI in a business function, rising to 32% on an employment-weighted basis; adoption is expected to reach 22% within six months. AI use is substantially higher in large firms and knowledge-intensive sectors, with use rates reaching 50%-60% (60%-70%, employment-weighted) for very large firms in the Information, Professional Services, and Finance sectors. Among adopting firms, the scope of use remains limited: 57% of users integrate AI in three or fewer business functions, most commonly Sales and Marketing (52%), Strategy and Business Development (45%), and IT (41%). In 23% (41%, employment-weighted) of firms, workers use AI in work-related tasks. Writing, document analysis, and information search are the leading Generative AI use in tasks, though 65% of firms limit use to three or fewer tasks. The evidence points to both top-down and bottom-up diffusion channels: worker task use sometimes occurs without formal firmlevel adoption, and firm-level adoption sometimes occurs without worker task use. Most users (66%) rely on AI solely to augment tasks, while AI-related employment decreases are rare, occurring in only 2% of firms. Employment declines tend to concentrate among large firms and firms that use AI to substitute for worker tasks. Regression analysis shows a robust positive correlation between firm commercial performance and the breadth of AI integration, including functional deployment, task-level use, and operational investment. A distinct divergence emerges, however, with respect to labor outcomes. Functional breadth and operational investment are positively associated with employment decreases, whereas worker-task integration shows no significant link to headcount reduction once functional integration and operational investment are taken into account.

Keyword*:* Artificial Intelligence, AI, Technology Diffusion, Generative AI

---

* This paper is a revised version of the paper that accompanied the Spring 2026 release of the public data based on the second AI supplement in Census Bureau's Business Trends and Outlook Survey (BTOS). Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data (Project No. P-7529868, Disclosure Review Board (DRB) approval number: CBDRB-FY26-0159). We thank Robert Sienkiewicz and Sean Wang for helpful comments. We thank the following for background on their respective surveys Brent Meyer (Survey of Business Uncertainty), Alexander Bick (Real-Time Population Survey), and Jeff Allen (comparisons across surveys). The first version of this paper was completed when Lucia Foster (retired) was a Census Bureau employee. John Haltiwanger was also a part-time employee of the Bureau of the Census at the time of the writing of this paper.
E-mails: {kathryn.bonney,cory.breaux,emin.m.dinlersoz}@census.gov, Lucia.Foster.economics@gmail.com, halt@umd.edu, apande@umd.edu

1 Introduction

A major impediment to understanding how artificial intelligence (AI) is diffusing through the economy is that measurement has not kept pace with the ways AI is deployed within organizations. Existing evidence has largely come in two separate streams: firm-level surveys that ask whether business uses AI and worker- or household-facing surveys that measure whether individuals use AI in their jobs.[1] While each approach is informative, both are incomplete in isolation.

Firm-level measures indicate whether AI is adopted but provide limited insight into where and how broadly it is used within organizations – for example, whether deployment spans multiple business functions or remains narrowly concentrated. Conversely, worker-reported measures capture task-level exposure but may only partially reflect firm-level adoption, particularly when use is uneven across functions or emerges through decentralized experimentation. Moreover, simple adoption rates reveal little about the organizational investments required to implement AI or about its effects on task structure, employment and capital use. As a result, we lack systematic evidence on the internal structure of AI diffusion: how use varies across functions and tasks, how worker use maps to organizational adoption, and how these layers relate to firm outcomes.

This measurement gap is reflected in the divergence between firm- and worker-based estimates of AI use.[2] On the firm side, estimates from the second AI supplement in U.S. Census Bureau's Business Trends and Outlook Survey (BTOS) indicate that about 18% of firms report using AI over the prior two weeks (32% when employment-weighted) – for the supplement reference period of Nov 2025-January 2026. On the worker side, Bick et al. (2026) report that, using their January-February 2026 Real-Time Population Survey wave, 43% of workers used Generative AI (GenAI) for work. While these estimates are not directly comparable due to differences in concepts, sampling frames, respondents, and reference periods, the gap suggests that worker-based measures capture a broader range of AI interactions than those reflected in formal firm-level adoption. This divergence underscores the need for a unified framework that links organizational adoption with worker task use.

This distinction motivates a conceptual separation between two diffusion channels that may operate simultaneously. In a "top-down" channel, firms deliberately implement AI through deployment in business functions, workflow integration, and complementary investments, with worker use aligned to these formal structures. In a "bottom-up" channel, diffusion begins with worker-level experimentation and task-level tool use that may be concentrated in particular roles or tasks and only later becomes codified into firm policies and broader functional deployment as practices scale. These channels may have different empirical signatures within firms. Consider, for example, adoption that is narrow across functions and weakly reflected in worker use versus adoption that is accompanied by extensive worker engagement. These patterns cannot be assessed using firm-only or worker-only measures. Moreover, these distinct channels of diffusion and use may have different connections with firm outcomes. For instance, formal firm-level deployment and organizational changes to accommodate AI may be more relevant for output

[1] For firm surveys, see, e.g., Acemoglu et al. (2022, 2025), Baslandze et al. (2026), Bonney et al. (2024a,b), Yotzov et al. (2026), and Zolas et al. (2020). For worker- or household-facing surveys, see, e.g., Bick et al. (2025, 2026) and Census Bureau's Household Trends and Outlook Pulse Survey (HTOPS).

[2] There are also substantial gaps in adoption rates between firm surveys as well. As discussed below, much of this is accounted for by distinguishing between firm surveys that are heavily represented by large firms in AI intensive sectors relative to the BTOS survey which is nationally representative with survey weights that reflect the universe of US firms.

and employment changes, whereas decentralized use in worker tasks may be more associated with worker-level productivity improvements.

In this paper, we provide an integrated view of AI diffusion by jointly measuring AI use at three interconnected layers: firm-level adoption, deployment across business functions, and worker-task use. We do so using newly collected data from the second BTOS AI supplement, combined with core BTOS measures of firm AI use. This supplement expands earlier data collection by incorporating detailed information on functional deployment, worker use, and generative AI applications across tasks, as well as on task-level changes (augmentation, substitution, and creation), capital replacement, organizational adjustments, employment effects, and barriers to adoption. By linking these dimensions within a single instrument, the data enable a comprehensive analysis of where AI is used within firms, how it is implemented, how it relates to worker activity, and how these patterns correlate with firm performance and workforce outcomes. An important limitation is that worker-use measures are reported by firm respondents rather than directly by employees and thus reflect firm-level assessments of worker behavior rather than independent worker reports.[3]

Empirically, we use these data to characterize both the extensive and intensive margins of AI diffusion. We begin by documenting adoption patterns across firm size, sector, and geography, and by relating firm-level adoption to worker-task use. We then examine the breadth of AI deployment across business functions and tasks, highlighting the degree of overlap and divergence between organizational and worker-level use. Finally, we analyze how different dimensions of AI integration—functional deployment, task-level use, and operational investment—relate to firm performance, employment, and other outcomes using regression-based summaries.

Quantitative Highlights

Given the extensive set of novel findings from the BTOS supplement, we provide a more in-depth summary here of the main quantitative findings.

For the supplement reference period (Nov 2025 - Jan 2026), the firm-level AI use rate is 18%, or 32% when employment-weighted, while on the worker side, 23% of firms (representing 41% of employment) report AI integration into tasks related to business functions. Although these two adoption types overlap significantly despite differing reference periods, a notable decoupling exists between institutional and operational use. Specifically, in 36% of firms where workers utilize AI, there is no indication of formal firm-level adoption, a pattern suggesting a bottom-up or grassroots trajectory. Conversely, 19% of firms with formal AI use show no evidence of worker-task use, potentially indicative of a top-down approach facing implementation lags. These findings demonstrate that both top-down and bottom-up trajectories are relevant for understanding the microstructure of AI diffusion.

We also find that AI use is concentrated in large firms and knowledge- and human-capital intensive sectors, such as Information, Professional Services, and Finance. These sectors are expected to approach or exceed employment-weighted adoption rates of 50% in the next six months, indicating that a majority of workers in these sectors will soon be working in firms using AI. Importantly, the diffusion rates rise

[3]The "ideal" survey would link firm and worker level measurements with the same firm. Absent this "ideal," it would be difficult to ensure representative coverage by linking existing firm-level and worker-level surveys. We return to this point in Section 4.

to 50%-60% range (60%-70%, employment weighted) when large firms in these sectors are considered. Moreover, if the respondents' titles are filtered on "top executives and owners" in very large firms in these sectors, the rates reach 80%.

Within business functions, Sales and Marketing exhibits the highest functional AI adoption rate at 52% (48% employment-weighted), followed by significant integration in strategy and business development, IT, and R&D. On the worker side, GenAI is most frequently utilized for writing and editing, occurring in 85% of firms (92% employment-weighted) that report task-level adoption, with information search and document analysis also prevalent. Despite these prominent use cases, the scope of diffusion remains narrow: approximately 57% of functional adopters utilize AI in at most three business functions, and nearly 65% of firms limit worker task use to three or fewer task types.

Latent Class Analysis identifies five distinct types of AI-using firms based on their functional profiles. Minimalist adopters (37% of functional AI users) represent the largest group, characterized by low use probabilities across all business functions, while comprehensive adopters (4%) exhibit high probabilities of use across all functions. The remaining population shows some specialization: marketing specialists (31%) focus primarily on sales and outreach, while administrative integrators (15%) and technical strategists (12%) tend to use AI in managerial/finance/human resources and R&D/IT/strategy functions, respectively. These patterns confirm that while adoption is expanding, comprehensive functional integration is currently the exception rather than the rule.

For AI adopters, questions on AI-driven changes in worker tasks, employment changes and capital provide a firm's reported effect of AI on outcomes. We interpret these estimates as the firm's subjective assessment of the causal effect of AI use; that is, the respondent's comparison of the realized outcome with what it believes would have occurred in the absence of AI. For AI-driven changes in tasks, we analyze the prevalence of task augmentation, substitution, and creation associated with AI use. Augmentation is the primary effect: 44% of AI-using firms report augmenting worker tasks, and 66% of firms experiencing task effects do so exclusively. In contrast, pure task substitution and creation are rare, each occurring in only 5% of firms. Only about 7% of firms indicate that they are both augmenting and substituting tasks. While substitution remains limited in scope, its intensity is rising; the share of substituting firms reporting a "large number" of replaced tasks increased from 2.5% to 7% since the initial BTOS supplement – and the share of firms reporting "small number" declined from 85% to 71%.[4] These patterns suggest AI currently functions primarily as a labor-augmenting technology, though the shift toward higher-intensity replacement indicates a gradual evolution.

Consistent with the observed task effects that indicate mostly augmentation and limited intensity of task replacement, AI-driven employment change instances remain modest, with only about 5% of AI-using firms reporting any headcount impact – distributed nearly equally between increases (2.3% firm-weighted and 3.7% employment-weighted) and decreases (2.0% firm-weighted and 2.4% employment-weighted). In contrast, AI-driven capital substitution is more prevalent, with 16% of AI-using firms replacing existing software and equipment with AI-integrated solutions. In other words, AI appears to be already altering the investment behavior of firms in terms of new capital installation and upgrades or expenditures on software.

[4] A caveat to this comparison is that both the core questions and the supplement questions have changed since the first supplement, though the wording of the question on the extent of task replacement remains the same.

Although AI-attributed employment changes remain uncommon, they are concentrated in particular sectors, among the largest firms, and in firms experiencing broader changes in worker tasks. Employment increases remain more common overall and are prevalent when AI augments existing tasks, whereas declines are more prevalent when AI substitutes for tasks or is associated with multiple types of task change. These patterns provide early, suggestive evidence about where AI-related labor adjustment is emerging, although differences across various groups are not always statistically significant.

To provide additional analysis of the relationship between AI and outcomes, we construct firm-level diffusion indices capturing AI use, implementation, and impact. We first summarize three distinct dimensions of adoption: business function use breadth, worker-task integration, and operational investments to implement AI. These are aggregated into an overall AI use index, while a separate AI impact index synthesizes reported effects on tasks, labor, and capital. Finally, these components are combined into a holistic AI integration index to assess the depth and breadth of adoption across the firm population.

We provide descriptive statistics and correlations for these indices. Utilizing some of these parsimonious metrics, we employ regression-style summaries to estimate conditional associations between specific aspects of AI use and firm outcomes (including employment, sales, and overall performance), while controlling for a robust set of fixed effects. Unlike our AI-driven outcome analysis, these regression estimates cannot be interpreted as subjective treatment effects of AI but we nevertheless find the statistical associations instructive. We find that the breadth of business function use, worker task use, and operational investments all have positive and statistically significant associations with the likelihood of above-average current and future performance, and current sales increases. In terms of the magnitudes, functional use breadth has generally the strongest association with these outcomes, compared to task use breadth and operational investment. Second, while the breadth of functional use and operational investment are also positively and statistically significantly associated with the likelihood of current and future employment declines, worker task use breadth has no statistically significant relationship with these outcomes.

A more detailed Shapley decomposition based on a regression including indicators of all functional-use, task-use, and organizational-change question responses shows that the explanatory importance of AI use varies substantially across outcomes. Business functions account for the largest share of explained variation in performance, sales, and broader employment outcomes, with strategy and innovation, commercial and customer relationships, and administrative functions featuring prominently. By contrast, employment declines explicitly attributed to AI are explained primarily by complementary human-capital, organizational, technical-infrastructure, and data investments. These descriptive results suggest that where firms use AI is most relevant for general business outcomes, while whether and how they take concrete steps to implement it is especially indicative of AI-attributed employment reductions.

These findings support the view that broader AI integration (in terms of functions, tasks, and investment) is connected to better firm performance. At the same time, the observed associations with employment outcomes suggest that functional breadth and operational investments may signify top-down, deeper structural re-design aimed at capital-labor substitution. Conversely, worker use of AI tools may function as a bottom-up output/productivity enhancer with no concrete firm strategy to replace roles with AI – consistent with the dominance of task augmentation and low incidence of employment decreases. In addition, these distinct associations have implications for the productivity J-curves associated with AI adoption and use at the firm level. While task-level use may provide immediate augmentation

and productivity gains, broader structural investments and functional integration are likely the primary drivers of long-term productivity takeoff, even if they depress measured performance in the short run. Further identification of these connections in a causal framework is an important goal for future work.

The BTOS AI use by firms statistics at first glance seem substantially lower than other firm-level surveys such as those reported in Yotzov et al. (2026). The latter paper uses a sample of about 1000 firms as part of the ongoing Survey of Business Uncertainty (SBU). Consistent with a recent paper (see Allen (2026)), we highlight the concentration of heavy AI use in large firms in specific sectors (e.g., Information, Finance, Professional Services) to reconcile these differences. The BTOS is a nationally representative survey of the universe of US firms where it is useful to highlight that about 75% of US firms have less than 10 employees. We show that if the BTOS sample is restricted to large firms in heavy AI use industries, then AI use patterns using the BTOS and SBU results are broadly similar. Relatedly but distinctly, BTOS AI worker use (by asking firms) on an employment-weighted basis yield broadly similar patterns to Bick et al. (2026). While there are other issues of comparability (e.g., wording of questions, sample and reference periods), the concentration of AI use in specific industries and at large firms as well as distinguishing between firm and worker use are the dominant factors to help reconcile differences in estimated AI use.

Roadmap

The paper is organized as follows. Section 2 describes the BTOS AI supplement and the mapping of survey items into firm-, function-, and task-level measures. Section 3 benchmarks firm-level adoption and expectations while documenting heterogeneity across firms, sectors, and geography. Section 4 examines the three layers of AI use (overall, functional, and worker-task), quantifying their overlap and identifying distinct firm types based on functional use patterns. Section 5 analyzes AI-driven changes to work through task augmentation, substitution, and creation, as well as broader labor and capital effects. Section 6 investigates organizational adjustments, operational investments, and barriers to adoption. Section 7 details the construction of AI use and impact indices and analyzes their properties. Section 8 utilizes these indices to estimate conditional associations between multi-layer AI deployment and firm-level performance and employment outcomes. Section 9 contextualizes the BTOS results with other recent firm and worker surveys, and Section 10 summarizes key findings and concludes with directions for future research.

2 Data and measurement

2.1 The Business Trends and Outlook Survey (BTOS)

This section describes the Business Trends and Outlook Survey (BTOS) instrument and the second AI supplement – see Appendix A for the BTOS collection schema and questionnaire with this supplement. We provide the measurement architecture that underpins the paper's analysis, mapping survey content into a three-layer framework: firm-level adoption, functional deployment by the firm, and worker-task integration. This multi-layered approach allows us to distinguish between the extensive margin (binary adoption) and the intensive margin (the breadth of deployment across functions and worker tasks, and

the depth of task-level integration – e.g. the intensity of task replacement). We also document critical design features and wording changes that inform the interpretation of the results, particularly regarding cognitive testing and the evolution of the core AI instrument.

The BTOS provides near real-time measurements of business economic activity, and with the AI supplement, timely information on AI use at the firm, function and worker-task-level.[5] The BTOS is the only nationally representative large-scale survey of firms in the U.S. with such timely data, which is critical given the rapid developments of AI. Beyond the timely tracking, the BTOS provides guidance on the type of information on AI use that can and should be collected by the U.S. statistical agencies. Lessons learned from the BTOS collection helps inform plans to include AI questions on other core business surveys, such as the Annual Integrated Economic Survey (AIES), that underlie key economic indicators. The BTOS findings and analysis on business AI use also provides essential guidance for what needs to be collected to track the impact of AI on the U.S. economy.

For any given BTOS sample year, there are four or five 12-week cycles comprised of 6 biweekly collection panels (see collection schema in Appendix A). In addition to the core questions asked over the entire sample year, the BTOS often includes a topical supplement that lasts for one complete cycle (that is, 6 biweekly collections comprising a 12-week cycle). For the AI supplement data, responses from the entire 1.2 million business sample were collected over all six biweekly panels (November 17, 2025 to February 8, 2026). Non-response adjustment was performed for each biweekly panel, but all responses were pooled together, and the weights were adjusted to reflect that the entire sample is included in estimation.

2.2 Content development

The BTOS has included two core AI questions since September 2023 on business use of AI in the last two weeks and whether a business expects to use AI in the next 6 months.[6] The first AI supplement administered in December 2023 through February 2024 asked additional questions on AI use and its impact within firms.[7] Between September 2023 and November 2025, the BTOS utilized adoption metrics that identify AI use within the context of "producing goods or services". This phrasing was meant to be both an umbrella term and a deliberate filter, designed to capture economically significant AI integration into main business activities related to provision of goods or services. By anchoring the measurement to core output activities, the survey delineated fundamental operational adoption from auxiliary, incidental, or peripheral uses that may not exert a measurable impact on firm-level productivity or total value added.[8] Since core business activities related to "producing goods and services" can vary from business

[5]Data are collected for two weeks after the reference period and released two weeks later. Estimates reflect business conditions approximately four weeks earlier. For example, the first biweekly collection for the revised AI core questions: in the field November 17-30, 2025, reference period November 3-16, 2025, and published on December 4, 2025.

[6]The core AI questions for the period Sep 2023-Nov 2025 were (see also BTOS Content V2 Supplement 11.29.2023): Q23. Between MMM DD – MMM DD, did this business use Artificial Intelligence (AI) in producing goods or services? (Examples of AI: machine learning, natural language processing, virtual agents, voice recognition, etc.) (Yes/No/Don't know) Q30. During the next six months, do you think this business will be using Artificial Intelligence (AI) in producing goods or services? (Examples of AI: machine learning, natural language processing, virtual agents, voice recognition, etc.) (Yes/No/Don't know)

[7]See Bonney et al. (2024a,b) for the findings from this first AI supplement.

[8]Such uses could include, for instance, smart HVAC systems (AI models predict building occupancy based on calendar invites and historical weather data to pre-cool or heat specific zones, reducing energy overhead), automated onboarding (AI

to business, the core questions left room for interpretation by the respondents.[9] This original wording, adapted from the 2019 Annual Business Survey (also featured in the 2023 Annual Business Survey), was optimized for a pre-GenAI landscape that transformed rapidly in 2023 after the release of ChatGPT in late 2022. Following standard procedures for new survey content, questions were subject to pretesting through cognitive interviews of businesses.[10]

Additional cognitive testing conducted in 2025 revealed that respondents engaging in intermediation-related, marketing-focused, or administrative-heavy business activities had difficulties in associating their AI use with the term "production". Testing demonstrated that while many respondents initially reported no AI use, subsequent interviews revealed significant integration in functions like marketing, accounting, project management, and R&D – functions where AI use likely increased substantially due to GenAI since 2023.

Effective November 17, 2025, the BTOS transitioned to an updated wording for its core AI questions based on expert feedback and cognitive testing (see Questions 23 and 33 in Appendix A – these are the only two core AI questions). The revised phrasing, which shifts from use in "producing goods or services" to use in "any business function", was designed to minimize potential respondent under-reporting by "casting a wider net" to measure formal use of AI at the firm level. The transition to functional language helps the survey capture AI's role as a General Purpose Technology (GPT) utilized in both core activities related to production of goods and services and other vital business functions.

Following the change in core AI questions, the second BTOS AI supplement (fielded November 2025 – February 2026) significantly expanded upon the initial 2023–2024 AI supplement.[11] While maintaining longitudinal continuity in numerous aspects, the current instrument adopts a different approach to measuring AI use in business functions to better capture the cross-departmental nature of modern AI. Central to this supplement are two functional matrices that assess AI use across 15 distinct business functions over the previous six months and projected use for the following six months (Questions 24 and 34).[12] Two functions (production of goods and provision of services, products or merchandise) are included to these questions to provide a bridge to the legacy core questions and allow use to assess whether firms narrowly interpreted the phrase "in producing goods and services" as strictly "production" activity.

Additionally, the supplement incorporates granular information on AI use in worker tasks, inquiring about employee-level use in business function-related tasks (Question 30) and the more general utilization of GenAI by workers for task assistance (Question 31) with yes, no, and do not know responses. The extent of GenAI use in 9 types of worker tasks is assessed in Question 32 to collect data on what tasks

chatbots helping new hires find the dental insurance form or explain how to set up their VPN), or general worker personal AI use not related to formal work tasks.

[9]For example, providing a legal service and producing a computer chip have substantially different business processes that lead to the final product.

[10]See discussion in Bonney et al. (2024a). The challenge of measuring AI in BTOS was discussed in the Federal Economic Statistics Advisory Committee meeting December 2024, see FESAC Update Dec16.

[11]See the BTOS questionnaire with the first AI supplement under the Historical Data tab here: Questionnaire with AI Supplement 2023-2024.

[12]The 15 business functions are production of goods (e.g., making or assembling products, construction); provision of services, products, or merchandise (e.g., providing services, products, or merchandise to customers); strategy and business development; finance and accounting; sales and marketing; customer service; research and development; information technology; human resources; public relations and communication; management and administration; sourcing, supply chains, and purchasing; quality management and control; distribution; and legal and compliance.

most commonly incorporate assistance from GenAI tools.[13]

A key challenge in measuring the diffusion of AI is the "visibility gap" created by its integration into standard software workflows and decentralized departmental use. The multi-layered measurement framework in the supplement traverses the organizational hierarchy, moving from aggregate firm-level adoption down to specific business functions, worker usage, and granular task domains. Additionally, it evaluates the impact of AI on tasks, employment, and capital through a series of questions (Questions 25–28). One question asks about the nature of the relationship between AI use and employee tasks in terms of replacing employees, augmenting employees' work, or introducing an entirely new task (Question 25).[14] For those respondents who select that AI is being used to perform a task previously done by an employee, they are then asked about whether the number of tasks was a "small", "moderate," or "large" number of tasks (Question 26). AI using respondents are asked whether they used AI to substitute for operations previously performed by existing software or equipment (Question 27) and the impact on the business's total employment – increased, decreased, no change (Question 28). The framework further incorporates data on organizational changes and operational investments to implement AI, and barriers to AI use (Questions 29 and 35). For the former, respondents are asked to select all that apply from a list of 9 possible changes. For the latter, respondents who indicated they will not (or did not know if they will) be using AI in the next six months are asked to select all that apply from a list of 11 possible reasons.[15]

This high-resolution approach serves as a vital cognitive function for survey respondents: while a broad, top-level question may fail to trigger a positive response due to a lack of centralized oversight or AI awareness, the subsequent functional and task-specific matrices prompt respondents to evaluate their operations at a more concrete level. By providing these multiple entry points for identification, the methodology is designed to capture AI integration that may be missed at the extensive margin (the firm-level yes/no). This multi-layer measurement structure aims to mitigate item non-response and under-reporting, potentially offering a more robust proxy for the underlying diffusion rate and a more nuanced characterization of AI's structural footprint within the firm.

[13] The 9 tasks, based on the tasks used in Bick et al. (2025), are: information processing, paperwork, or filing; writing or editing documents, emails, or communications; interpreting, analyzing, translating, or summarizing documents; customer support; developing or researching new projects, processes, or products; searching for information or technical help; software coding or debugging, data analysis or visualization, tutoring, training, or learning. A write-in option for "other" tasks was also provided.

[14] "In the last six months, did this business use Artificial Intelligence (AI) to do any of the following?" and directs respondents to select all that apply from the following choices: Perform a task previously done by an employee; Supplement or enhance a task performed by an employee, Introduce a new task not previously done by an employee, and None of the above.

[15] Question 29 responses are: Trained current staff to use AI; Hired staff trained in AI; Purchased computing power or specialized equipment or software; Purchased cloud services or cloud storage; Changed data collection or data management practices; Developed new workflows; Used vendors or consulting services to install or integrate AI; Other (please describe); and This business did not make any changes to use AI.

Question 35 responses are: Too expensive; AI is not a mature enough technology yet; Lack of knowledge on the capabilities of AI; Concerns about privacy/security; Concerns about bias; Lack of skilled workforce; Lack of required data; Laws and regulations prevent or restrict use of AI; Previous or current use of AI did not meet expectations; AI is not applicable to this business; Other (please describe).

2.3 Respondent characteristics

A caveat to the worker task-related questions in the supplement is that the survey responses are often provided by a centralized firm-level respondent, typically an owner, high-level manager, or executive, rather than by individual workers – though in some cases technical or other personnel also provide input.[16] The survey structure thus generally relies on firm-level respondents to provide a high-level overview of AI integration. While this centralized reporting aims to identify formal organizational use, it may not capture every instance of idiosyncratic worker-led adoption. Furthermore, while the methodology is designed to encourage consistency across survey modules, we recognize that the actual respondent or the sources they consult may vary across different sections of the supplement. This structure also introduces a potential visibility bias: decentralized AI use by individual employees may be under-reported. While this might lead to a more conservative estimate of total task-level AI activity, it provides a more reliable measure of formal AI integration, which is more likely to drive the firm's long-term capital and labor adjustments.

2.4 Measurement goals

Beyond the AI-specific questions, the BTOS instrument also captures qualitative data on firm performance, inputs, and economic outcomes. These include an overall assessment of firm performance (levels relative to the average) and directional indicators (i.e. increase, decrease, no change) for revenue, total employment, demand, worker hours, and price (for inputs and outputs). We leverage some of these indicators to assess the relationship between different measures of AI use intensity and both realized and expected firm-level performance.

In summary, the rich data collected in the AI supplement enable us to:

1. Quantify alignment: Measure the overlap or divergence between firm adoption and worker-level use within the same enterprise. Firm-level use may signify more formal and centralized adoption, whereas worker-task level use may include decentralized and less formal adoption.

2. Map diffusion breadth: Identify the specific business functions and task domains where AI penetration across firms is most prevalent and classify firms into major types based on functional adoption patterns.

3. Design extensive and intensive measures of adoption: Construct measures that summarize various dimensions of AI adoption and impact at the micro level by combining survey questions.

4. Link patterns to outcomes: Analyze how heterogeneous patterns of functional and worker-task use correlate with reported effects on employment and firm performance.

For the analysis, we pool all firms surveyed during the supplement period, creating a sample of more than 117,000 distinct firms for the analysis – no firm was sampled more than once in this period. This large sample offers statistical precision in the granular analyses we undertake based on the detailed supplement questions.

[16] See Appendix Table C.3 for a broad classification of the BTOS respondent types. In cognitive testing, we observed that some respondents sought input from technical personnel more knowledgeable about AI implementation to answer some of the questions.

3 Patterns of firm AI use

We begin by analyzing time-series and cross-sectional patterns of AI adoption based on the BTOS core questions for current and expected use (Questions 23 and 33). The time-series analysis tracks the evolution of adoption rates and shifts following survey revisions, while the cross-sectional analysis identifies key diffusion patterns across industrial sectors, firm size classes, and geography.

3.1 AI use rate over time

Figure 1a illustrates the time series for estimated firm-level AI adoption.[17] Over time, the current firm-weighted use rate rose from 3.5% to 10% pre-revision, reaching 18% in the post-revision phase. Similarly, six-month expectations grew from 7.5% to 14% before increasing to 22% post-revision. A discrete jump in late 2025, coinciding with the commencement of the new BTOS cycle, can be attributed to three primary factors: (i) revised core question wording, (ii) continued adoption during the lapse in federal funding, during which data collection did not occur, and (iii) the introduction of the second AI supplement, which provided additional context and likely increased the propensity to report use.[18]

Employment-weighted estimates (Figure 1b) reveal significantly higher adoption intensities, with current use reaching approximately 35% and six-month expectations exceeding 40%. To evaluate the nature of these expectations, Appendix Figure C.1 compares forward-shifted expectations with realized outcomes 6-months ahead, adjusting crudely for the differences in reference periods. Although expectations tend to trend above realized use (likely due to optimism bias or differing reference frames), the two series follow similar trajectories, suggesting that firm-level expectations serve as a directional indicator for aggregate AI diffusion.

Although survey revisions preclude a direct comparison of aggregate pre- and post-revision adoption rates, business function use questions (Questions 24 and 34) provide a proxy to AI use "in producing goods or services"; specifically, the union of the responses for two business functions, "Production of goods (e.g., making or assembling products, construction)" and "Provision of services, products, or merchandise (e.g., providing services, products, or merchandise to customers)". This combination yields use rates of 6% for the prior six months and 7% for the next six months, lower than the most recent pre-revision estimates of 10% and 14%, respectively. Thus, the original core AI questions likely captured AI use beyond that in production and service provision.[19]

Table 1a utilizes contingency matrices to connect current and expected AI use, illustrating that both adoption and non-adoption are characterized by significant inertia. Conditional on current use (Panel B), 77.4% of firms plan to continue in the next six months, while 10.2%, likely representing experimenters, project-based users, or de-adopters, expect to stop. For current non-users, persistence is even higher, with 71.5% intending to remain non-adopters and 19.1% with no clear plans; only 9.4% plan to switch. Employment-weighted estimates in Table 1b (Panel B) reveal higher user persistence (85.0%), and a transition rate for non-users that nearly doubles to 17.7%. This pattern indicates that while aggregate AI use is sticky, larger firms appear to be more proactive in both maintaining and initiating adoption.

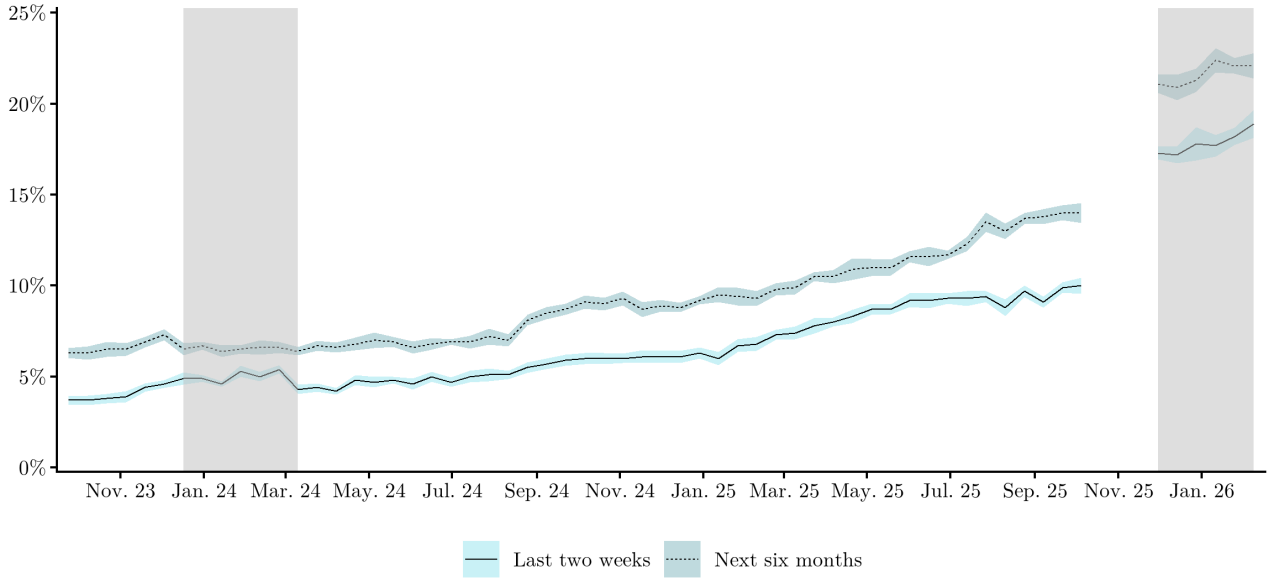
[17] For this exercise, we use the entire time-series, not just the second supplement period.

[18] A similar effect was observed around the introduction of the first BTOS AI supplement.

[19] Note that the 6% figure pertains to "last six months" as opposed to "last two weeks", indicating that the "last two weeks" use rate could be even lower.

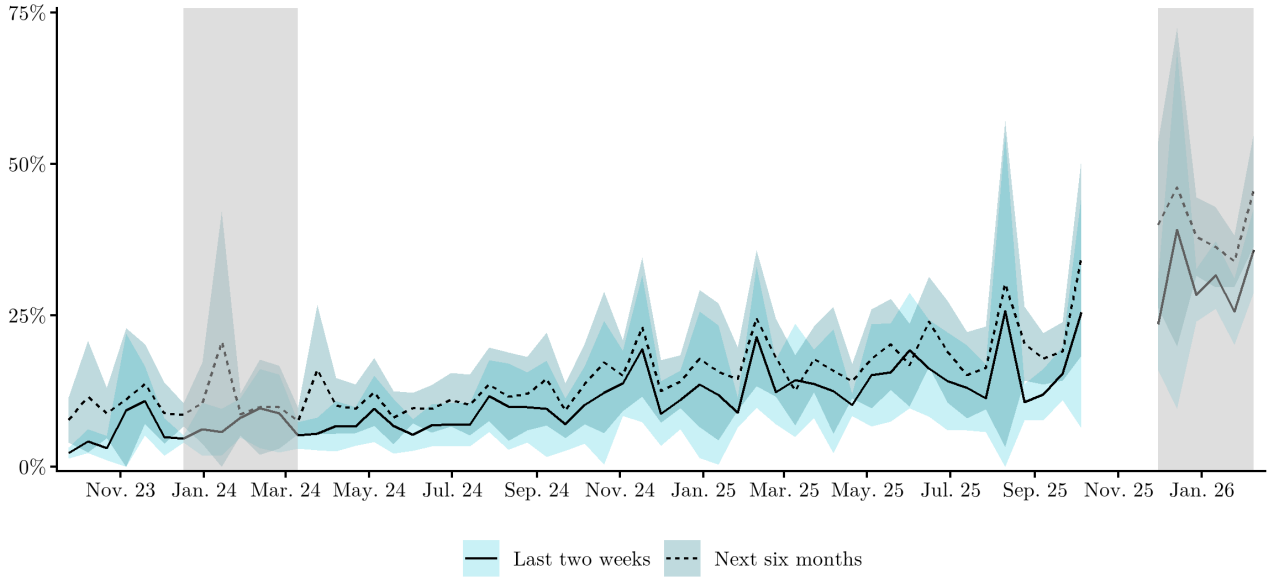
Figure 1: Firm AI use rates over time

(a) Firm-weighted



Note: Shaded areas indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Grey shaded areas represent AI supplement collection periods.

(b) Employment-weighted



Note: Shaded areas indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Grey shaded areas represent AI supplement collection periods.

Panels C in Table 1a and 1b indicate that future users are dominated by firms that persist in AI use, which account for 63.4% of future user firms (65.0%, employment-weighted).

3.2 Sectoral patterns

An analysis of the sectoral patterns in Figure 2 reveals substantial heterogeneity, characterized by a concentration of AI deployment within knowledge- and human capital-intensive industries. Prior to the revision of the core questions, adoption was also primarily concentrated in these industries – see Bonney et al. (2024a). While this pattern remains, the introduction of the new BTOS core AI questions in late 2025 triggered a significant, discrete level shift across sectors.

Under the revised instrument, the five sectors exhibiting the highest current firm-weighted AI adoption rate are Information (38%), Professional, Scientific, and Technical Services (34%), Educational Services (31%), Finance and Insurance (30%), and Real Estate (24%). In addition, firms spanning multiple sectors (indicated by "XX" in Figure 2) also have high use rates (24%). When contextualized against the nationwide use rate of 18%, the disparity highlights a profound digital divide; the Information sector's current adoption rate is more than double the national mean. Conversely, sectors relying on physical output, such as Manufacturing, Construction, and Transportation and Warehousing, and trade sectors (Wholesale and Retail), have much lower adoption rates.

The anticipated AI use rates over a six-month horizon uniformly exceed current utilization, though the magnitude of this "expectation gap" varies by industry. Expected future use reaches higher levels among the knowledge sectors: Information (43%), Professional, Scientific, and Technical Services (37%), followed closely by Educational Services (35%) and Finance and Insurance (34%). Multi-sector firms (XX) also indicate a higher future use rate of 34%.

Employment-weighted estimates reveal substantially higher rates. In the Information sector, the current use rate is 54% and is expected to rise to about 67% in the next six months. In other high-use sectors (Finance and Insurance, Professional Services, Education) and in multi-sector firms (XX), employment-weighted use rates are expected to approach or exceed 50%. At the same time, some lagging sectors, such as Manufacturing and Retail, are expected to experience significant increases in employment-weighted use rates over the next six months.

3.3 Patterns by firm size class

During the pre-revision period, the growth in current AI use was relatively modest among the smallest enterprises.[20] Firms with 1 to 4 employees saw their current use slowly drift from an initial baseline of approximately 4% to roughly 10%. In contrast, the largest size class (250 or more employees) demonstrated a steeper initial diffusion curve, with current adoption increasing from roughly 5% to 13% over the same time frame.

The introduction of the new core AI questions in late 2025 induced a significant level shift up across all size classes. Figure 3 shows the use rates by size class for the supplement period. Current firm-weighted AI use among the largest firms (250+) surged to 31% from 13%. Notably, the smallest classes also experienced a sharp upward revaluation, with current use for firms under 20 employees around 18%,

[20] See BTOS Public Data by Firm Size Class for these pre-revision patterns.

Table 1: Contingency tables for AI use by firms

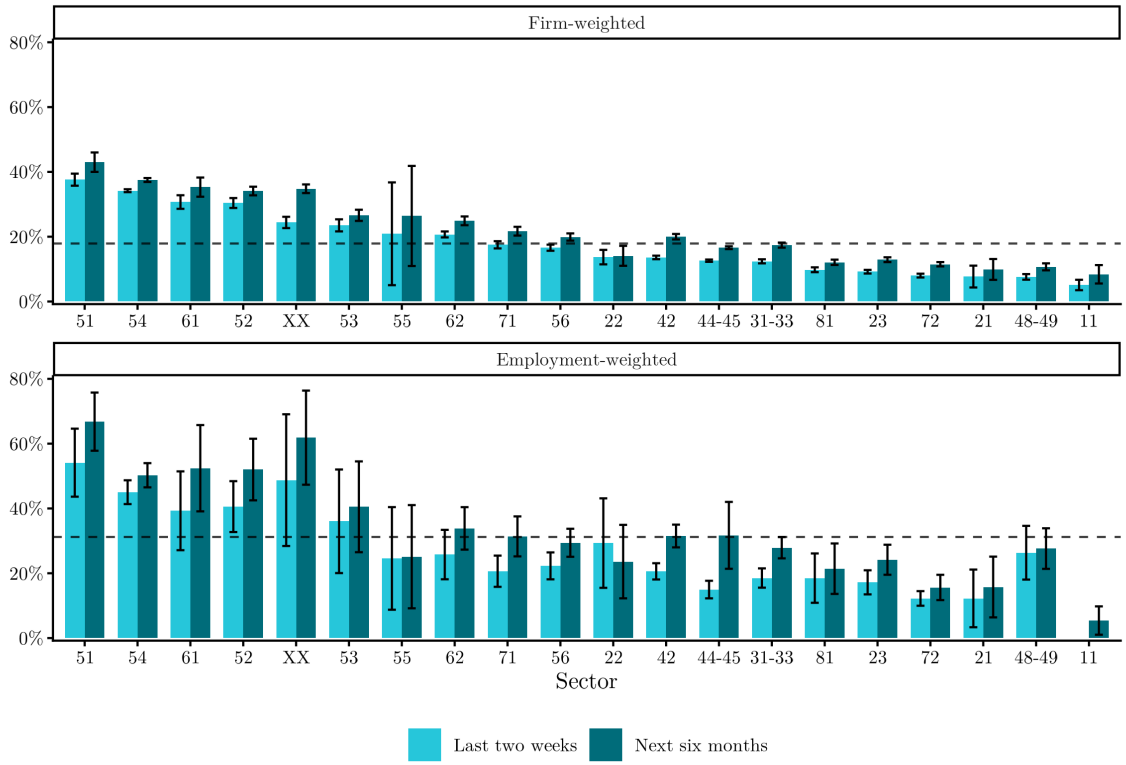
(a) Firm-weighted

		AI use, next six months			
		Yes	No	Don't know	Total
Panel A. Unconditional Shares					
AI use, last two weeks	Yes	13.7%	1.8%	2.2%	17.7%
	No	6.7%	51.2%	13.7%	71.6%
	Don't know	1.2%	3.3%	6.3%	10.8%
	Total	21.6%	56.3%	22.2%	100%
Panel B. Conditional on Current AI Use (rows)					
AI use, last two weeks	Yes	77.4%	10.2%	12.4%	100%
	No	9.4%	71.5%	19.1%	100%
	Don't know	11.1%	30.6%	58.3%	100%
Panel C. Conditional on Expected AI Use (columns)					
AI use, last two weeks	Yes	63.4%	3.2%	9.9%	
	No	31.0%	90.9%	61.7%	
	Don't know	5.6%	5.9%	28.4%	
	Total	100%	100%	100%	

(b) Employment-weighted

		AI use, next six months			
		Yes	No	Don't know	Total
Panel A. Unconditional					
AI use, last two weeks	Yes	25.6%	1.8%	2.6%	30.1%
	No	8.8%	27.7%	13.2%	49.7%
	Don't know	5.0%	1.8%	13.5%	20.3%
	Total	39.4%	31.3%	29.3%	100%
Panel B. Conditional on AI use last two weeks (rows)					
AI use, last two weeks	Yes	85.0%	6.0%	9.0%	100%
	No	17.7%	55.7%	26.6%	100%
	Don't know	24.6%	8.9%	83.5%	100%
Panel C. Conditional on expected AI use next six months (columns)					
AI use, last two weeks	Yes	65.0%	5.8%	9.2%	
	No	22.3%	88.4%	44.8%	
	Don't know	12.7%	5.8%	45.9%	
	Total	100%	100%	100%	

Figure 2: AI use rates by sector



Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors.
Employment-weighted, last two weeks estimate suppressed for Sector 11 due to data quality concerns.
Charts sorted by firm-weighted, current use.
Dashed line represents nationwide AI use rate (last two weeks).

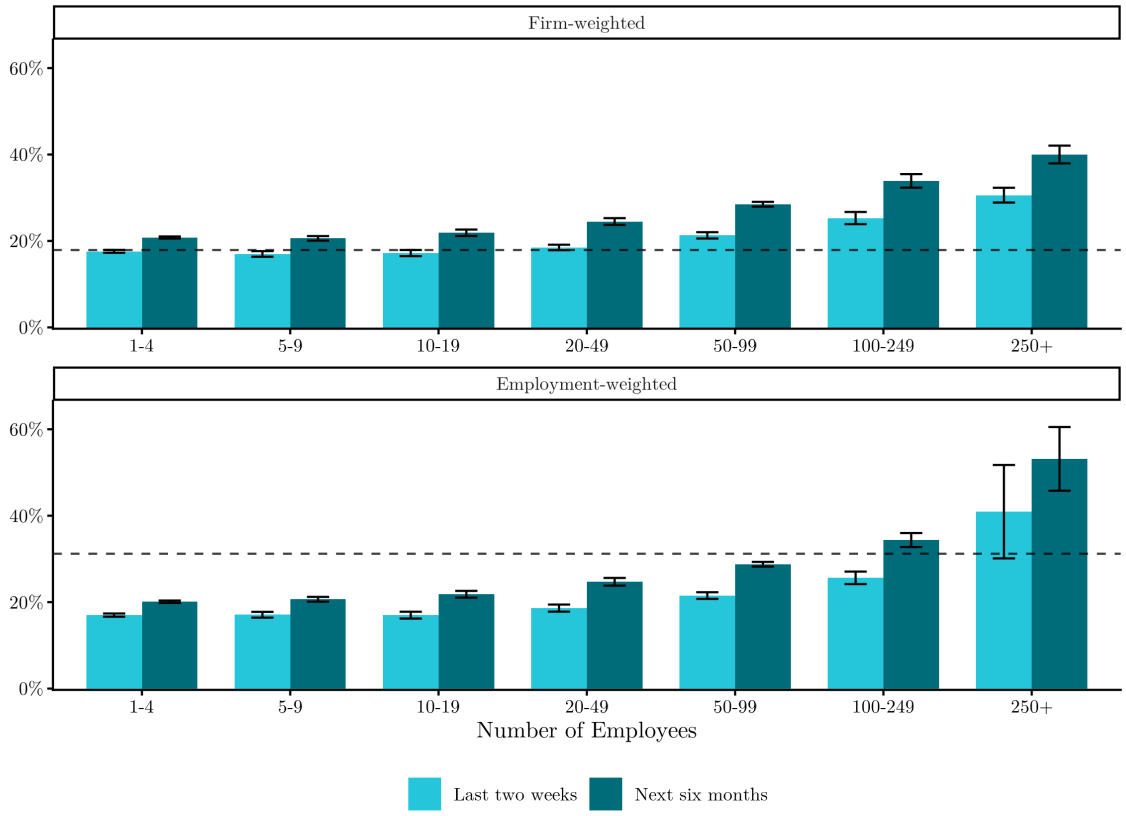
NAICS sectors: (11) Agriculture, Forestry, Fishing and Hunting; (21) Mining, Quarrying, and Oil and Gas Extraction; (22) Utilities; (23) Construction; (31–33) Manufacturing; (42) Wholesale Trade; (44–45) Retail Trade; (48–49) Transportation and Warehousing; (51) Information; (52) Finance and Insurance; (53) Real Estate and Rental and Leasing; (54) Professional, Scientific, and Technical Services; (55) Management of Companies and Enterprises; (56) Administrative and Support and Waste Management and Remediation Services; (61) Educational Services; (62) Health Care and Social Assistance; (71) Arts, Entertainment, and Recreation; (72) Accommodation and Food Services; (81) Other Services (except Public Administration); (92) Public Administration; (XX) Multi-sector Firms

up from 10%. This indicates that while the size-adoption gradient remains intact, even micro-enterprises are engaging in higher AI use rates.

A nuance emerges when comparing these current utilization levels with anticipated integration over a six-month horizon, revealing a widening "expectation gap" that scales with firm size. For the smallest cohort (1 to 4 employees), expected future use under the revised instrument sits at roughly 21%, only slightly exceeding their current use rate of 18%. Conversely, the data reveals an aggressive forward-looking posture among larger enterprises. Firms with 100 to 249 employees expect future adoption to reach 34% compared to a current use rate of 25%, while the largest firms (250+) project a surge to 40% expected use against their 31% current baseline. In other words, while small businesses may be adopting a lower trajectory for adding AI capabilities, large and medium-sized enterprises are systematically preparing for more AI implementation in the immediate future.

These patterns are generally mimicked by the employment-weighted estimates in Figure 3. Notably, the estimates indicate that more than 50% of employment in the largest size class will be accounted for by AI-using firms in the near future. At the same time, small to medium size classes have employment-weighted use patterns that are very similar to the corresponding firm-weighted ones.

Figure 3: AI use rates by firm size



Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Dashed line represents nationwide AI use rate (last two weeks).

3.4 Geographic patterns

State-level analysis reveals regional disparities in AI integration. From September 2023 until the late-2025 survey revision, adoption growth was concentrated in specific regional hubs and among firms operating across state lines. Multi-state firms, designated by the "XX" classification, established the highest baseline of current use and sustained the most aggressive initial growth prior to the wording change. Among individual states, a group of technologically dense states, such as Utah, Colorado, Texas, and Washington, closely tracked this multi-state cohort, demonstrating robust early adoption trajectories.

The implementation of the revised BTOS core AI question in late 2025 induced a nationwide level shift, yet the magnitude of this structural break was asymmetric across regions. The maps in Figure 4 show use rates across states. Under the revised instrument, Panel A indicates that multi-state firms (that tend to be larger) continue to dominate the distribution, reporting the highest current firm-weighted use rate at approximately 29%. Among individual states, Colorado, Arizona, Nevada and Florida register the highest current rates, clustering between 20% and 24%. Conversely, a cluster of lagging states, including North Dakota, West Virginia, Arkansas, Louisiana, Mississippi, Alabama anchors the bottom of the distribution. These regions have current adoption rates of 9% to 12%, potentially indicating slower adoption in certain geographies. Employment-weighted estimates in Panel B indicate more evenly distributed use rate across states.

Anticipated use rates across states over the next six months are shown in the maps in Appendix Figure C.2a and C.2b (firm- and employment-weighted, respectively). Most states are poised to experience increases in their use rates, even the ones that are currently lagging – resulting in a somewhat more evenly distributed adoption rate across states. Leading cohorts are forecasting further integration: multi-state firms anticipate future firm-weighted use rates approaching 38% (up from 29%), while firms in states like Arizona, Nevada, Utah, Texas, Florida, Delaware, and Colorado expect their use rate to surge to 25% or higher. While multi-state enterprises and tech-centric states (particularly in the West) appear to be systematically preparing for a widespread AI diffusion, some states in the South, Midwest, and Central U.S. will continue to lag in the near future despite the anticipated increases in adoption rates.

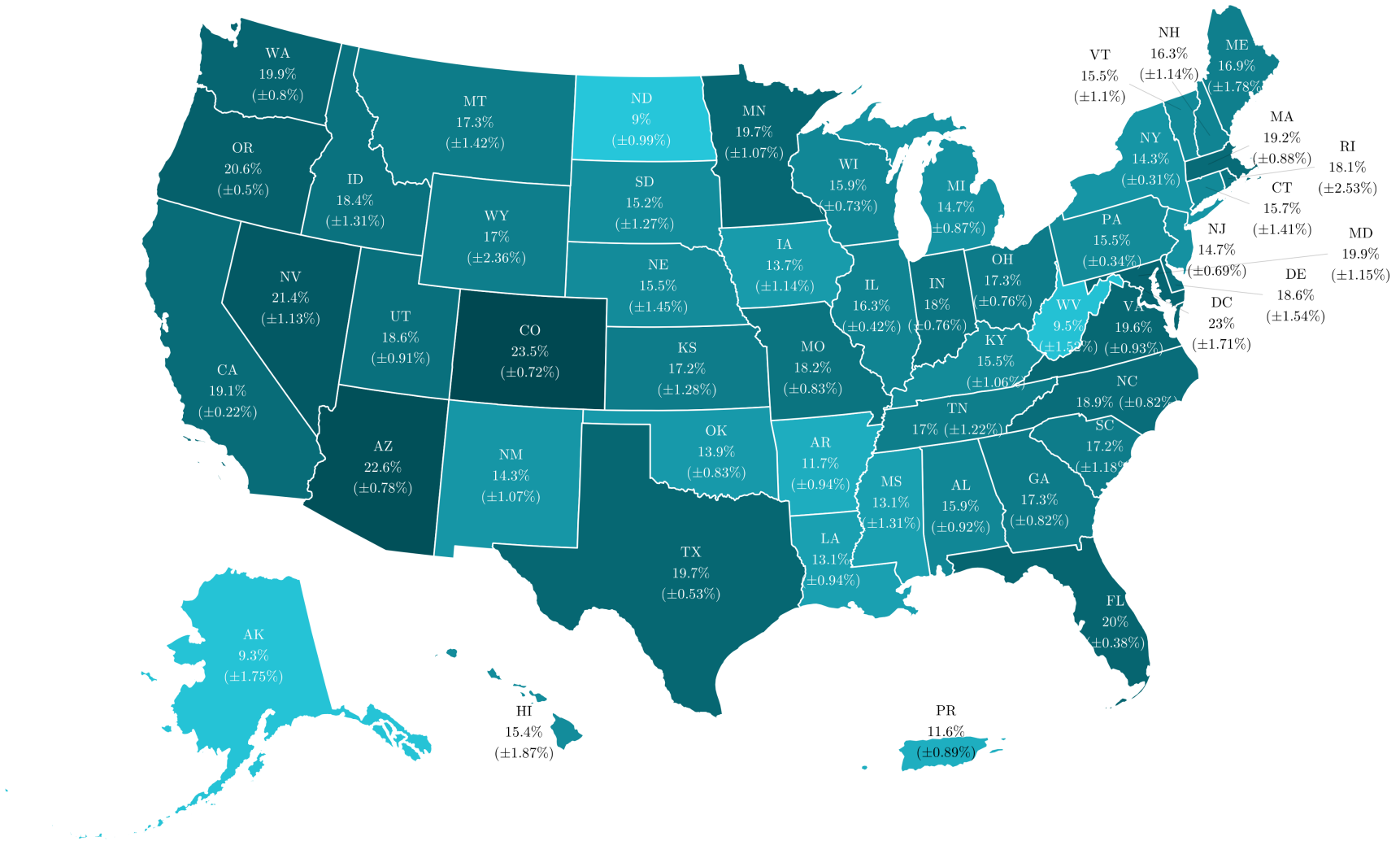
4 AI use within firms

Next, we describe where AI is used inside firms and how the layers of use relate. To fix concepts, Figure 5 provides a representation of AI adoption within the firm for a business function. At the foundational layer, the firm may implement AI as an autonomous infrastructure (such as background data processing or automated optimization engines) that operates with little or no individual worker intervention. Consequently, a firm may have high levels of AI utilization in a specific business function while the associated worker-level tasks remain largely manual. In this scenario, a worker (such as Worker 1) acts as a passive recipient of AI-generated output rather than an active user of the technology. Conversely, the collaborative case (Worker 2) represents an integrated state where workers utilize AI copilots or interfaces to augment their specific tasks, creating a bidirectional flow between labor and AI. There could also be cases where workers adopt external tools independently of firm infrastructure (Worker 3), and workers whose tasks remain untouched by AI despite the firm's broader digital transformation (Worker 4).

This framework demonstrates that worker-level use does not necessarily translate into firm-level AI

Figure 4: AI use rates across states (last two weeks)

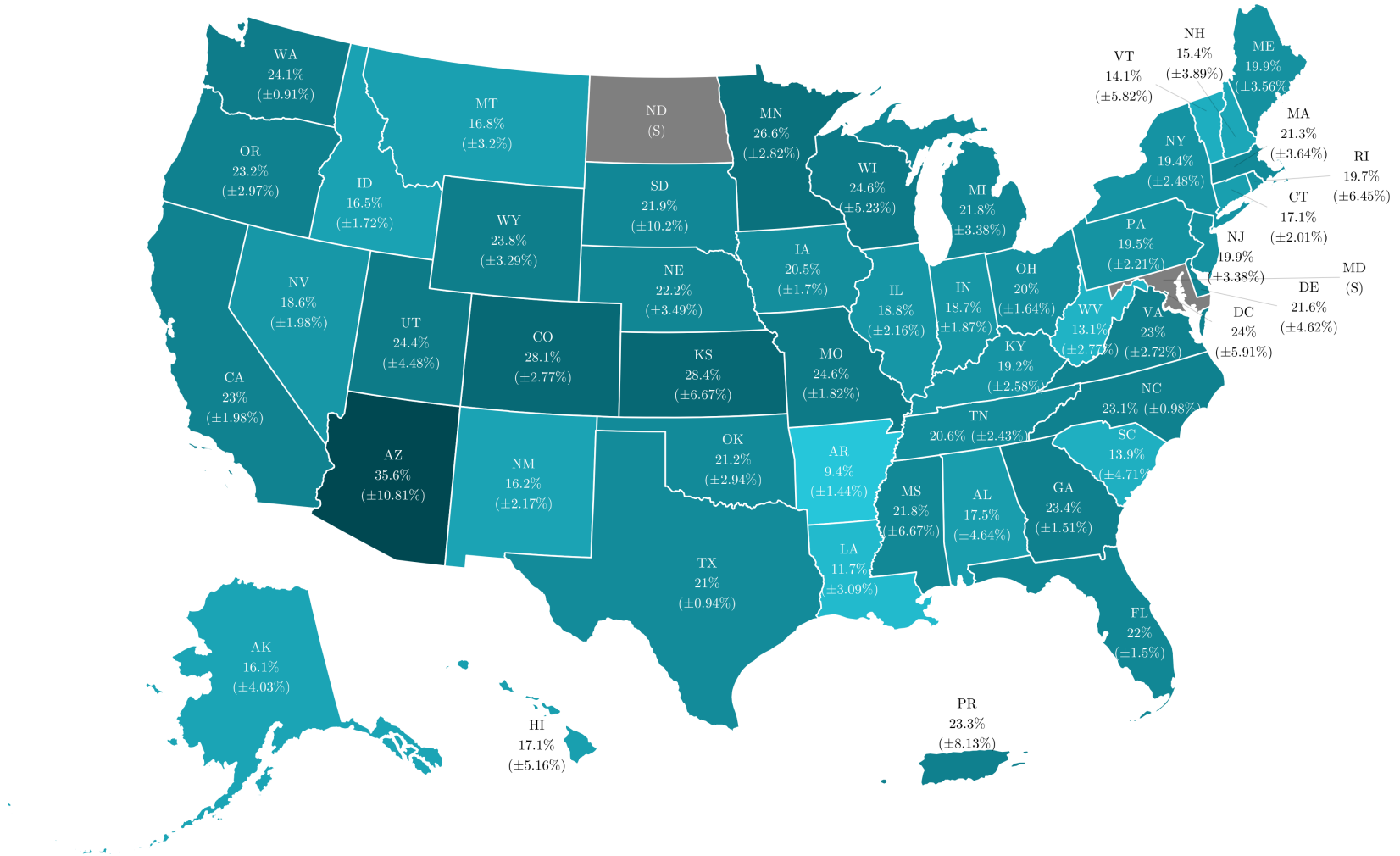
(a) Firm-weighted



Note: Values after estimates are ± standard errors.
Multi-State Firms: 29.3% (±0.99%)

Figure 4: AI use rates across states (continued)

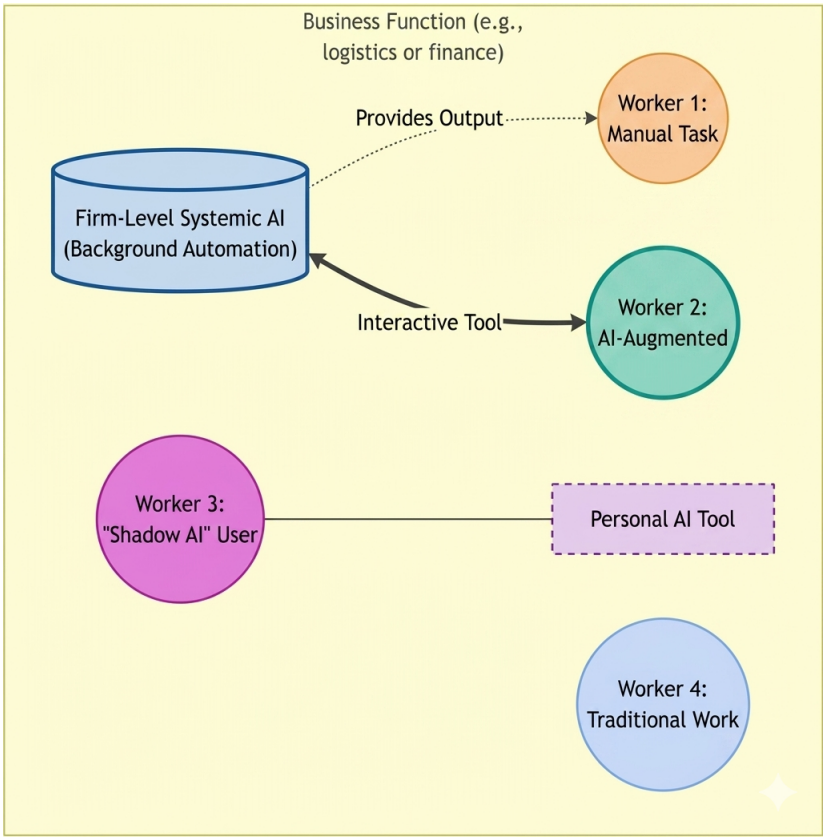
(b) Employment-weighted



Note: Values after estimates are ± standard errors.
Multi-State Firms: 46.7% (±7.6%)
(S) = Estimate suppressed due to quality concerns.

adoption. Similarly, firm-level centralized or formal adoption in a business function does not necessarily imply that workers use AI tools in daily work-related tasks. As we noted in the Introduction, the ideal measure would provide both a firm-level and worker-level perspective in order to capture these different patterns.

Figure 5: Generic AI use patterns in a business function



4.1 The relationship between firm AI use and worker AI/GenAI use

We start by documenting how firm-level AI use differs from worker task use. As mentioned earlier, a methodological constraint is that the survey was not sent separately to individual workers in a firm. The firm and worker AI use questions were thus likely answered by the same respondent in the firm – though consultations may have been made with workers in preparing a response. In some cases, this can generate positive correlation between the responses: in a firm that uses AI intensely, a respondent may believe workers are also using it, even though AI may be part of background automation that does not interact with workers (see Figure 5). In other cases, discrepancies may arise due to the respondent's lack of information on individual workers' practices (visibility bias) – though knowledge of at least one worker using AI would be sufficient for an affirmative response for the worker use questions. Discrepancies may also stem from differing reference periods: two weeks for firm-level use versus six months for worker tasks. Crucially, our worker-level AI use rate denotes the incidence of task-level AI use within firms rather than the share of AI use in the individual worker population.

Figure 6 utilizes Venn diagrams to detail the structural relationship between formal adoption and

operational execution. While formal firm-level AI use stands at 18%, worker-task use is notably higher at 23%, indicating that AI activity often diffuses through channels outside of centralized adoption. A breakdown of this overlap reveals that while 14.4% of firms utilize both layers, 6.8% exhibit worker-level adoption in the absence of reported firm-level use – a signal of bottom-up diffusion. Conversely, only 2.5% of firms maintain formal adoption without corresponding use in worker tasks. Among firms with any AI presence, 61% demonstrate integrated use, whereas 29% exhibit solely worker-level use and 10% only firm level implementation. The union of these layers implies an aggregate AI incidence of nearly 24%, a figure significantly higher than the firm-level adoption rate of 18%.

These patterns suggest that worker-task driven AI adoption is a significant contributor to overall AI use rate among firms. As a result, both the top-down and bottom-up adoption appear relevant in the diffusion. Notably, 39% of AI-using firms exhibit adoption through a single channel alone, with 29% driven by worker-level task use and 10% by formal firm-level adoption.

Next, we assess worker use of AI in work-related tasks in support of business functions (Question 30) versus worker use of GenAI more generally in any work-related task (Question 31). Workers use AI in 23% of firms, larger than the rate of use of GenAI that sits at 21%. While the overlap in these two types of use is substantial, in 2.8% of firms, workers use GenAI, but not AI. This discrepancy may not be surprising. The question on worker AI use (Question 30) explicitly states employee use of AI "to assist in any work-related tasks that support business functions", whereas the GenAI question (Question 31) only specifies "to assist in any work-related tasks" – to capture broader GenAI use patterns among workers. This difference in wording may have induced some respondents to consider, in their responses to Question 31, certain work-related tasks that are not strictly part of a business function. Another potential explanation may be a "brand name" fallacy: some respondents may identify AI with GenAI only.[21] In other words, some workers may be using sophisticated AI tools, but the respondent only recognizes GenAI because it is more visible in the processes leading to the final output.

Additionally, 4.1% of firms report worker-level AI use without GenAI integration, likely reflecting specialized applications such as Agentic AI for workflow automation or machine learning for predictive analytics. Within the 23.9% of firms reporting any worker-level AI or GenAI activity, 71% exhibit an overlap between the two categories, while 11% report GenAI use exclusively and 19% utilize non-generative AI only.

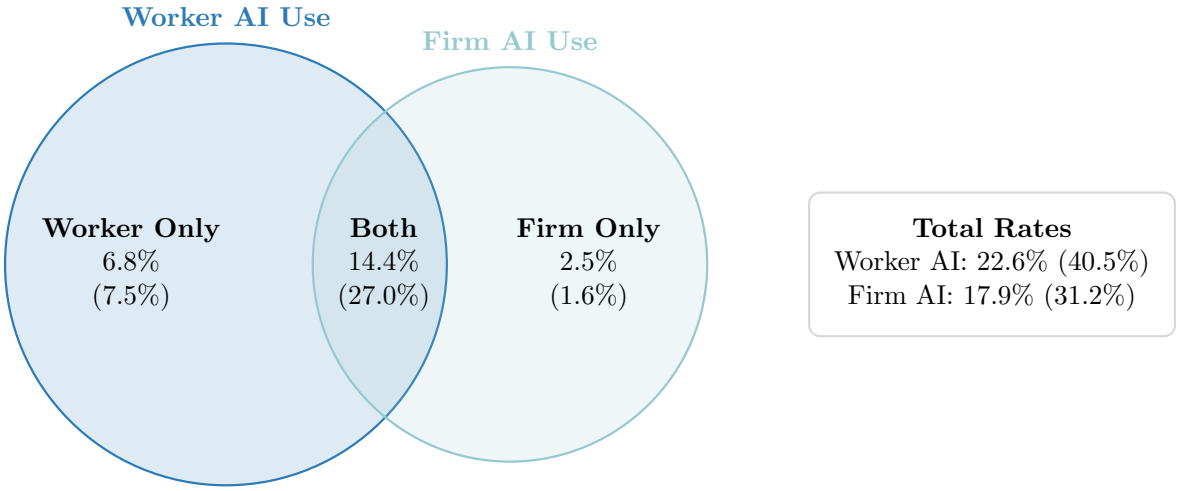
Table 2 presents conditional AI use rates, illustrating the estimated probability that a firm adopting one layer of AI also integrates the other two. These probabilities highlight a notable asymmetry in adoption trajectories: while 80.8% of firms with formal adoption report worker-level integration, only 64.0% of firms with active worker-task use have established formal firm-level use (Panel A). Furthermore, conditional on formal firm-level use, worker GenAI adoption (71.8%) lags behind general worker AI use (80.8%).

Employment-weighted rates in Panel B paint a similar picture. Worker AI use rates conditional on firm use remain higher than the firm use rates conditional on worker AI use. These findings reinforce the salience of the bottom-up channel as a critical component of aggregate AI penetration.

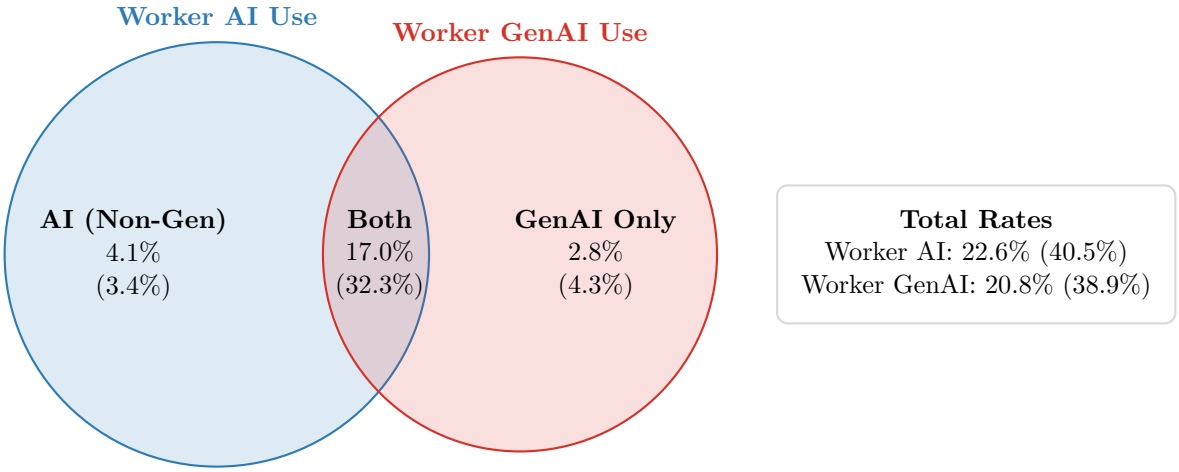
[21] For example, in a marketing firm a few workers may use AI-driven Programmatic Advertising (which uses complex AI to bid on ad space) and many others may rely on, e.g., Midjourney, for image generation. When faced with the questions, the firm-level respondent (e.g., a manager) might think "We are not a tech company, so we don't use AI for technical purposes, but we use GenAI for our marketing art."

Figure 6: Firm AI use, worker AI use, and worker GenAI use

Panel A. Firm versus worker AI use



Panel B. Worker AI versus GenAI use



Note: Parentheses denote employment-weighted shares. The Venn diagrams exclude "don't know" responses. Because of this restriction, the sum of the Venn diagram segments will not match the corresponding total adoption rates calculated across all firms (shown inside the boxes).

Table 2: Conditional firm and worker AI use rates

Panel A. Firm-weighted				
		Firm AI Use	Worker AI Use	Worker GenAI Use
Conditional on:	Firm AI use	100.0%	80.8%	71.8%
	Worker AI use	64.0%	100.0%	75.5%
	Worker GenAI use	61.8%	82.1%	100.0%
Panel B. Employment-weighted				
		Firm AI Use	Worker AI Use	Worker GenAI Use
Conditional on:	Firm AI use	100.0%	86.5%	76.0%
	Worker AI use	66.6%	100.0%	79.8%
	Worker GenAI use	60.8%	83.0%	100.0%

4.2 AI use in business functions

Figure 7 presents the recent (last six months) and expected future (next six months) AI use rates by business function based on Questions 24 and 34. The expected future use rate is based on Question 34, which is prompted conditional on reporting expected future AI. To make the recent and future rates comparable, we condition the recent use rates on firm AI use in at least one function.

Firm-weighted estimates indicate that sales and marketing is the most common business function where AI is used – by 52% of the firms using AI in at least one business function. Strategy and business development comes next (45%), followed by information technology (41%) and R&D (40%). Other highly used functions are administrative ones (public relations and communication, management and communication, legal and compliance), as well as customer service, finance and accounting and provision of services and products. Use in production and supply-related functions, quality control, and distribution are relatively less common. One reason for these low rates is these functions may be specific to some sectors and business types (e.g. supply-chain and quality management in manufacturing).

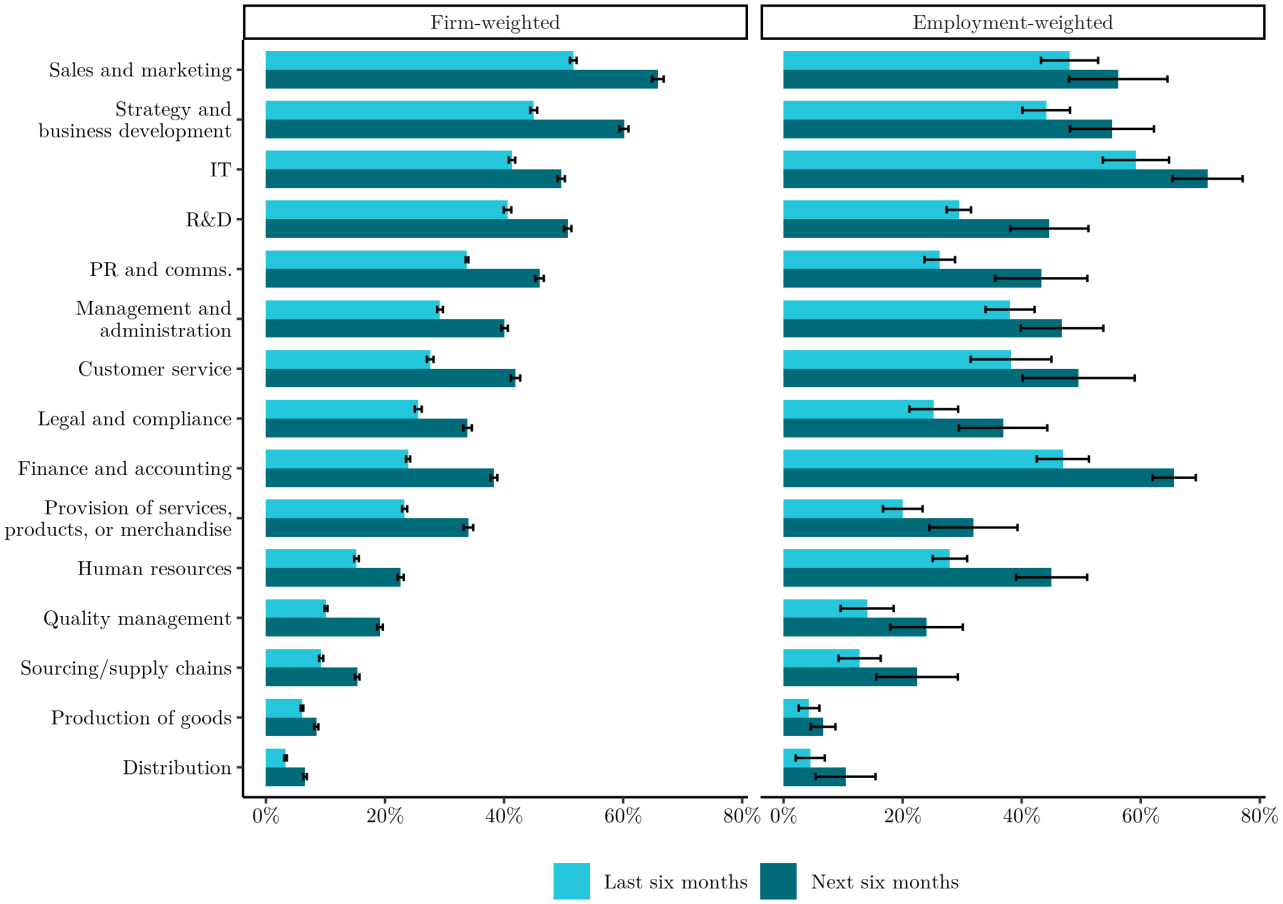
Firm-weighted expected future use rates indicate an increase in use across all business functions. Largest percentage-point increases are anticipated in sales and marketing, strategy and business development, customer service, and finance and accounting.

Employment-weighted estimates reveal that IT is the most commonly used function, followed by finance and accounting and sales and marketing. Notable percentage-point increases in employment-weighted estimates are expected in finance and accounting, human resources, R&D, and public relations in the near future.

An important measure of within firm AI diffusion is the number of functions integrating AI. Figure 8 shows that 24% (firm-weighted) of firms with functional use indicate they use AI in only one business function. The rates decline nearly monotonically as the number of functions increases, with about 1% of AI-using firms using it in all 15 functions. Employment-weighted shares exhibit a similar pattern overall but also point to somewhat higher shares for large number of functions compared to the firm-weighted shares.

Overall, functional use patterns indicate concentrated use in sales and marketing, strategy and busi-

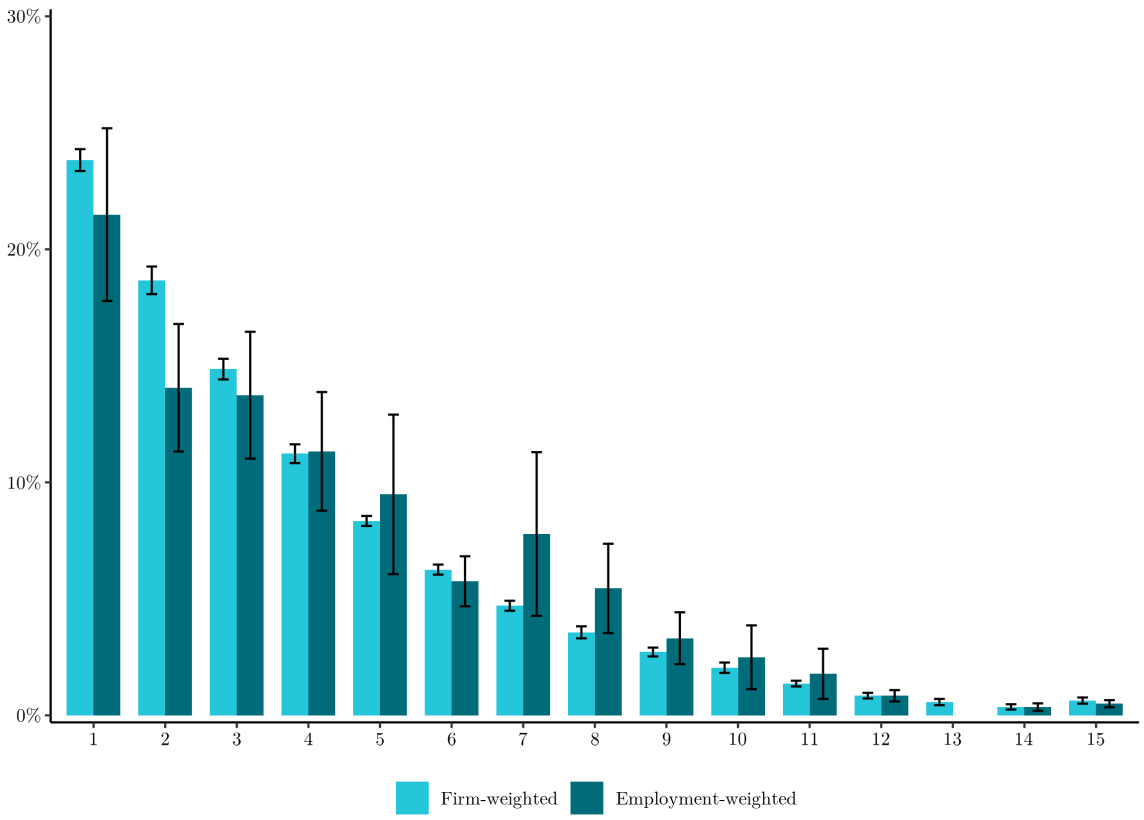
Figure 7: AI use rate by business function – conditional on use in at least one function



Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Response rates are conditional on answering 'Yes' to AI use for at least one business function.

ness development, finance and accounting, and IT functions. At the same time, conditional on functional use, most firms use AI in relatively small number of functions – 57% of users (firm-weighted) utilize it only in 1-3 functions.

Figure 8: Histogram of the number of AI-integrated business functions



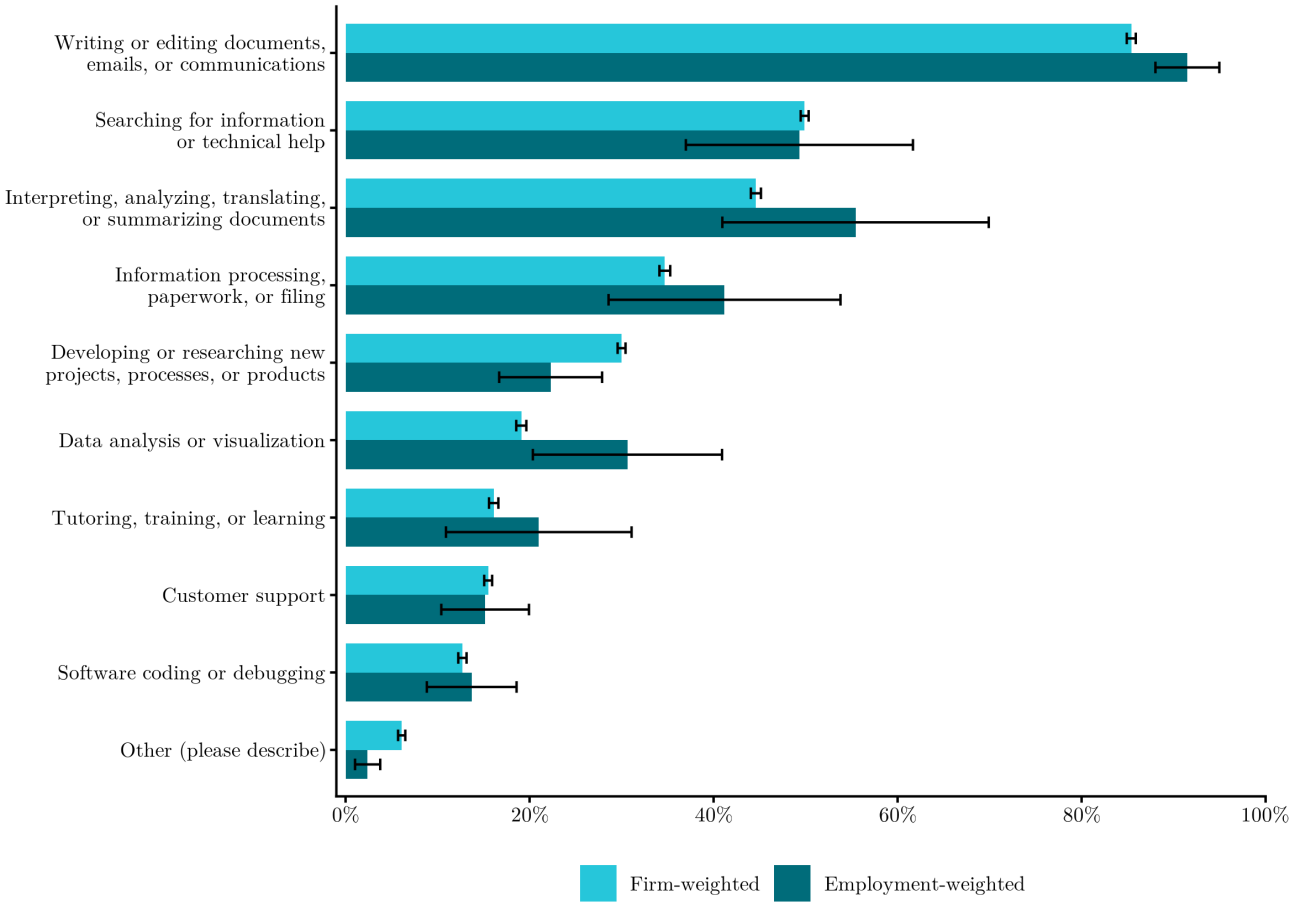
Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Employment-weighted, n=13 suppressed due to quality concerns.

4.3 GenAI use in worker tasks

Analysis of GenAI use patterns in tasks (Question 32), conditional on affirmative worker-level use, indicates that writing and editing has the highest rate of GenAI use, reported by 85% of firms (Figure 9). Information search follows at a distant second (50%), with document analysis, information processing, and R&D representing other high-use cases. Specialized tasks, such as data visualization, customer support, and software debugging, exhibit lower incidence, likely due to their concentration in specific industries/firms or a continued requirement for rigorous human oversight.

A measure of the extent of within-firm AI integration for worker tasks is the number of distinct worker tasks where GenAI is used (Figure 10). Conditional on GenAI use in work-related tasks, about 26% of firms use it in only 1 task, and around 2% of firms use GenAI in 9-10 tasks (including the "other" task category). Employment-weighted rates show a similar profile, but with somewhat higher rates at 4 and 5 tasks, and generally higher shares in larger number of tasks relative to the firm-weighted rates. As in the case of business functions, we observe concentrated use in worker tasks, with writing and editing and

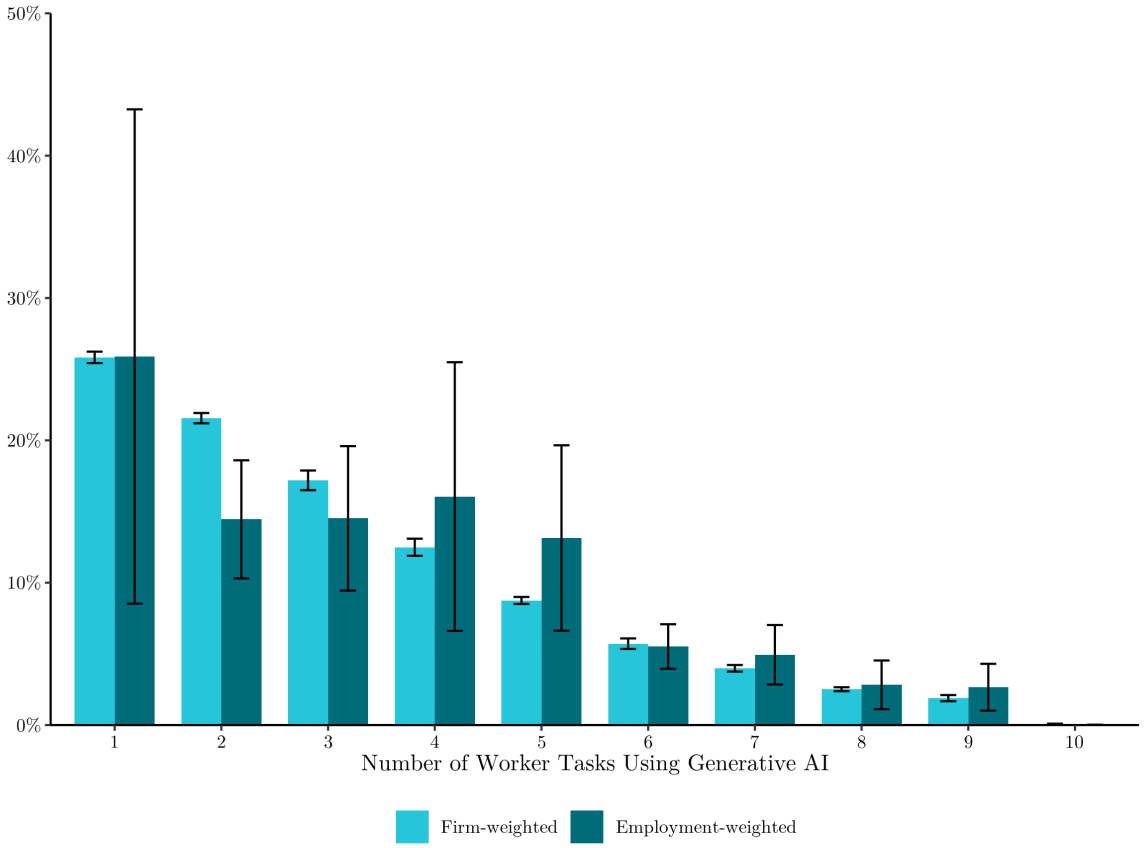
Figure 9: Worker GenAI use rates by task – conditional on use in at least one task



Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Q32 is conditional on responding 'Yes' to Q31: Generative AI use by employees.

information search leading the task-level adoption. In nearly 65% of GenAI use cases (firm-weighted), use is limited to 1-3 worker tasks.

Figure 10: Histogram of the number of AI-integrated worker tasks



Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors.

4.4 Firm types by functional AI use

An important question is what types of AI-using firms are emerging as the technology is integrated into critical business functions. Are there firm types with specialized use in certain business functions or is the functional adoption profile relatively similar across all firms? To uncover the salient firm types based on functional AI deployment, we employ a Latent Class Analysis (LCA) to categorize firms based on their adoption patterns across 15 distinct business functions – conditional on adoption of at least one function. Unlike traditional clustering methods such as K-means, which relies on geometric distance, LCA is a model-based approach that assumes the observed functional use patterns are driven by an underlying, unobserved (latent) categorical variable representing a firm's adoption type.[22]

A challenge in analyzing survey data from the Business Trends and Outlook Survey (BTOS) is the sampling design. To ensure that the identified types are representative of the broader business population, we implement a weighted LCA approach that incorporates sampling weights directly into the likelihood function.[23]

[22]See Goodman (1974), Vermunt (2003), and Magidson and Vermunt (2004) for the basics of LCA.

[23]See Vermunt and Magidson (2007) and Vermunt (2017). By treating the weights as part of the probabilistic framework,

We evaluated model fit across solutions ranging from two to ten classes, utilizing the Bayesian Information Criterion (BIC) to assess class separation.[24] The five-cluster solution was selected to balance between parsimony and descriptive power. While the scree plot in Figure C.3 in the Appendix indicates that two classes may provide the sharpest separation, we opted for 5 classes to give a more nuanced picture of functional AI use patterns, as the two classes yield a relatively uninformative separation: "high" versus "low" functional use likelihood across all functions.

The five classes identified reveal that AI integration is not monolithic but a series of distinct strategic paths. The classification allows us to differentiate between generalists, who deploy AI across many departments, and specialists, who concentrate AI use in a narrow set of functions.[25] The five classes can be described as follows – see Figure 11 for the use probabilities for each business function by class:[26]

- **Minimalist Adopters (Class 1):** This class consists of firms with low probability of use across all functions. 10% of all firms (or 37% of business function AI users) are in this category.
- **Marketing Specialists (Class 2):** These firms are characterized by a high probability of use in Sales and Marketing, but lower probabilities in other functions. These firms appear to also specialize in customer-centric and public-facing AI applications, such as customer service and public communications. They constitute about 9% of all firms or 31% of business function AI users.
- **Technical Strategists (Class 3):** This class has high probability of use in R&D, Strategy, IT, and Sales and Marketing. Firms in this class are likely innovators with growth orientation. 3% of all firms (12% of business function AI users) fall into this category.
- **Institutional-Administrative Integrators (Class 4):** Firms in this class exhibit high probability of use in Management, Strategy, Sales/Marketing, Finance, Human Resources, Public Relations, and Legal/Compliance. They integrate AI into many administrative and managerial core functions. They can also be described as governance specialists. These firms constitute 4% of all firms (15% of business function AI users).
- **Comprehensive Adopters (Class 5):** These are power users that have a high probability of use across almost the entire functional spectrum. Only 1% of all firms (or 4% of business function AI users) are in this class.

Table 3 summarizes firm characteristics across classes, revealing distinct relationships between firm size and integration intensity. Institutional-Administrative Integrators are the largest on average (41 employees), while Marketing Specialists are the smallest (18), matching the size profile of non-users and suggesting that small firms leverage GenAI primarily for outreach. Notably, Minimalist Adopters average 30 employees that exceed the average size of non-users, which indicates this class includes larger firms

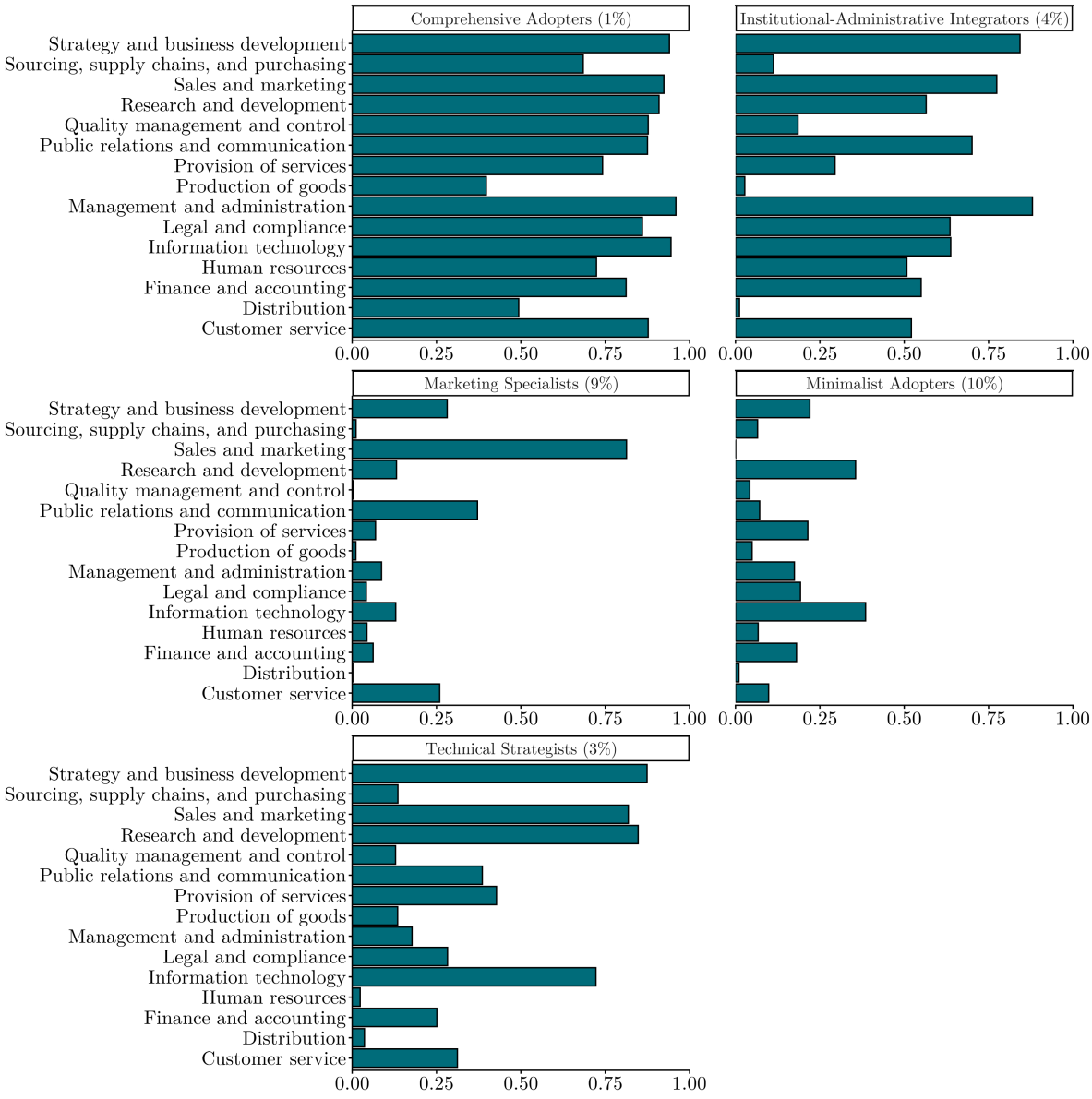
the model avoids the sample bias inherent in unweighted clustering, where small-firm patterns might be drowned out by over-represented strata. The random component of the initialization ensures the model avoids local maxima, providing a more stable and replicable classification of firms across the functional landscape.

[24] See Nylund et al. (2007) for a study on the selection of the number of classes in LCA.

[25] See Short et al. (2008) for a similar application of LCA to determine organizational types.

[26] To make the interpretation simple, for each firm we used the "modal" probability across classes to assign the firm to a given class. Within each class, we then calculate the rates of use for each function (firm-weighted).

Figure 11: Business function use probabilities by firm class



that have yet to scale AI beyond a narrow functional scope. Technical Strategists occupy a middle ground with 22 employees.

Functional and task-level integration are highest among Comprehensive Adopters, who utilize AI across 12 of 15 surveyed functions and five distinct worker tasks. In contrast, functional breadth is the lowest (2) for both Minimalist Adopters and Marketing Specialists. Across all classes, worker task breadth correlates positively with functional breadth. While Comprehensive Adopters, Integrators, and Technical Strategists also report the highest number of organizational and operational investments, these adjustments remain at an early stage, averaging fewer than three per firm across all classes.

Table 3: Firm characteristics by firm class

	Share of all firms		**Average number of:**			
Firm Class	**Firm-weighted**	**Employment-weighted**	**Employees**	**Business functions using AI**	**Worker tasks using AI**	**Org. changes**
Comprehensive Adopters	1.2%	2.1%	34.03	12.02	5.02	2.38
Technical Strategists	3.4%	3.8%	22.25	5.56	3.20	1.25
Inst.-Admin. Integrators	4.2%	8.6%	41.24	7.25	3.74	1.43
Marketing Specialists	8.6%	7.6%	17.82	2.32	1.36	0.51
Minimalist Adopters	10.2%	15.1%	30.10	2.13	1.28	0.54
Non-Users	72.3%	62.9%	17.68			

Note: Averages are firm-weighted.

Figure 12 illustrates the sectoral concentration within each adoption class relative to the overall firm population, where a value of 1 indicates equal representation. Across all classes, knowledge-intensive sectors, specifically Information (51), Professional Services (54), Finance (52), and Education (61), are significantly over-represented, often at twice the population share. Notably, Professional Services firms are most concentrated among technical specialists, while multi-sector entities gravitate toward institutional-administrative and comprehensive adoption. In contrast, non-users are characterized by a distinct under-representation of these AI-heavy industries.

Figure 13 shifts the perspective to within-sector composition, showing the share of each class within a given sector.[27] While sectors such as Information (51), Professional Services (54), and Management (55) demonstrate a higher propensity for comprehensive adoption, others like Retail (44), Real Estate (53), and Arts and Entertainment (71) are primarily characterized by marketing specialization. Despite this sectoral variation, the broad picture remains clear: across almost all industries, AI-using firms are most likely to fall into the minimalist or marketing specialist categories.

These patterns suggest that functional AI adoption is non-uniform and dependent on firm characteristics. The interaction of sector and size produces the following patterns. At the larger end of the spectrum, Institutional Integrators (41 employees) and Comprehensive Adopters (34 employees) leverage AI to manage organizational complexity and achieve broad-spectrum integration, particularly in sectors like Information, Finance and Education. In contrast, smaller firms averaging 18 employees (in Real Estate, Retail, the Arts, Education, Finance) exhibit marketing specialization. Technical Strategists

[27]Note that the sector use rates (heights of the bars) do not match the sector use rates in Figure 2, because Figure 13 is based on the business function use question whereas Figure 2 is based on the core firm AI use question.

(22 employees) represent a mid-sized, knowledge-intensive cohort concentrated in Professional Services, where adoption is driven by uses in strategy, IT, and R&D functions. Finally, the profile of Minimalist Adopters (30 employees) reveals a notable segment of relatively large firms that maintain a limited AI footprint across various sectors, suggesting that while scale enables adoption, it may, in some cases, coincide with organizational inertia or a cautious approach to integration. On the other hand, comprehensive, high-propensity use across all business functions remains a rare event. For the vast majority of current adopters, AI integration is a narrow, specialized endeavor rather than a broad organizational overhaul.

5 What AI does to work

Next, we focus on the mechanisms through which AI affects work by analyzing firms' reports of task substitution, task augmentation, and task creation. The section also measures task replacement intensity (e.g., the extent of tasks replaced). The section then considers reported AI-drives changes in employment and capital use patterns.

We interpret a firm's reported effect of AI on an outcome (e.g. employment or capital use) as its subjective assessment of the causal effect of AI use; that is, the respondent's comparison of the realized outcome with what it believes would have occurred in the absence of AI, rather than necessarily as changes in the outcome only.[28] The average reported effect provides a subjective analogue of the average treatment effect on the treated. The reported standard errors capture sampling variability in the estimated population shares but not errors in firms' counterfactual assessments or attribution of outcomes to AI. For example, when AI adoption coincides with organizational restructuring and changing demand, respondents may have difficulty determining how much of an employment change is attributable specifically to AI.

5.1 Task substitution, augmentation, and creation

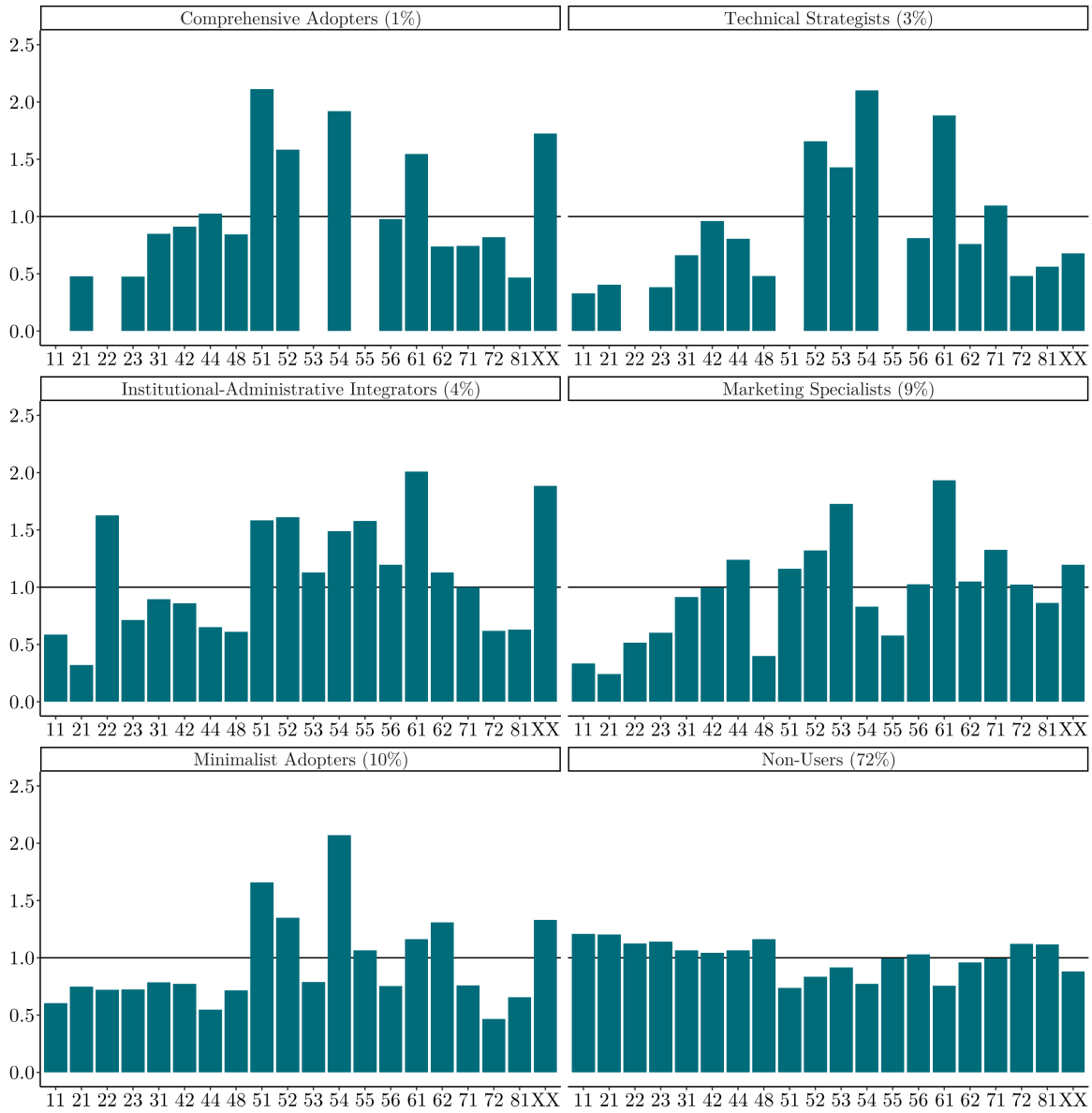
Figure 14 shows the fraction of AI-using firms by type of task effect – note that a firm can report more than one effect (Question 25). About 44% (54%, employment weighted) of firms use AI to supplement a task performed by an employee (task augmentation). About 10% (12%, employment weighted) use AI to perform a task previously done by an employee (task substitution/replacement). Similarly, about 11% (8%, employment weighted) introduce a new task (task creation). A large fraction (52%) of AI-using firms do not experience any of these three types of task effects.

The patterns in Figure 14 already suggest that task augmentation is the most common practice among AI-using firms. Conditional on a task effect (i.e. excluding "none of the above" group), an overwhelming fraction (91%) of firms with an AI-driven task effect use AI for augmentation only or in combination with substitution or creation. To isolate different types of task effect combinations, Figure 15 considers the prevalence of 7 mutually-exclusive task effect types, conditional on firm experiencing some task effect.

By far, the most common effect is task augmentation alone: about 66% (70%, employment weighted) of firms only augment tasks. Other task effect combinations occur in much smaller fractions of firms.

[28] Suppose a firm employed 100 workers six months ago and now employs 105 after adopting AI. The firm nevertheless believes it would have employed 110 workers without AI because growing demand would otherwise have required five additional hires. It could therefore report that AI "decreased" employment: employment increased by five over time, but AI reduced employment by five relative to the firm's perceived no-AI counterfactual.

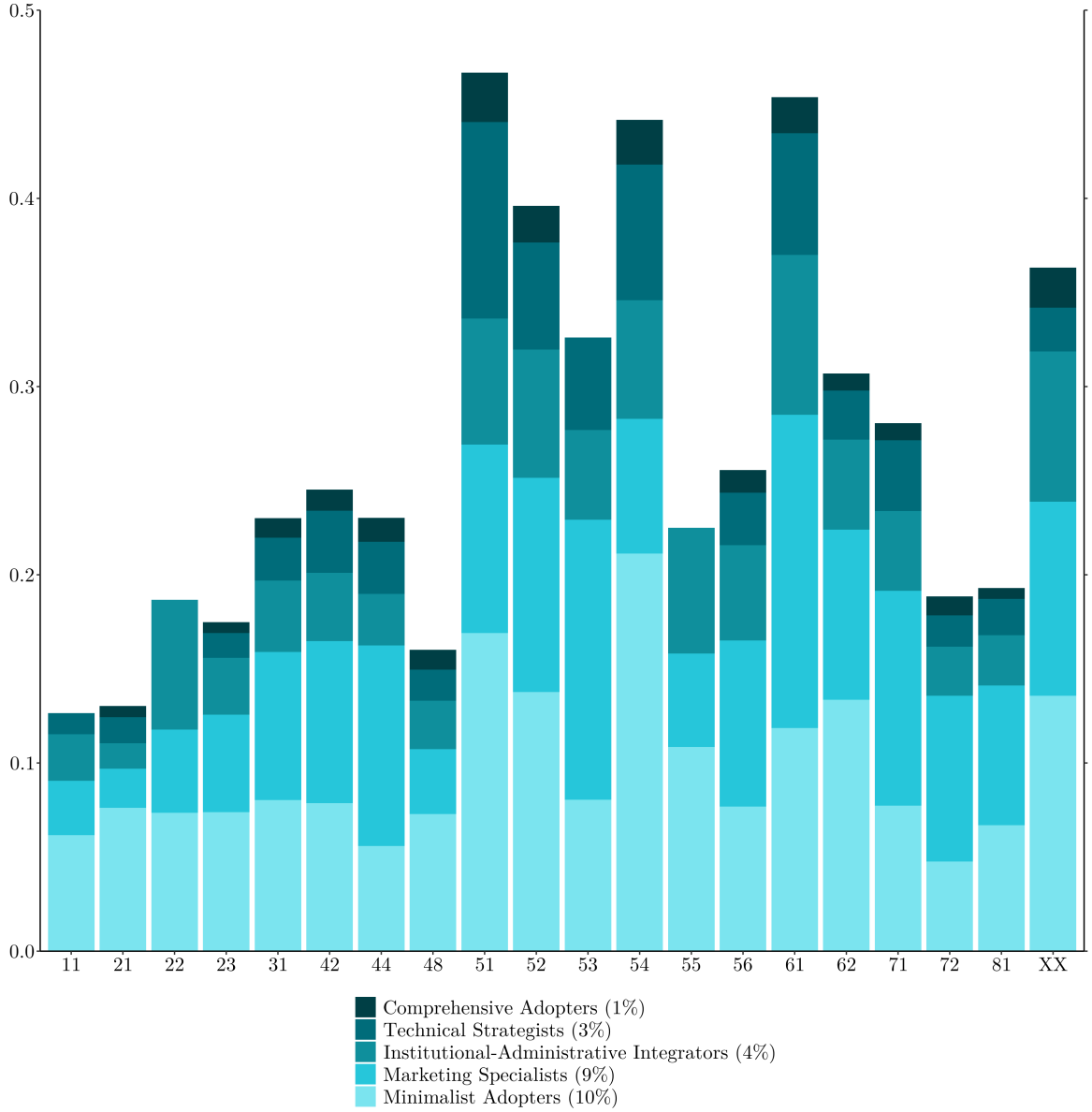
Figure 12: Normalized relative sectoral representations by firm class



Note: The horizontal line at 1 represents parity between the sectoral distribution for a given class and the economy-wide sectoral distribution.

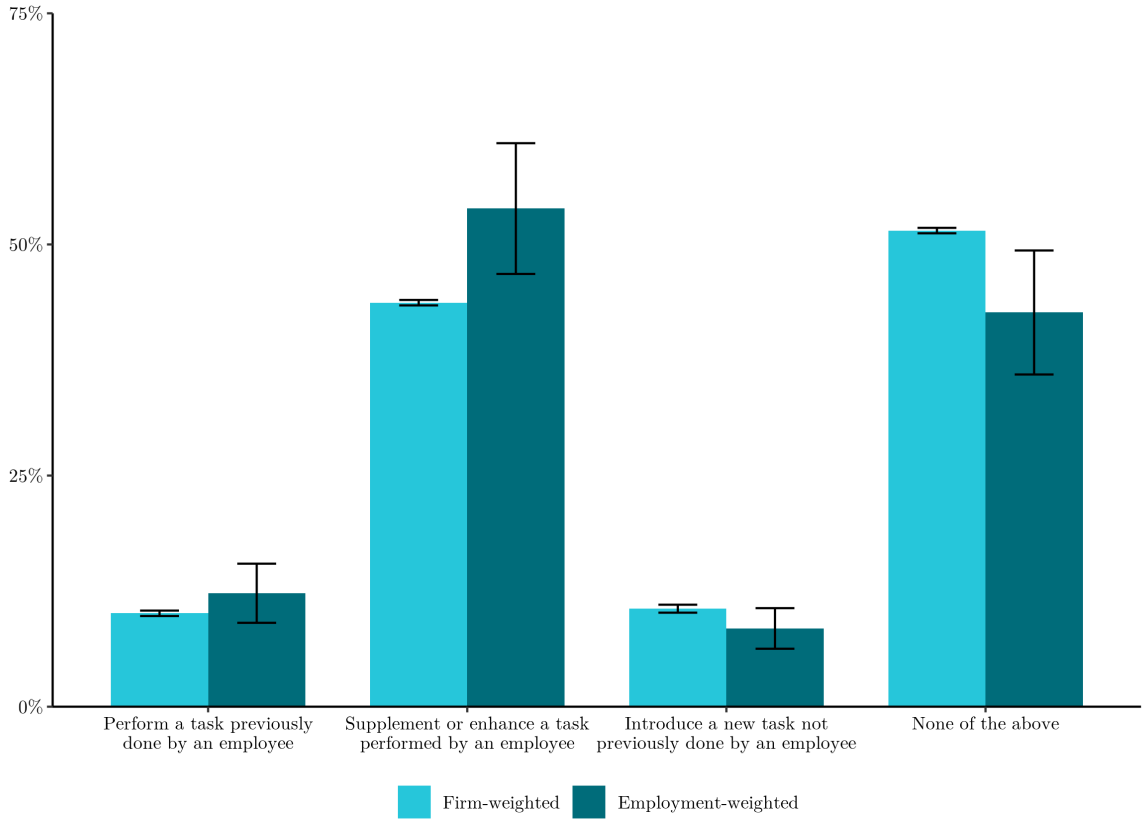
NAICS sectors: (11) Agriculture, Forestry, Fishing and Hunting; (21) Mining, Quarrying, and Oil and Gas Extraction; (22) Utilities; (23) Construction; (31–33) Manufacturing; (42) Wholesale Trade; (44–45) Retail Trade; (48–49) Transportation and Warehousing; (51) Information; (52) Finance and Insurance; (53) Real Estate and Rental and Leasing; (54) Professional, Scientific, and Technical Services; (55) Management of Companies and Enterprises; (56) Administrative and Support and Waste Management and Remediation Services; (61) Educational Services; (62) Health Care and Social Assistance; (71) Arts, Entertainment, and Recreation; (72) Accommodation and Food Services; (81) Other Services (except Public Administration); (92) Public Administration; (XX) Multi-sector Firms

Figure 13: Firm class shares by sector



NAICS sectors: (11) Agriculture, Forestry, Fishing and Hunting; (21) Mining, Quarrying, and Oil and Gas Extraction; (22) Utilities; (23) Construction; (31–33) Manufacturing; (42) Wholesale Trade; (44–45) Retail Trade; (48–49) Transportation and Warehousing; (51) Information; (52) Finance and Insurance; (53) Real Estate and Rental and Leasing; (54) Professional, Scientific, and Technical Services; (55) Management of Companies and Enterprises; (56) Administrative and Support and Waste Management and Remediation Services; (61) Educational Services; (62) Health Care and Social Assistance; (71) Arts, Entertainment, and Recreation; (72) Accommodation and Food Services; (81) Other Services (except Public Administration); (92) Public Administration; (XX) Multi-sector Firms

Figure 14: AI-driven task effects

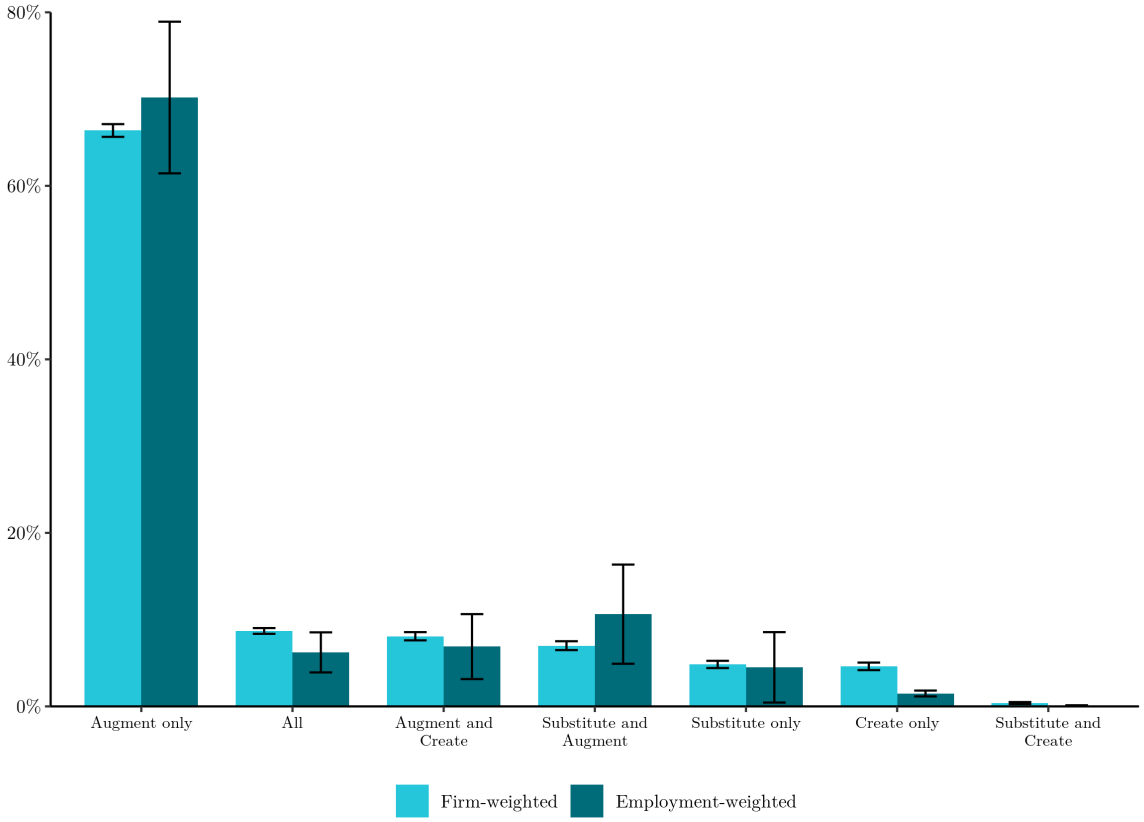


Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Q25 is conditional on AI use in at least one business function (Q24).

Only about 9% of firms experience all three effects, another 8% both augment and create, about 5% create only, and another 5% substitute only. The least common combination is substitution and creation (1% of firms). Employment-weighted estimates do not change this picture substantially and indicate that most of the employment in AI-using firms with a task effect is in firms that only augment.

Overall, task effects of AI are currently dominated by augmentation, rather than substitution or creation. While this dominance of augmentation suggests a currently worker-centric adoption path, rapid advances in AI can lead to changes in task effect profiles in the future. As technological constraints recede, the augmentation-to-substitution threshold is likely to shift, potentially transitioning current assistive roles of AI into automated ones as AI capabilities mature.

Figure 15: AI-driven task effects – mutually exclusive combinations

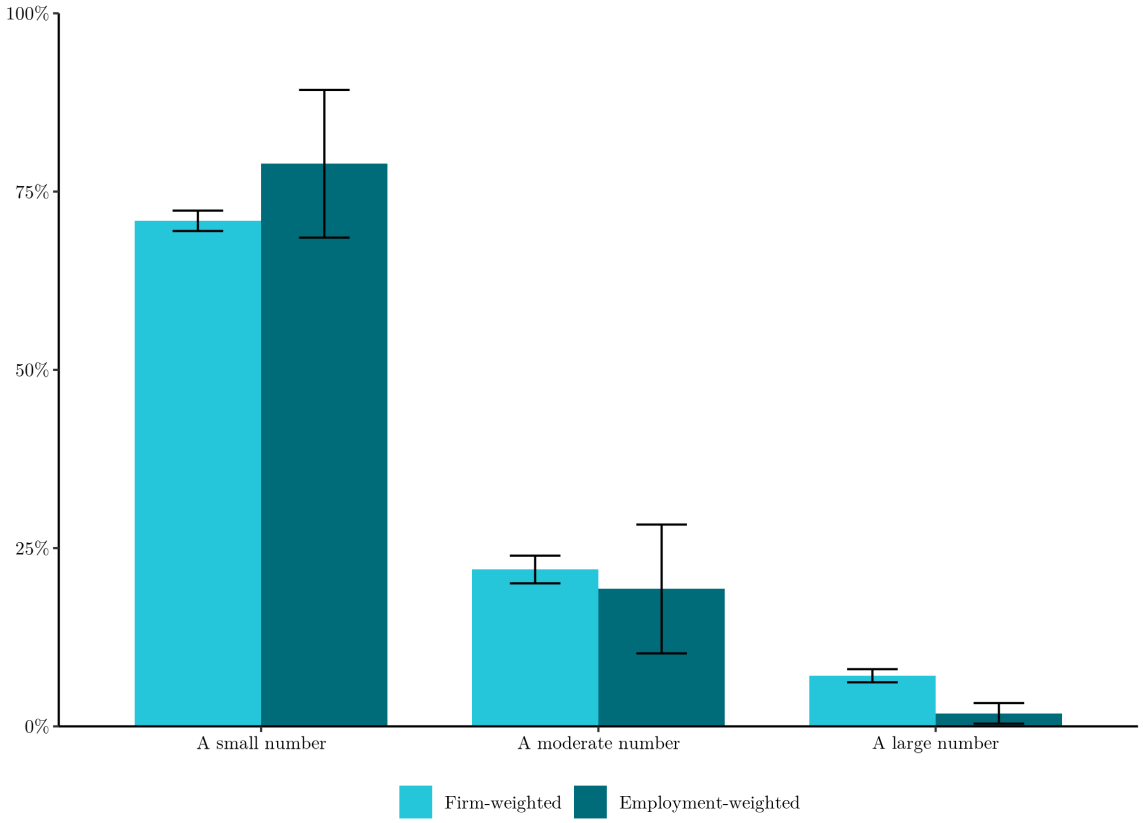


Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Q25 is conditional on AI use in at least one business function (Q24).

While task substitution is not yet widespread, it is important to know how intense the task substitution effect is (Question 26). Figure 16 shows that a large fraction of task-replacing firms (71%) do so for a small number of tasks, while only 7% replace a large number of tasks.[29] The incidence of AI-driven large-scale task replacement is thus limited, but still much more prevalent than the one observed in the first BTOS AI supplement: as of early 2024, 85% of firms substituting tasks replaced a small number of tasks, and only 2.5% replaced a large number of tasks – see Bonney et al. (2024a,b).[30]

[29] One caveat to these figures is that firms may have interpreted the quantity of tasks differently: some may have considered "per worker" number of tasks as the margin of adjustment, and others may have focused on the total number of tasks replaced

Figure 16: Intensity of task substitution



Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Q26 is conditional on responding Yes to Q25: 'AI was used to perform a task previously done by an employee'.

5.2 Changes to labor and capital

Building on task-level mechanisms, we next broaden the focus to employment effects and capital substitution patterns by analyzing reported net employment change and replacement of software and equipment (Questions 27 and 28). The limited extent of task replacement and the high incidence of augmentation already hints that full replacement of workers with AI may not be a common practice and AI-driven net employment decline instances may be limited.

As Table 4 reports, the fraction an overwhelming fraction of AI-users (95%, firm-weighted) do not experience any change in employment. The ones that experience an increase or decrease each constitute a small fraction of firms (about 2% each) – and these fractions remain small with employment-weights, though increases are more common than decreases. These patterns are very similar, qualitatively and quantitatively, to the ones reported in the first BTOS supplement (see Table 6 in Bonney et al. (2024a)), indicating that in the last two years since the first supplement, the rate of AI-driven employment change incidences has remained limited among firms – despite an increase in task replacement intensity. This finding aligns with the limited employment effects observed in recent studies based on administrative and survey data.[31] However, our analysis does not provide any information on the magnitudes of changes or the gross worker flows – e.g. AI-driven changes to hiring practices. Future data collection could focus on understanding these gross flows, especially the hiring dimension.[32]

Table 4: Share of AI-using firms by AI-driven employment change (last six months)

	Firm-weighted		**Employment-weighted**	
	share	**s.e.**	**share**	**s.e.**
Increased	2.3%	0.08%	3.7%	1.51%
Decreased	2.0%	0.09%	2.4%	0.52%
No Change	95.7%	0.14%	93.9%	1.35%

Compared to the incidence of net employment change, Figure 17 indicates a high incidence of AI-driven capital substitution. While no substitution is the most common pattern, 16% of AI-using firms replace equipment and software with AI. Taken together, the findings in this section suggest that in the current phase of diffusion, AI is serving more frequently as a substitute for legacy capital than for tasks and labor.

5.3 Where are AI-attributed employment changes concentrated?

Although employment changes attributed to AI use remain uncommon, their distribution may provide early evidence about where AI-related labor adjustment is emerging. The following comparisons describe

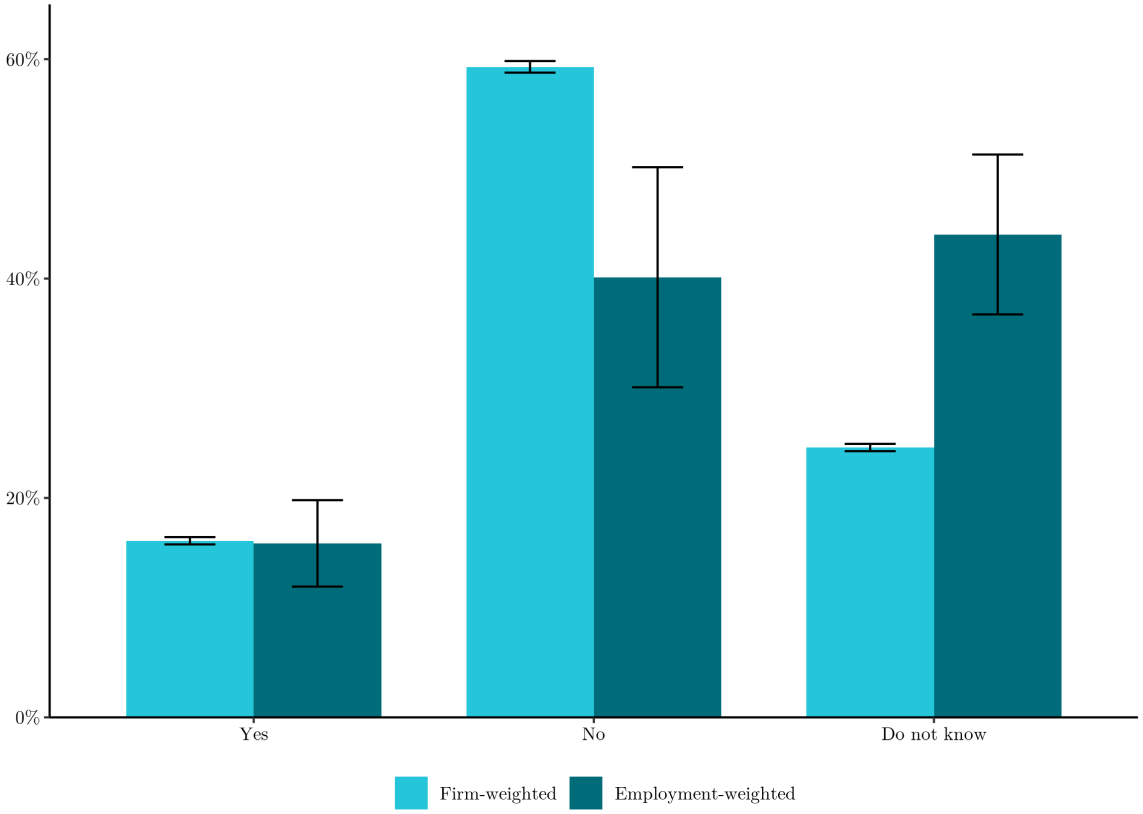
across all workers.

[30] We note here again that the revisions in the core and supplement questions may have influenced the responses to the task replacement intensity question.

[31] For instance, Humlum and Vestergaard (2026) find little effects of AI use on worker hours and earnings using administrative data from Denmark, Yotzov et al. (2026) observe that 9 out of 10 executives report no AI-driven employment change in the Survey of Business Uncertainty, despite high rates of AI use reported by these executives for their companies, and Baslandze et al. (2026) reach a similar conclusion from a different survey of corporate executives.

[32] BTOS is planning questions on the effects of AI on hiring for the next AI supplement (pending approval from the Office of Management and Budget).

Figure 17: AI-driven capital substitution

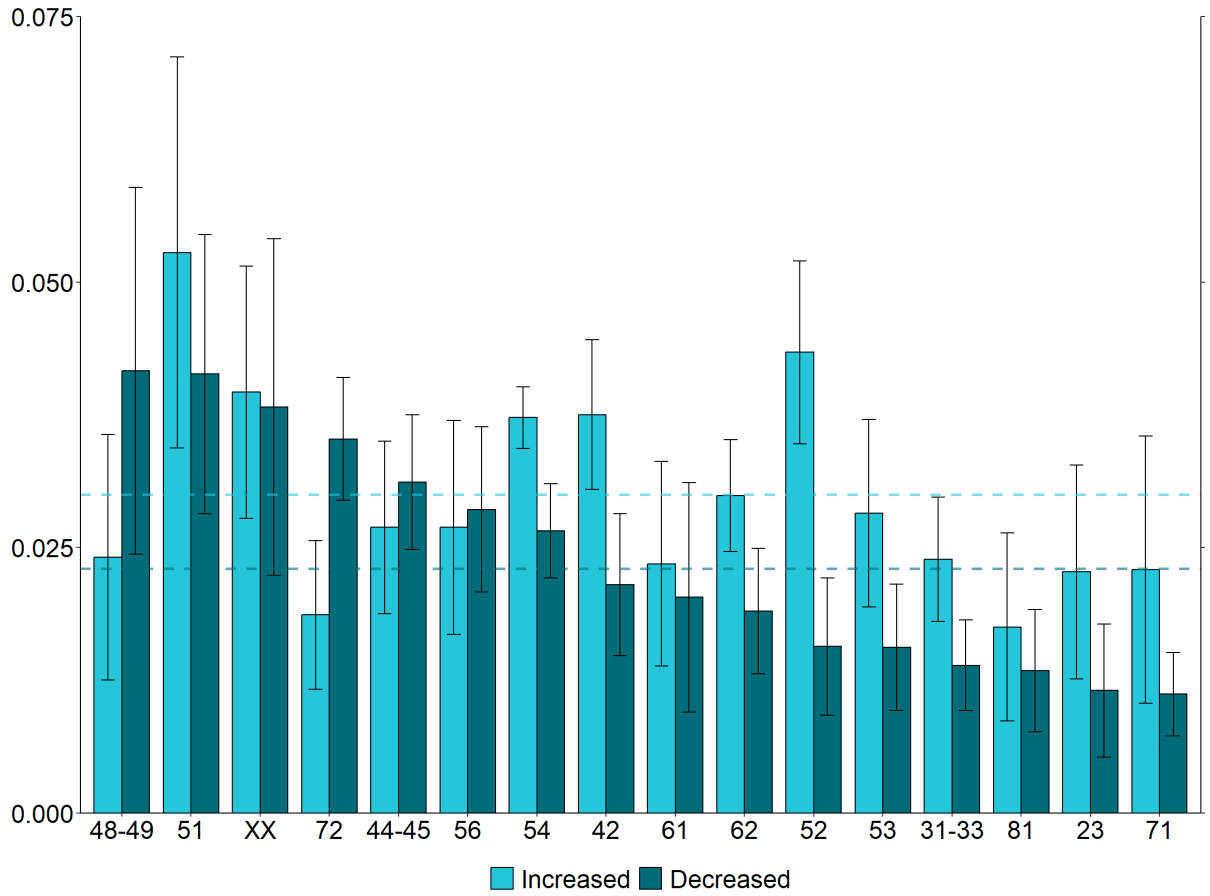


Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Q27 is conditional on AI use in at least one business function.

broad patterns in the point estimates. However, many confidence intervals overlap due to smaller samples underlying these detailed estimates, so differences and rankings discussed below should be interpreted cautiously. It is important to emphasize that the wording of the AI question asks for impacts on "total" (*net*) employment at the firm level–this can mask simultaneous AI-related job creation and destruction at the same firm. Finally, the sample underlying the figures in this section includes only those firms that use AI in at least one business function. Hence, reported rates of increase (3.0%) and decrease (2.3%) are slightly higher than the full sample.

Figure 18 shows substantial sectoral heterogeneity. The estimated decline rate is relatively high in Transportation & Warehousing, Information, multi-sector firms, and Accommodation & Food Services, at approximately 3.5–4.2%. Increases are more common than decreases in many sectors, with particularly large differences in Finance & Insurance, Wholesale Trade, and Health Care & Social Assistance. By contrast, the point estimates suggest that decreases dominate in Transportation & Warehousing and Accommodation & Food Services.

Figure 18: AI-driven employment change by sector



NAICS sectors: (11) Agriculture, Forestry, Fishing and Hunting; (21) Mining, Quarrying, and Oil and Gas Extraction; (22) Utilities; (23) Construction; (31–33) Manufacturing; (42) Wholesale Trade; (44–45) Retail Trade; (48–49) Transportation and Warehousing; (51) Information; (52) Finance and Insurance; (53) Real Estate and Rental and Leasing; (54) Professional, Scientific, and Technical Services; (55) Management of Companies and Enterprises; (56) Administrative and Support and Waste Management and Remediation Services; (61) Educational Services; (62) Health Care and Social Assistance; (71) Arts, Entertainment, and Recreation; (72) Accommodation and Food Services; (81) Other Services (except Public Administration); (92) Public Administration; (XX) Multi-sector Firms

Figure 19 indicates that there is no clear monotonic pattern in firm size. Nevertheless, both types of employment adjustment are relatively prevalent among the largest firms. Among firms with 1,000 or more employees, approximately 6.2% report an increase and 4.2% report a decrease. Firms with 500–999 employees also have a relatively high estimated decline rate of approximately 4.4%. These estimates are based on small samples and are therefore less precise than those for smaller firms but suggest that AI-related employment adjustment is concentrated among the very largest firms.

Figure 19: AI-driven employment change by size

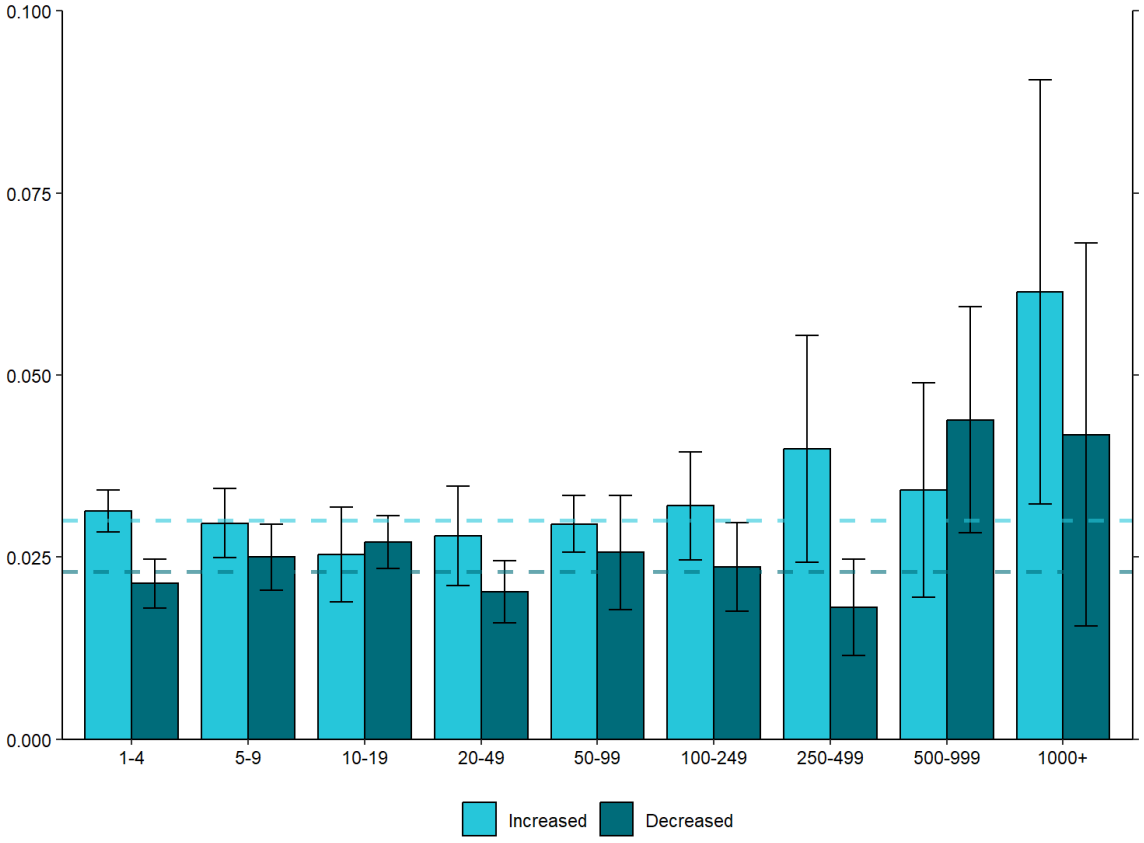
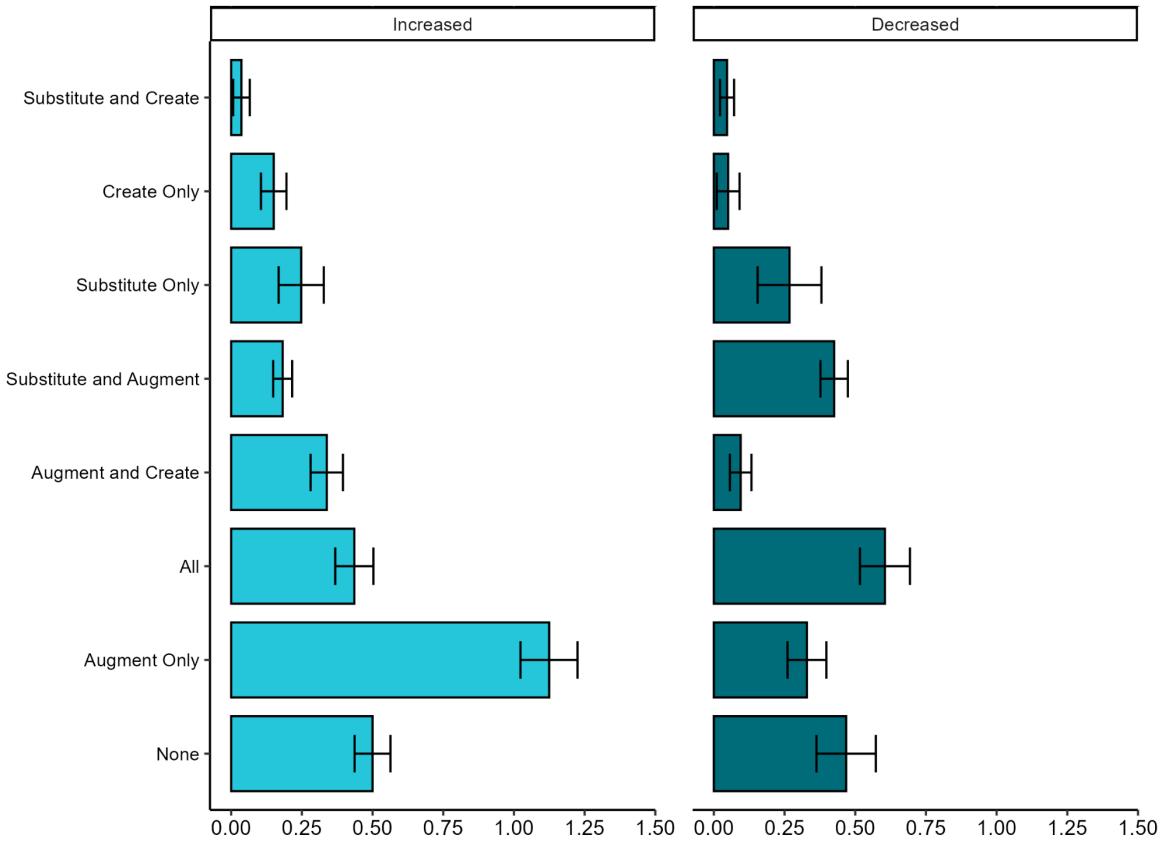


Figure 20 breaks down the total AI-related employment change–3.0% increase, 2.3% decrease–into mutually exclusive sets of firms based on reported AI impact on tasks (the same categories as in Figure 15). A plurality of firms experiencing employment increase (1.1% of AI-using firms, or about one-third of firms experiencing AI-driven employment increases) report AI only augmenting tasks. By contrast, employment decreases mostly come from firms reporting all three task effects or those reporting that AI both substitutes for and augments tasks. Such a pattern is consistent with the view that AI-driven increases in employment are driven by task augmentation while employment contraction is driven by task substitution.

Overall, the evidence suggests that early AI-attributed employment adjustments are concentrated in particular sectors, among the largest firms, and in firms experiencing broader changes in worker tasks. The results do not point to a uniformly labor-reducing effect: employment increases remain more common overall, particularly when AI primarily augments workers' existing tasks.

Figure 20: AI-driven employment change by task effect type

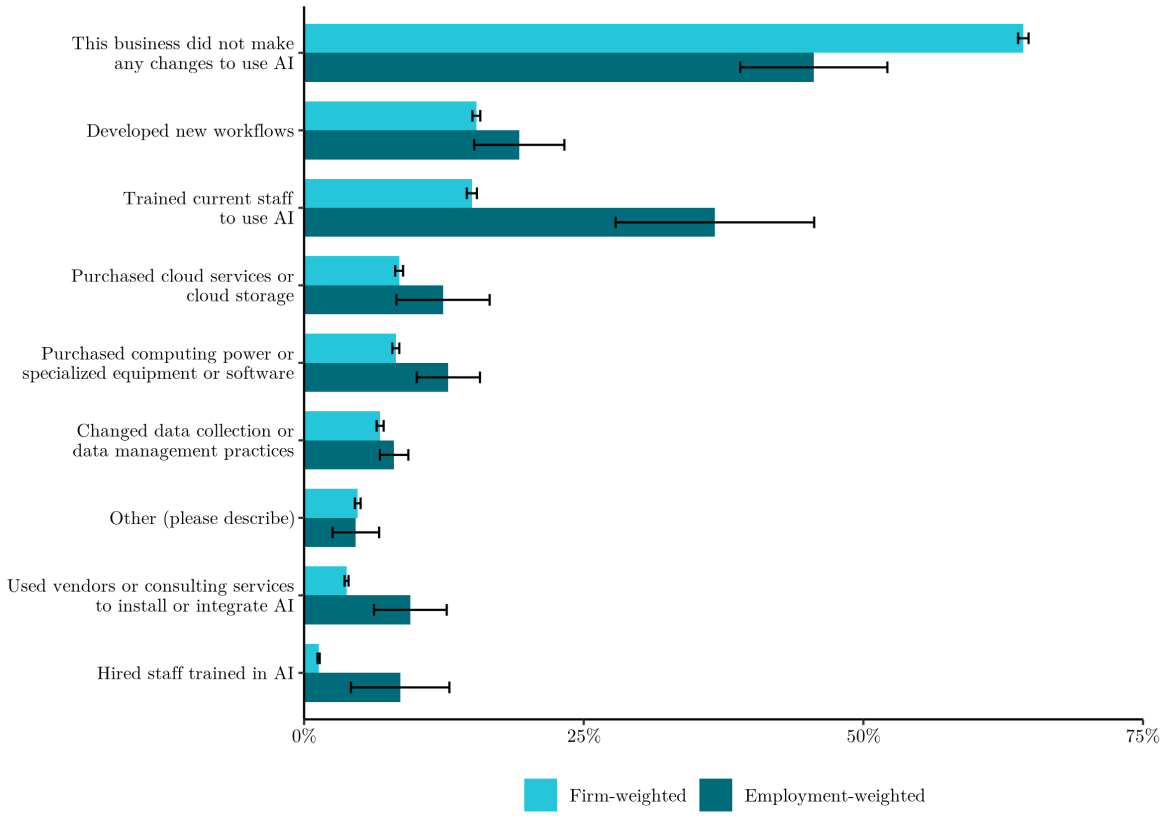


6 Organizational adjustments to use AI and barriers to adoption

This section considers organizational changes and investments to implement and operationalize AI, and barriers to AI adoption (Questions 29 and 35). Organizational adjustments and complementary investments are critical precursors to the "productivity J-curve", representing the structural groundwork necessary to translate AI adoption into tangible gains.[33] These changes serve as a proxy for the depth of integration, signaling whether a firm is fundamentally retooling its operations or merely adopting AI as a peripheral enhancement. They can also be precursors of performance, task, and employment effects resulting from such integration.

Figure 21 reveals that over half of AI-using businesses (64%) report no institutional adjustments, suggesting a reliance on "off-the-shelf" AI/GenAI tools or a significant lag in organizational restructuring. The most common adjustments are training staff and developing new workflows – each applicable to about 15% of AI-using firms. Deeper capital-related shifts, such as changes to data management and storage practices and complementary capital investments, take place in a smaller fraction of firms (in the 7-8% range). The least common adjustment is hiring staff trained in AI – consistent with low rates of AI-driven new task creation or employment increase.

Figure 21: Organizational adjustments to use AI



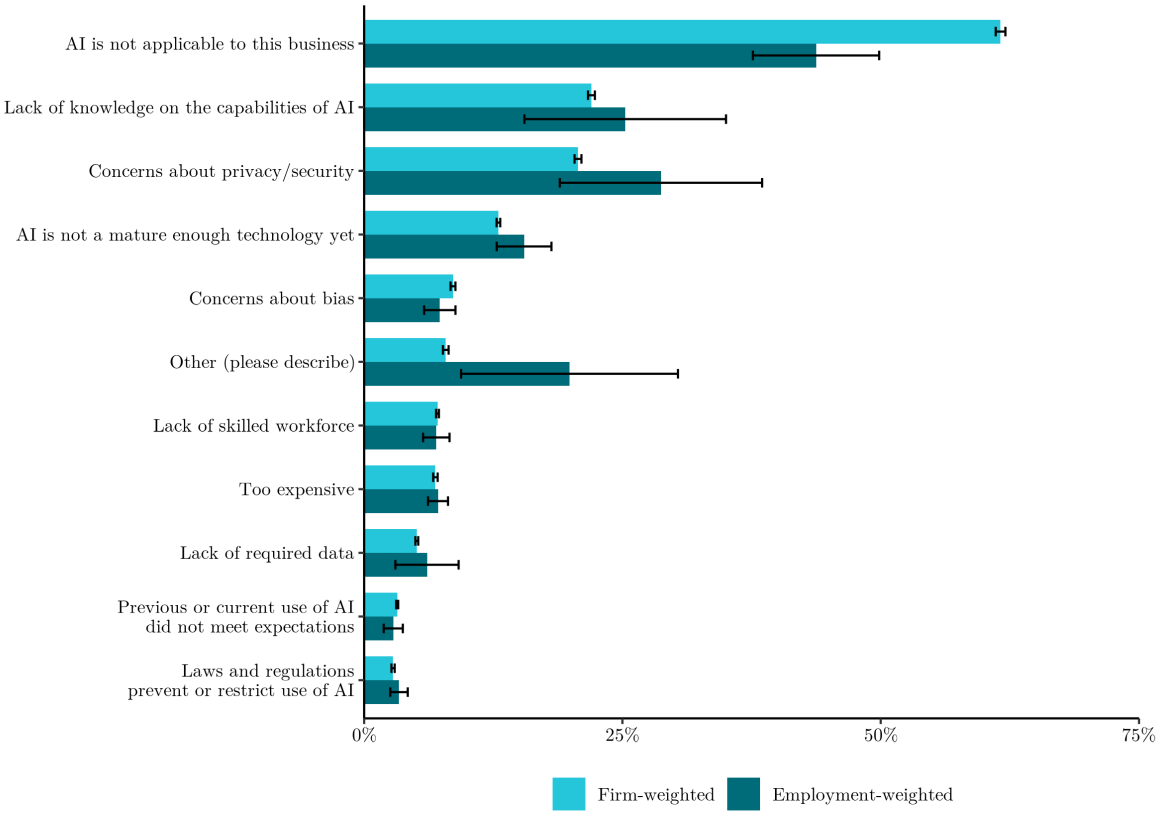
Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Q29 is conditional on AI use in at least one business function.

[33] See, e.g., Brynjolfsson et al. (2021) for a general analysis of productivity J-curves. In the context of AI, McElheran et al. (2025) and Dinlersoz et al. (2026) provide analysis and discussion related to productivity J-curves.

The AI supplement also asks firms not planning near-term AI use about their reasons for not adopting the technology (Question 35). Figure 22 indicates that the most common barrier is that AI is not applicable to the business (65% of firms, firm-weighted). Despite the expanding functional versatility of AI, many respondents still perceive the technology as inapplicable to their specific operations. This perceptual lag persists even as AI's utility across diverse business functions continues to broaden, reflecting a potential disconnect between latent technological potential and realized managerial awareness.

Lack of knowledge of the capabilities of AI comes at a distant second (22%), followed by concerns about privacy/security (20%). The least common barriers are using vendors or consulting services to implement AI and laws and regulations that prevent or restrict use of AI. While specific industries (e.g. healthcare, finance, and legal services) navigate sectoral or state-level constraints,[34] the absence of broad federal regulation as of early 2026 ensures that regulatory friction remains a minor factor in aggregate non-adoption.[35]

Figure 22: Barriers to AI use



Note: Error bars indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Q35 is conditional on non-AI use in Q33 (core).

[34]For instance, Colorado's SB 24-205 (effective early 2026) applies to insurance and financial services, requiring firms to use reasonable care to avoid algorithmic discrimination in lending, and annual impact assessments and public disclosures of how algorithms influence consumer premiums and coverage. For legal services, the American Bar Association has issued formal ethics opinions that mandate human verification of all AI-generated legal citations.

[35]On March 20, 2026, a new federal AI framework was released by the administration. While the accompanying America AI Act is currently in the legislative process, the Framework provides guidance on mandatory bias audits for high-risk systems and introduces the NO FAKES Act protections to prevent unauthorized AI-generated digital replicas of professionals, which is particularly relevant for the entertainment and healthcare sectors.

7 Firm-level indices of AI use and impact

While the BTOS provides high-resolution data across several granular AI use indicators, analyzing these variables in isolation risks obscuring the broader organizational patterns of technology integration. Constructing composite indices of AI use intensity and impact serves three primary methodological purposes. First, it addresses the dimensionality problem by distilling high-dimensional survey responses into stable, continuous metrics (diffusion indices) that facilitate cross-firm comparison and regression analysis. Second, by aggregating responses across functionally related questions, these indices mitigate the measurement error inherent in individual qualitative survey items, providing a more robust signal of a firm's latent technological capability. Finally, the index approach allows for a more nuanced characterization of the intensive margin of diffusion, moving beyond binary adoption indicators to capture the degree to which AI has been structurally embedded within the firm architecture. The latent class analysis has already demonstrated that there are relatively intensive AI users in terms of multiple business functions and tasks. The indices we construct provide a metric for quantifying the distribution on multiple dimensions.

We first define the overall AI integration index $S_{integration} \in [0, 1]$, a firm-level holistic composite metric designed to quantify the depth and breadth of AI integration within a firm. It aggregates survey data into four mutually exclusive components: functional breadth index (S_{firm}), worker-task integration index (S_{worker}), operational investment index (S_{invest}), and impact index (S_{impact}).

Among these four components, $S_{firm} \in [0, 1]$ reflects the breadth of AI use in business functions—the number of business functions where AI is used, scaled by the total number of business functions in the survey question. Similarly, $S_{worker} \in [0, 1]$ is the (scaled) number of worker tasks that integrate AI, and $S_{invest} \in [0, 1]$ is the (scaled) number of operational investments and changes undertaken by the firm to implement AI. Finally, $S_{impact} \in [0, 1]$ is a composite index based on the survey questions that capture task substitution/augmentation/creation effects, the extent of task replacement, and labor and capital effects. Appendix B details the construction of all indices.[36]

As index values increase from 0 to 1, the breadth of use or the depth of impact increases. For instance, $S_{firm} = 0$ means the firm uses AI in none of the 15 business functions, whereas $S_{firm} = 1$ corresponds to use in all functions. A similar interpretation applies to S_{worker} and S_{invest}. S_{impact} serves as a composite measure of the realized impact intensity of AI on a firm's internal operations. Higher values indicate a broader and more intensive footprint of AI across the production process: a higher number of distinct task effects (substitution/augmentation/creation), greater extent of task substitution (small/medium/large number of tasks replaced), a change in employment (increase or decrease) as opposed to no change, or the presence of capital substitution as opposed to no substitution.

The holistic AI integration index $S_{integration}$ is calculated as a weighted average:

$$S_{integration} = \frac{1}{4} \left(S_{firm} + S_{worker} + S_{invest} + S_{impact} \right).$$

We maintain a neutral stance on the components and use an equal weighting approach. This index measures the general footprint of AI in a firm in terms of both use and impact. The components S_{firm}, S_{worker}, and S_{invest} each measure different aspects of AI use and implementation, and they can be

[36] We note that these indices are not the only way to summarize the information in the survey and other indices can be constructed to emphasize different aspects of AI use and impact.

combined into a "use" index:

$$S_{use} = \frac{1}{3}\left(S_{firm} + S_{worker} + S_{invest}\right).$$

Figure 23 shows the histogram of $S_{integration}$ using both firm and employment weights. The distribution is highly positively skewed owing to the large fraction of non-users ($S_{integration} = 0$) and has a very small fraction of firms in the right tail. This pattern suggests that the firms where AI has the highest integration constitute a small fraction of firms and employment in the economy.

Table 5: Characteristics of firms by ranges of AI integration index ($S_{integration}$)

Index value	**Share of all firms**		**Average number of:**				**Avg. value of S_{impact}**
	Firm-weighted	**Employment-weighted**	**Employees**	**Business functions using AI**	**Worker tasks using AI**	**Org. changes**	
0	67.9%	46.0%	13.78	0	0	0	0
$(0, 0.25]$	25.1%	38.4%	31.03	2.25	1.22	0.37	0.08
$(0.25, 0.5]$	5.7%	12.6%	44.81	6.39	4.40	1.75	0.30
$(0.50, 0.75]$	1.2%	2.5%	43.72	9.97	7.15	3.35	0.56
$(0.75, 1.00]$	0.1%	0.5%	87.38	13.42	8.72	5.37	0.82

Note: All statistics are firm-weighted.

Table 5 shows characteristics of firms by value ranges. 68% of firms (46% employment-weighted) have zero value for the index. These firms tend to be small, with 14 employees on average. 25% of firms (38% employment-weighted) have low index value (in the range $(0, 0.25]$) and have more than double the average employment of the firms with index value zero. These firms use AI on average in 2 business functions and 1 worker task. They also undertake little organizational changes to integrate AI and do not show a high AI impact (low average value of S_{impact}).

As the index values increase, we observe a generally monotonic increase in the averages of all characteristics. Firms in the highest index range have on average 87 employees (nearly six times the average size of firms with no AI use) and use AI on average in 13 out of the 15 business functions and nearly in all worker tasks. They also undertake on average 5 distinct types of organizational change to accommodate AI and experience a high degree of AI impact (average value of $S_{impact} = 0.82$ for these firms).

Table 6: Descriptive statistics for AI indices

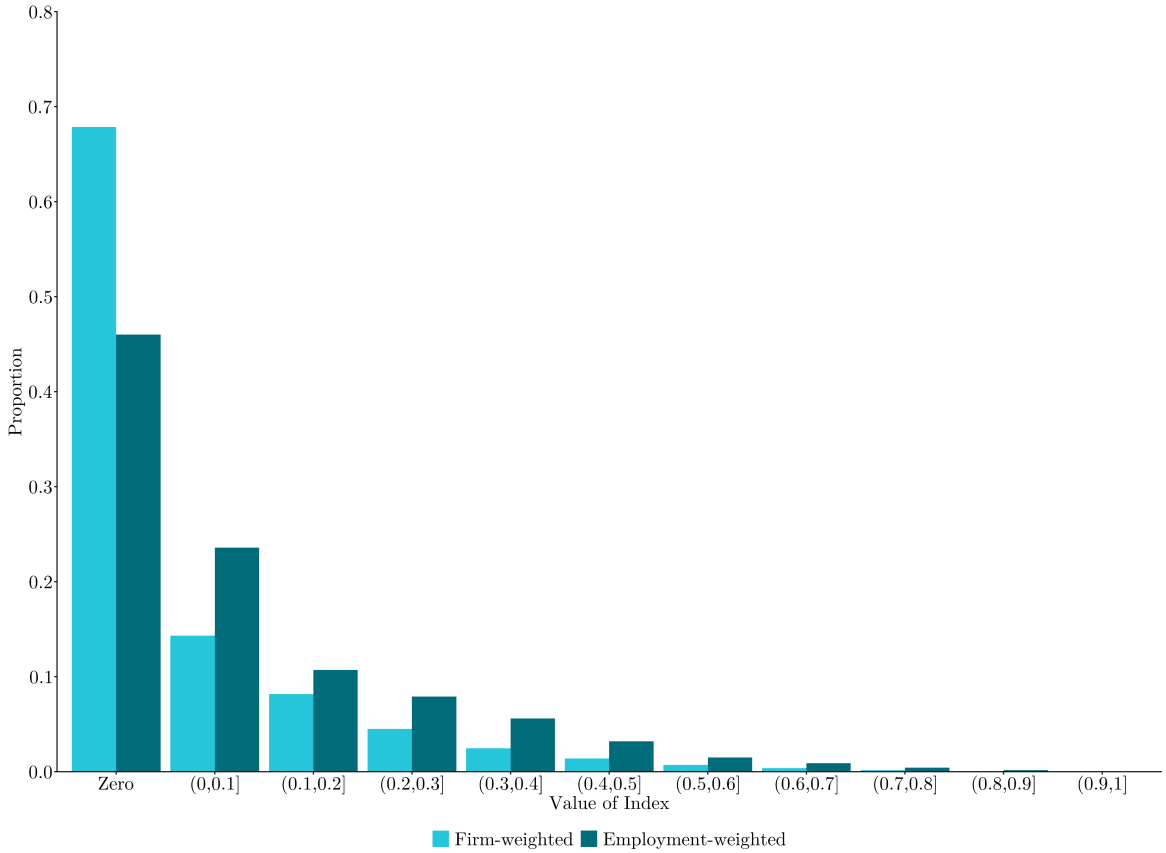
	Unconditional			**Conditional on $S_{integration} > 0$**		
Index	**Mean**	**S.D.**	**C.V.**	**Mean**	**S.D.**	**C.V.**
$S_{integration}$	0.053	0.113	2.150	0.164	0.147	0.898
S_{use}	0.055	0.118	2.150	0.172	0.154	0.898
S_{impact}	0.043	0.120	2.669	0.140	0.177	1.269
S_{firm}	0.071	0.154	2.171	0.220	0.201	0.914
S_{worker}	0.065	0.158	2.424	0.203	0.223	1.100
S_{invest}	0.030	0.091	3.068	0.093	0.142	1.532

Note: All statistics are firm-weighted.

Table 6 shows descriptive statistics for all indices. When firms with zero value for the AI integration

index ($S_{integration} = 0$) are included, the average index values are, not surprisingly, low, since such firms constitute a large fraction of the firm population. S_{firm} has the highest average value, followed by S_{worker}, S_{impact}, and S_{invest}, in descending order. The indices exhibit high variation relative to their means, with coefficients of variation (CV) exceeding 2. For firms with positive index values (the right panel), the average index values are much higher, but still low in an absolute sense. They also exhibit higher standard deviation, but lower CVs, since the means are higher relative to the left panel.

Figure 23: Histogram of AI integration index ($S_{integration}$)



Note: (0.9,1], firm- and employment-weighted suppressed due to data quality concerns.

Table 7 considers the correlations between the indices. Considering all firms (the left panel), the four components of $S_{integration}$ are highly correlated; some of this is by construction. Similarly, the high correlation between S_{use} and $S_{integration}$ is also largely expected. However, the individual components of AI use have much lower correlations with each other. While S_{firm} and S_{worker} have a strong positive correlation of 0.70, they remain far from collinear, suggesting that firm-level functional adoption and worker-level task integration do not always move in lockstep. Similarly, S_{firm} and S_{worker} show somewhat lower correlations with S_{invest} (0.65 and 0.62, respectively), indicating that the breadth of functional and task use is not a definitive predictor of the depth of operational investment. Furthermore, the generally lower correlations of (S_{firm}, S_{worker}, S_{invest}) with S_{impact} suggest that broader adoption and higher investment levels do not automatically translate into significant organizational effects at this stage of the diffusion cycle.

The observed correlations moderate significantly when the analysis is restricted to AI integrators (Ta-

ble 7, right panel). The higher coefficients in the full sample are driven largely by the high frequency of non-users, whose zero values across all indices create a strong, albeit mechanical, statistical association. Among firms with $S_{integration} > 0$, the indices become more decoupled, reflecting greater heterogeneity in how firms operationalize the technology. Within this population, the pairwise correlations between the four constituent components fall to the 0.46–0.55 range. This indicates that for active users, functional breadth, task integration, and operational investment do not scale in a uniform or linear fashion, suggesting that firms are making distinct strategic choices about where and how deeply to integrate AI.

The moderate correlations among S_{firm}, S_{worker}, and S_{invest} suggest that these components capture some distinct and non-overlapping dimensions of AI implementation and use. This allows us to move beyond binary adoption metrics and explore which specific aspects of diffusion—functional breadth, task-level integration, or operational investment—correlate most strongly with performance. The following section analyzes these associations, maintaining a descriptive rather than causal interpretation.

Table 7: Correlations of AI indices

	Unconditional					**Conditional on $S_{integration} > 0$**				
	$S_{integration}$	S_{use}	S_{impact}	S_{firm}	S_{worker}	$S_{integration}$	S_{use}	S_{impact}	S_{firm}	S_{worker}
S_{use}	0.98					0.97				
S_{impact}	0.85	0.75				0.78	0.61			
S_{firm}	0.90	0.91	0.69			0.81	0.83	0.52		
S_{worker}	0.88	0.90	0.63	0.70		0.80	0.85	0.46	0.51	
S_{invest}	0.81	0.81	0.66	0.65	0.62	0.76	0.75	0.55	0.50	0.47

Note: All statistics are firm-weighted.

8 AI use and firm outcomes

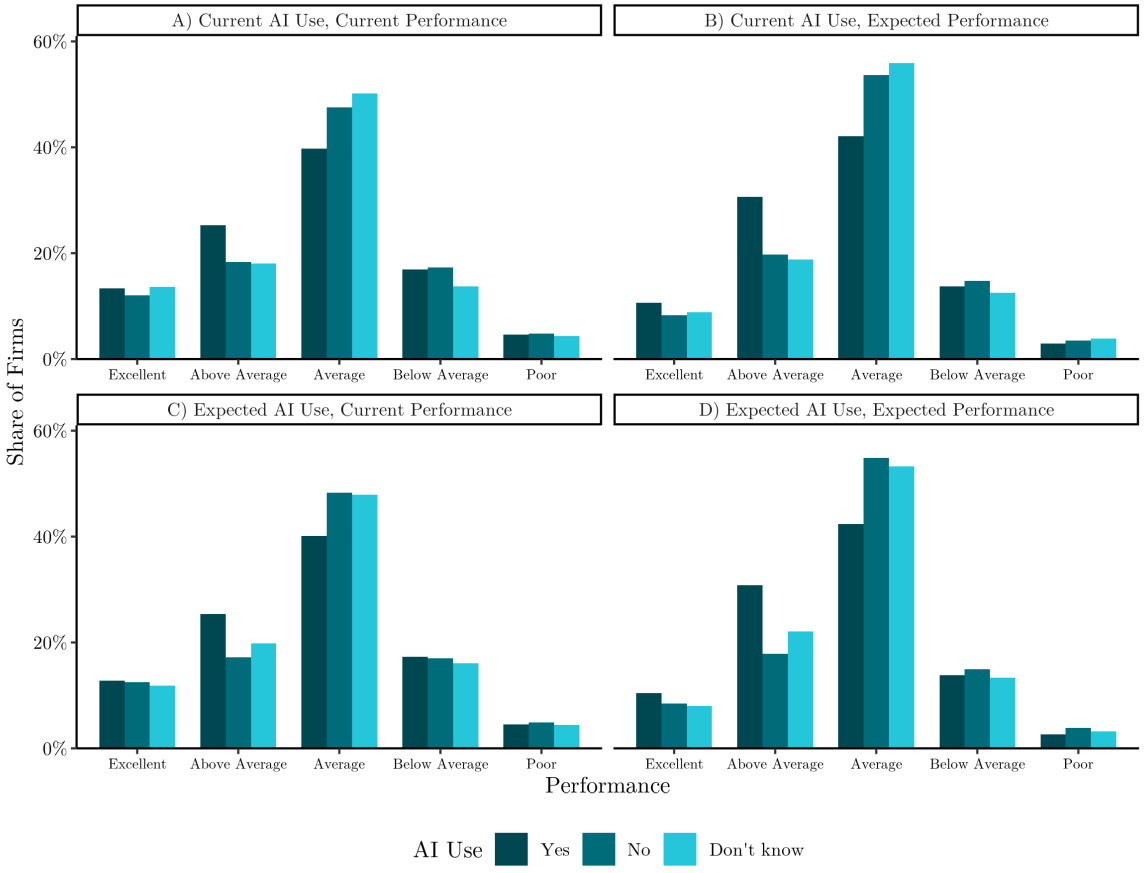
This section connects multi-layer AI use to business outcome indicators. The analysis explores whether extensive and intensive-margin measures of AI use are informative about firm outcomes in regression-style conditional associations.

8.1 Unconditional associations

Figure 24 shows the relationships between current/future AI use and current/future overall performance outcomes – based on overall performance questions (Questions 3 and 15) and core firm AI use questions (Questions 23 and 33). In all panels, there is a larger fraction of AI-using firms in "above average" and "excellent" categories. These patterns are broadly similar to the corresponding findings from the first BTOS AI supplement (see Figure 8a in Bonney et al. (2024a)) and indicate a positive relationship between AI use and performance.

Table 8 provides summary statistics for various firm outcome indicators by AI use. The outcome indicators are based on other questions in the BTOS outside the AI supplement: current overall performance (Question 3), change in sales in the last two weeks (Question 4), change in employment in the last two weeks (Question 5), expected overall performance 6-months ahead (Question 15), and expected

Figure 24: Distribution of overall performance outcomes by AI use



employment change 6-months ahead (Question 16). The qualitative responses to these questions are converted into indicator variables summarizing the directional changes:

- **Above-average performance indicator:** 1 if overall performance is above average or excellent, 0 otherwise; based on the response categories (poor, below average, average, above average, excellent),
- **Employment decrease indicator:** 1 if the firm indicates a decrease, 0 otherwise; based on the response categories (increase, no change, decrease),
- **Sales increase indicator:** 1 if the firm indicates an increase, 0 otherwise; based on the response categories (increase, no change, decrease).

Based on Table 8, AI users exhibit superior performance and sales increase likelihood compared to non-users and "don't know" respondents. However, these firms do not have substantially higher propensity of employment decline – either recently or in the near-term. However, these statistics are unconditional. Next, we consider a regression framework that controls for key observables, such as firm size, detailed industry (3-digit NAICS), state, and survey period, and utilize intensive measures of AI use based on the indices constructed in the previous section.

Table 8: Firm outcomes by AI use

		Firm outcomes				
		Last two weeks			Next six months	
AI use, last two weeks	**Share of firms**	**Performance > Average**	**Employment Decrease**	**Sales Increase**	**Performance > Average**	**Employment Decrease**
Yes	17.8%	38.9%	9.8%	18.9%	41.6%	9.1%
No	71.4%	30.6%	9.9%	11.6%	28.3%	8.8%
Don't know	10.7%	32.3%	10.0%	10.7%	28.1%	7.7%
All	100.0%	32.2%	9.9%	12.8%	30.6%	8.7%

Note: For firm outcome columns, cells show firm-weighted averages of firm outcome indicators as percentages.

8.2 Regression analysis

To examine the relationship between a firm-level outcome measure y_i and the measures of AI use at the firm and worker level, we pool all firms sampled in the supplement periods to estimate OLS regressions (or Linear Probability Models) with the AI use index S_{use}, and another with S_j ($j = firm, worker, invest$) as the main independent variables:

$$y_i = \alpha + \beta_{use} S_{use,i} + C_i + I_i + G_i + T_i + \epsilon_i \tag{1}$$

$$y_i = \alpha + \sum_j \beta_j S_{j,i} + C_i + I_i + G_i + T_i + \epsilon_i \tag{2}$$

The outcome variables y_i are the performance indicators defined in the previous section (see also Table 8). We add fixed effects for firm size categories (C_i), 3-digit NAICS industries (I_i), geography (state)

G_i, and survey periods T_i. Controls for firm size and industry are especially important, as the analysis so far has shown that AI use is concentrated in certain sectors and larger firms. To obtain population estimates, regressions (1) and (2) are firm-weighted.

Specification (1) is a restricted version of specification (2) that constrains the coefficients of S_j to be equal – by definition of S_{use}. We check whether the more flexible specification (2) provides additional insight by testing the pairwise equality of coefficients in specification (2) and examining the relative fit and information content of (2) with respect to (1). Specification (1) is parsimonious and robust to high correlation between the components, S_j, but can be biased if the restriction it imposes is not valid. Specification (2) is more flexible and useful especially in assessing the unique associations of different aspects of AI use with outcomes, albeit at the cost of wide confidence intervals if S_j are strongly correlated. However, Table 7 indicates that the correlations among S_j are moderate.

8.2.1 Regressions with the extensive AI-use margin

Table 9 examines the extensive margin using indicators for positive values of S_{use} and its components, S_j. In Specification (1), Panel A, $S_{use} > 0$ is positive and statistically significant for every outcome. AI use is associated with 3.2 and 8.3 percentage point increases in the probabilities of above-average current and future performance, respectively, and a 6.4 percentage point increase in the probability of sales growth (see also Table 12). These differences are economically meaningful relative to baseline rates of approximately 31 percent for performance and 13 percent for sales (Table 8). The corresponding associations with current and future employment decline are smaller in absolute terms—1.7 and 1.8 percentage points, respectively—against baseline rates of approximately 9–10 percent.

Specification (2), Panel B, separates the functional-use, worker-use, and investment margins. The indicator $S_{firm} > 0$ is positive and statistically significant for all outcomes, with coefficients of 1.9 percentage points for current performance, 5.2 for future performance, and 3.5 for current sales growth. Its coefficients for current and future employment decline are smaller, at approximately 0.8–0.9 percentage points. The largest coefficients on $S_{worker} > 0$ are 3.8 percentage points for future performance and 3.4 for current sales growth; its associations with current and future employment decline are not statistically significant. By contrast, $S_{invest} > 0$ is positive and statistically significant throughout, and its coefficients for current and future employment decline—2.0 and 1.6 percentage points—are the largest among the three component indicators.

Adjusted R^2 and AIC generally favor Specification (2), although the improvement varies across outcomes.[37] Equality tests reject five of the 15 pairwise comparisons among the component coefficients. Four rejections concern employment outcomes and indicate larger coefficients for S_{invest} than for S_{firm} or S_{worker}.

Overall, the presence of functional AI use, worker AI use, and supporting investments is positively associated with aspects of performance or sales, but their relationships with employment decline differ: functional use has comparatively small associations, worker use has no statistically significant association, and operational investments have the largest coefficients.

[37] Disclosure-related rounding causes some reported AIC values to appear identical across panels even when the underlying values differ.

Table 9: OLS regression (LPM) of firm outcomes on the indicators of AI use

	Performance and sales outcomes			**Employment outcomes**	
	Current Performance > Average	**Future Performance > Average**	**Current Sales Increase**	**Current Employment Decrease**	**Future Employment Decrease**
Panel A. Specification (1)					
$S_{use} > 0$	0.032***	0.083***	0.064***	0.018***	0.017***
	(0.004)	(0.004)	(0.003)	(0.002)	(0.002)
N	111,300	111,200	111,200	111,200	111,200
R^2	0.0494	0.0538	0.0255	0.0208	0.0151
R^2 Adj.	0.0480	0.0524	0.0240	0.0194	0.0136
R^2 Within	0.00099	0.00683	0.00743	0.00074	0.00072
R^2 Within Adj.	0.00099	0.00683	0.00743	0.00073	0.00071
AIC	144600	142200	82440	51150	42190
Panel B. Specification (2)					
$S_{firm} > 0$	0.019***	0.052***	0.035***	0.009**	0.008**
	(0.005)	(0.005)	(0.004)	(0.003)	(0.003)
$S_{worker} > 0$	0.010*	0.038***	0.034***	−0.002	0.002
	(0.005)	(0.005)	(0.004)	(0.003)	(0.003)
$S_{invest} > 0$	0.016**	0.031***	0.025***	0.020***	0.016***
	(0.006)	(0.006)	(0.004)	(0.003)	(0.002)
N	111,300	111,200	111,200	111,200	111,200
R^2	0.0495	0.0549	0.0269	0.0210	0.0153
R^2 Adj.	0.0480	0.0535	0.0254	0.0195	0.0138
R^2 Within	0.00106	0.00802	0.00885	0.00094	0.00088
R^2 Within Adj.	0.00103	0.00800	0.00882	0.00091	0.00085
AIC	144600	142100	82280	51140	42160

$+\ p < 0.1$, $*\ p < 0.05$, $**\ p < 0.01$, $***\ p < 0.001$

Note: Robust standard errors in parentheses. All specifications include fixed effects for firm size class, 3-digit NAICS industry, state, and survey period. Observation counts (N) are rounded in accordance with disclosure rules.

8.2.2 Regressions with the extensive and intensive AI-use margins

Table 10 repeats the analysis using the continuous indices, including zero values and thus combining the extensive and intensive margins. As in Table 9, Panel A shows that S_{use} is positively associated with every outcome. A one-standard-deviation increase corresponds to increases of 1.5 percentage points in the probability of above-average current performance, 4.1 points for above-average future performance, 2.9 points for sales growth during the preceding two weeks, and approximately 1.0 point for both current and future employment decline.[38]

Specification (2), in Panel B, separates the three component indices. A one-standard-deviation increase in S_{firm} is associated with increases of 1.7 percentage points in the probability of above-average current performance, 3.2 points for above-average future performance, and 1.9 points for current sales growth. The corresponding increases in the probabilities of current and future employment decline are 1.0 and 0.5 percentage points, respectively.

The coefficients on S_{worker} are generally smaller, mostly statistically insignificant, and sometimes negative. A one-standard-deviation increase is associated with changes of approximately 0.9 percentage points in the probabilities of current sales growth and above-average future performance, while its associations with current and future employment decline are near zero and statistically insignificant. By contrast, S_{invest} is positively and significantly associated with current sales growth and both current and future employment decline. Its one-standard-deviation effects are below 1 percentage point for every outcome, reflecting the index's more limited variation. Nevertheless, its association with future employment decline, at 0.6 percentage points, is slightly larger than the corresponding 0.5 percentage point estimate for S_{firm}.

Adjusted R^2 is uniformly higher and AIC generally lower in Panel B than in Panel A, indicating that Specification (2) provides a better overall fit with less information loss. Equality tests reject 10 of the 15 pairwise comparisons among its component coefficients at conventional significance levels.

8.2.3 Regressions with the intensive AI-use margin

Table 11 examines the intensive margin by restricting the sample to firms with positive AI use, $S_{use} > 0$. It also introduces the outcome indicating whether a firm experienced an employment decline *attributable to* AI use during the preceding six months (Question 28), which is asked only of firms using AI.

In Specification (1), Panel A, S_{use} is positive and statistically significant for every outcome. Conditional on positive AI use, a one-standard-deviation increase in the index is associated with increases of 1.1 and 3.5 percentage points in the probabilities of above-average current and future performance, respectively, and 2.3 points in the probability of current sales growth. The corresponding increases are 1.4 percentage points for current employment decline, 1.1 points for future employment decline, and 2.0 points for current employment decline attributed to AI.

Specification (2), Panel B, separates the three component indices. As in Tables 9 and 10, S_{firm} is positive and statistically significant for every outcome. A one-standard-deviation increase is associated with increases of 1.8 percentage points in current performance, 3.1 points in future performance, and 1.4 points in current sales growth. The corresponding estimates are 1.2 percentage points for current em-

[38] The standard deviations used to calculate these magnitudes are reported in Table 6.

Table 10: OLS regression (LPM) of firm outcomes on AI use indices

	Performance and sales outcomes			Employment outcomes	
	Current Performance > Average	**Future Performance > Average**	**Current Sales Increase**	**Current Employment Decrease**	**Future Employment Decrease**
Panel A. Specification (1)					
S_{use}	0.122***	0.342***	0.248***	0.091***	0.080***
	(0.014)	(0.014)	(0.011)	(0.009)	(0.009)
N	111,300	111,200	111,200	111,200	111,200
R^2	0.0494	0.0544	0.0253	0.0215	0.0155
R^2 Adj.	0.0479	0.0530	0.0239	0.0199	0.0140
R^2 Within	0.00093	0.00751	0.00728	0.00125	0.00105
R^2 Within Adj.	0.00092	0.00750	0.00728	0.00124	0.00104
AIC	144600	142100	82410	51080	42130
Panel B. Specification (2)					
S_{firm}	0.113***	0.206***	0.120***	0.064***	0.032**
	(0.016)	(0.016)	(0.012)	(0.011)	(0.011)
S_{worker}	−0.020	0.060***	0.059***	−0.008	0.003
	(0.015)	(0.015)	(0.012)	(0.010)	(0.009)
S_{invest}	0.018	0.041^{+}	0.057**	0.040**	0.062***
	(0.024)	(0.024)	(0.019)	(0.015)	(0.015)
N	111,300	111,200	111,200	111,200	111,200
R^2	0.0497	0.0549	0.0255	0.0216	0.0156
R^2 Adj.	0.0482	0.0534	0.0240	0.0201	0.0141
R^2 Within	0.00124	0.00798	0.00742	0.00148	0.00117
R^2 Within Adj.	0.00122	0.00795	0.00740	0.00145	0.00114
AIC	144600	142100	82400	51080	42130

$+\ p < 0.1$, $*\ p < 0.05$, $**\ p < 0.01$, $***\ p < 0.001$

Note: Robust standard errors in parentheses. All specifications include fixed effects for firm size class, 3-digit NAICS industry, state, and survey period. Observation counts (N) are rounded in accordance with disclosure rules.

ployment decline, 0.5 points for future employment decline, and 0.8 points for AI-attributed employment decline.

The coefficients on S_{worker} are less consistently significant, with positive and statistically significant associations only for future performance and current sales growth. Its one-standard-deviation effects are below 1 percentage point in absolute value for every outcome and are largest, at approximately 0.7 points, for future performance and current sales. Estimates for S_{invest} are generally positive and significant, including for current sales growth. Its largest association is with AI-attributed employment decline, at 1.7 percentage points, followed by future employment decline at 0.9 points.

Adjusted R^2 is higher in nearly every case and AIC generally lower in Panel B than in Panel A, indicating better overall fit and less information loss for Specification (2). Equality tests reject 11 of the 18 pairwise comparisons among the component coefficients at conventional significance levels, indicating that their associations with the outcomes differ materially.

8.2.4 Summary of regression magnitudes

Table 12 summarizes all regression magnitudes discussed above, along with the outcome means. Table 13 puts these magnitudes into further perspective by expressing them as percentages of the outcome means. The heat map indicates that commercial performance and labor outcomes exhibit varying degrees of correlation depending on the function, worker task, or investment level measures. Across all indices, the breadth of functional adoption (S_{firm}) correlates most strongly with performance and revenue-related metrics. Worker-task integration (S_{worker}) comes next, followed by operational investment (S_{invest}) – both generally exhibit more moderate magnitudes for performance and sales increase compared to S_{firm}.

Labor outcomes reveal a distinct divergence in how institutional retooling and task-level integration relate to employment levels. Operational investments (S_{invest}) and functional breadth (S_{firm}) are more strongly associated with employment decreases. In contrast, worker-level task integration (S_{worker}) is uniquely characterized by negligible and sometimes negative relative magnitudes for employment decreases. This indicates that while structural changes such as functional expansion and capital investment are positively associated with labor substitution, task-level assistance by workers correlates more with augmentation than with headcount reduction. Future expectations for employment decreases maintain a stronger link to S_{invest} compared to other use measures. Note also that AI-driven current employment decrease is most strongly associated with S_{invest} followed by S_{firm}, and the link to S_{worker} is weak.

8.2.5 Shapley decompositions

Table 14 reports the Shapley decomposition of the variation explained by the three AI-use indices for the regression in Panel B of Table 11. It is important to note that this contribution (i.e., within R^2) is small in all regressions, so the indices together do not explain a large fraction of the variation in outcomes to start with. The firm-use index, S_{firm}, accounts for the largest share of the explained variation in current and future performance, sales growth, and current employment decline. Its contribution ranges from 43.6 percent for sales growth to 80.1 percent for current performance. By contrast, the investment index, S_{invest}, is the largest contributor to future and AI-attributed employment decline, accounting for 59.9 and 61.8 percent of the explained variation, respectively. The worker-use index, S_{worker}, generally makes

Table 11: OLS regression (LPM) of firm outcomes on AI use indices, conditional on $S_{use} > 0$

	Performance and sales outcomes			**Employment outcomes**		
	Current Performance > Average	**Future Performance > Average**	**Current Sales Increase**	**Current Employment Decrease**	**Current Employment Decrease (AI-driven)**	**Future Employment Decrease**
Panel A. Specification (1)						
S_{use}	0.070***	0.228***	0.151***	0.092***	0.129***	0.073***
	(0.019)	(0.019)	(0.015)	(0.013)	(0.009)	(0.012)
N	39,020	39,030	39,030	39,030	34,990	39,030
R^2	0.0497	0.0491	0.0242	0.0311	0.0285	0.0223
R^2 Adj.	0.0458	0.0452	0.0202	0.0272	0.0240	0.0183
R^2 Within	0.00049	0.00521	0.00363	0.00211	0.01800	0.00140
R^2 Within Adj.	0.00047	0.00518	0.00360	0.00208	0.01797	0.00137
AIC	52920	53720	37500	20420	−33830	18700
Panel B. Specification (2)						
S_{firm}	0.083***	0.141***	0.062***	0.057***	0.038***	0.022^{+}
	(0.018)	(0.018)	(0.014)	(0.012)	(0.007)	(0.012)
S_{worker}	−0.029^{+}	0.035*	0.034**	−0.002	0.005	0.003
	(0.015)	(0.016)	(0.012)	(0.011)	(0.006)	(0.010)
S_{invest}	0.019	0.045^{+}	0.059**	0.048**	0.120***	0.066***
	(0.024)	(0.024)	(0.019)	(0.016)	(0.011)	(0.015)
N	39,020	39,030	39,030	39,030	34,990	39,030
R^2	0.0503	0.0496	0.0243	0.0316	0.0333	0.0227
R^2 Adj.	0.0464	0.0457	0.0203	0.0276	0.0288	0.0186
R^2 Within	0.00112	0.00578	0.00370	0.00259	0.02289	0.00175
R^2 Within Adj.	0.00104	0.00570	0.00363	0.00251	0.02280	0.00167
AIC	52920	53690	37500	20420	−34030	18700

$+ \; p < 0.1$, $* \; p < 0.05$, $** \; p < 0.01$, $*** \; p < 0.001$

Note: Robust standard errors in parentheses. All specifications include fixed effects for firm size class, 3-digit NAICS industry, state, and survey period. Observation counts (N) are rounded in accordance with disclosure rules.

Table 12: Summary of regression magnitudes and outcome means

	Performance and sales outcomes			**Employment outcomes**		
	Current performance > avg.	**Future performance > avg.**	**Current sales increase**	**Current employment decrease**	**Future employment decrease**	**Current employment decrease (AI-driven)**
Independent Variables	Table 9: All firms – indicators of positive index values as regressors *percentage point changes going from indicator = 0 to indicator = 1*					
$S_{use} > 0$	3.21	8.31	6.38	1.79	1.68	
$S_{firm} > 0$	1.92	5.18	3.50	0.92	0.82	
$S_{worker} > 0$	1.02	3.84	3.45	−0.18	0.21	
$S_{invest} > 0$	1.57	3.06	2.52	2.00	1.58	
Mean of Dep. Var.	32.21%	30.62%	12.79%	9.86%	8.75%	
	Table 10: All firms – index values as regressors *percentage point changes corresponding to 1 SD increase in index values*					
S_{use}	1.09	4.06	2.94	1.09	0.95	
S_{firm}	1.74	3.17	1.84	0.99	0.49	
S_{worker}	−0.31	0.95	0.93	−0.13	0.04	
S_{invest}	0.17	0.37	0.52	0.37	0.56	
Mean of Dep. Var.	32.21%	30.62%	12.79%	9.86%	8.75%	
	Table 11: Firms with $S_{use} > 0$ – index values as regressors *percentage point changes corresponding to 1 SD increase in index values*					
S_{use}	1.07	3.53	2.33	1.43	1.13	2.00
S_{firm}	1.67	2.83	1.26	1.14	0.44	0.77
S_{worker}	−0.65	0.78	0.76	−0.05	0.07	0.12
S_{invest}	0.27	0.63	0.84	0.68	0.93	1.71
Mean of Dep. Var.	38.86%	41.57%	18.88%	9.80%	9.13%	2.00%

Table 13: Regression magnitudes as percentages of outcome means

	Performance and sales outcomes			Employment outcomes		
	Current performance > avg.	**Future performance > avg.**	**Current sales increase**	**Current employment decrease**	**Future employment decrease**	**Current employment decrease (AI-driven)**
Independent Variables	Table 9: All firms – indicators of positive index values as regressors *percentage point changes going from indicator = 0 to indicator = 1, as % of dep. var. mean*					
$S_{use} > 0$	10.0%	27.1%	49.9%	18.2%	19.2%	
$S_{firm} > 0$	6.0%	16.9%	27.4%	9.3%	9.4%	
$S_{worker} > 0$	3.2%	12.5%	27.0%	−1.8%	2.4%	
$S_{invest} > 0$	4.9%	10.0%	19.7%	20.2%	18.0%	
Mean of Dep. Var.	32.21%	30.62%	12.79%	9.86%	8.75%	
	Table 10: All firms – index values as regressors *percentage changes corresponding to 1 SD increase in index values, as % of dep. var. mean*					
S_{use}	3.4%	13.3%	23.0%	11.0%	10.8%	
S_{firm}	5.4%	10.3%	14.4%	10.1%	5.7%	
S_{worker}	−1.0%	3.1%	7.3%	−1.3%	0.5%	
S_{invest}	0.5%	1.2%	4.0%	3.7%	6.4%	
Mean of Dep. Var.	32.21%	30.62%	12.79%	9.86%	8.75%	
	Table 11: Firms with $S_{use} > 0$ – index values as regressors *percentage changes corresponding to 1 SD increase in index values, as % of dep. var. mean*					
S_{use}	2.8%	8.5%	12.3%	14.6%	12.4%	99.9%
S_{firm}	4.3%	6.8%	6.6%	11.6%	4.8%	38.7%
S_{worker}	−1.7%	1.9%	4.0%	−0.5%	0.8%	6.0%
S_{invest}	0.7%	1.5%	4.5%	6.9%	10.2%	85.3%
Mean of Dep. Var.	38.86%	41.57%	18.88%	9.80%	9.13%	2.00%

a smaller contribution, although it accounts for approximately one quarter of the explained variation in sales growth.

8.2.6 Regressions with functional-use, worker-task, and organizational-change indicators

The aggregate AI-use indices characterize how firm outcomes vary with the extent of AI use but do not identify the specific business functions, worker tasks, or implementation changes associated with these outcomes. We therefore estimate specifications that jointly include indicators for AI use in 15 business functions and 10 worker tasks, as well as 9 indicators for organizational and operational changes. These specifications include the same fixed effects as the preceding regressions.

Because the resulting 34 indicators are potentially correlated, their coefficients are difficult to interpret individually. We instead present the results of a Shapley decomposition that allocates the within R^2 across predictors. Conducting an exact decomposition at the level of the individual indicators is computationally prohibitive because the number of predictor subsets increases exponentially with the number of regressors.[39] We therefore use an LLM-assisted classification to organize the indicators into nine conceptually related groups: four business-function groups, three organizational-change groups, and two worker-task groups. We review and revise this classification based on the similarity of the underlying functions, tasks, and organizational changes and then apply the Shapley decomposition across the nine groups rather than 34 indicators.[40]

Table 15 summarizes the three largest contributors to each regression's within R^2, while Appendix Table C.1 provides the complete decomposition. We note that, as in the case of the regressions with the indices, within R^2 values are small, even with the 34 individual indicators included as regressors. Collectively, business functions account for the largest share of explained variation in five of the six outcomes, ranging from 46.3 percent for current sales growth to 58.6 percent for future employment decline. The exception is AI-driven employment decline, for which implementation changes and investments account for 50.2 percent.

The leading individual contributors in Table 15 vary across outcomes. Specialized business, technical, and analytical tasks contribute most to current performance, whereas strategy, innovation, and technical functions lead for future performance. Commercial, customer, and external relationship functions are the largest contributors to current sales growth and current employment decline, while administration and governance leads for future employment decline. Administrative functions also rank among the two largest contributors to current performance, future performance, and both current and future employment decline. General knowledge tasks rank among the three largest contributors to future performance, sales growth, and current employment decline, indicating that both specialized and broadly applicable worker uses of AI help account for differences in firm outcomes.

Organizational and operational investments are more prominent for forward-looking or explicitly AI-attributed employment outcomes. Technical infrastructure and data investments are the second-largest contributor to future employment decline. For AI-driven employment decline, human capital and organizational investments and technical infrastructure investments are the two leading contributors. Thus,

[39] With K predictors, an exact Shapley decomposition requires evaluating contributions across as many as 2^K subset models. For 34 indicators, this represents more than 17 billion possible subsets.

[40] The definitions and the content of the groups can be found in Table C.2.

Table 14: Shapley decomposition of AI-use indices and firm outcomes (for Panel B in Table 11)

	Performance > average		Sales increase	Employment decrease		
AI-use index	**Current**	**Future**	**Current**	**Current**	**Future**	**AI-driven**
S_{firm}	**80.1**	**62.2**	**43.6**	**57.2**	28.2	25.2
S_{invest}	13.2	17.9	29.5	33.0	**59.9**	**61.8**
S_{worker}	6.7	19.9	26.9	9.8	12.0	13.0
Within R^2	0.0011	0.0058	0.0037	0.0026	0.0017	0.0229

Notes: Entries for the three AI-use indices report their percentage contributions to the within R^2 allocated by the Shapley decomposition of the regression in Panel B of Table 11; consequently, the entries in each column sum to 100 percent, subject to rounding. The final row reports the within R^2 allocated across the three indices. Boldface indicates the largest contribution within each outcome.

business functions and worker tasks are generally more prominent in explaining performance, sales, and broader employment outcomes, whereas complementary organizational and technical investments are especially important for employment reductions explicitly attributed to AI. These findings are overall consistent with the earlier findings from the Shapley decomposition in section 8.2.5 based on the regressions with the indices.

Table 15: Leading contributors to explanatory power for the regression with the full set of indicators

Outcome	**First**	**Second**	**Third**
Current performance	Specialized tasks	Administrative functions	Commercial and customer relationship functions
Future performance	Strategy, innovation, and technical functions	Administrative functions	General knowledge tasks
Current sales increase	Commercial and customer relationship functions	Human capital and organizational investments	General knowledge tasks
Current employment decrease	Commercial and customer relationship functions	Administrative functions	General knowledge tasks
Future employment decrease	Administrative functions	Technical infrastructure and data investments	Commercial and customer relationship functions
AI-driven employment decrease	Human capital and organizational investments	Technical infrastructure and data investments	Specialized tasks

Notes: The table reports the three largest Shapley contributions to the within R^2 of each regression. Black indicates business functions, red indicates operational investments, and brown indicates worker tasks. The rankings describe contributions to explanatory power and should not be interpreted as causal effects.

These results describe contributions to regression explanatory power and should not be interpreted causally. Unobserved firm characteristics, such as management quality, may simultaneously facilitate AI adoption and improve firm performance. Reverse causality is also possible: firms with stronger performance or sales may have greater resources to finance and implement sophisticated AI systems.

Establishing causal relationships and identifying the underlying mechanisms remain important directions for future research.

9 A comparison with AI use rate estimates in related literature

In this section, we place our findings in the context of other recent work on business adoption of AI. A recurring issue in the large and growing body of papers estimating business AI use concerns the comparability of such estimates; for example, Crane et al. (2025) compare results across 16 sources of estimates (including six firm surveys) and find that differences in estimates are attenuated when considering sampling differences and whether or not the estimates are employment-weighted.[41] We do not attempt a similar "survey of surveys" here but instead start with an illustrative example to help provide context for the BTOS results; we use a firm-level survey for these exercises and then focus our attention on a worker-level survey.

Common themes emerge in comparing AI use survey estimates. Yotzov et al. (2026, p. 4) reviewing differences in estimates point to "differences in survey timing, question wording, sample composition, and the position of survey respondents within their business."[42] Similarly, Bick et al. (2026) point to framing of the adoption question, reference period, target respondent, and sample characteristics. Likewise, Allen (2026) highlights differences in sampling distributions, unit of analysis (worker versus firm), question framing and materiality of reported usage, and target respondent.[43] While we discuss these themes individually below, it is important to remember how their interaction determines how we interpret the results across different surveys. Allen (2026) pulls together some of these themes in comparing Yotzov et al. (2026) and Bick et al. (2026) to BTOS.

We use the Survey of Business Uncertainty (SBU) analyzed in Yotzov et al. as our illustrative example of a national, firm-level survey conducted over a similar time period, allowing us to examine these potential reasons in a series of empirical exercises.[44] We use the Real-Time Population Survey (RPS) conducted by Bick et al. as our illustrative example for a worker-level survey because it is a national survey conducted over the same time-period. All three collections are intended to be repeated so understanding their differences will continue to be relevant. The SBU team has partnered with central banks from around the world to standardize the AI collection, conduct the AI collection twice a year (SBU, henceforth June and December), and publish statistics for the U.S. and other countries. The RPS team has partnered with international academic colleagues to field versions in six European countries (coordinated with a US collection). Given the repeated nature of these surveys, an added consideration is potential collaborations to coordinate collection to better understand differences in estimates. The

[41] The six firm-level surveys analyzed by Crane et al. (2025) include the BTOS but not the SBU. Bonney et al. (2024a) compare results from the first BTOS AI supplement to two waves of the Annual Business Survey (ABS) and McKinsey and Company survey results.

[42] Yotzov et al. (2026) reference estimates from the Annual Business Survey (ABS), BTOS, McKinsey for the US, their primary source of estimates for the US is the Survey of Business Uncertainty but they also include a limited set of exercises using a worker-level survey, the Survey of Working Arrangements and Attitudes (SWAA).

[43] Allen (2026) compares the BTOS, SBU, and RPS.

[44] See section 2.2.3 of Bick et al. (2026) for their comparison of BTOS to the EU-ICT-Firm Survey. Note some of our discussion about the SBU applies to the EU-ICT-Firm Survey since the two surveys have similar reference periods (both use "current"), target respondents (decision makers), and sample characteristics (specifically size, since EU-ICT-Firm does not include firms with fewer than 10 employees).

Census Bureau has discussed this potential with the Atlanta Federal Reserve Bank to coordinate a future collection between the BTOS and SBU. Less formally, the Census Bureau drew lessons learned from the RPS in editing the task question on BTOS and, more recently, in working with the Bureau of Labor Statistics on AI questions on the Household Trends and Outlook Pulse Survey (HTOPS).

Comparison to SBU

We start with a brief overview of the SBU. The SBU is a monthly online national survey of CEOs, CFOs, and senior executives at U.S. firms.[45] Senior executives are recruited for the sample by telephone to confirm their position and company—those that agree are then added to the survey panel. Thus, the respondent to the SBU has been identified as a decision-maker in the firm and has some informal relationship to the Atlanta Federal Reserve Bank. Yotzov et al. (2026) note that "the sponsorship of the Atlanta Fed facilitates the recruitment of senior management (p. 6)." The November 2025 sample includes about 1,100 respondents. About 80% of the respondents are "decision-makers" (CEOs, CFOs, or senior management).

The November 2025 SBU asks a few questions about use of AI; Yotzov et al. (2026) focus on four of these questions. The first AI use question asks:

> "On average, how frequently do you **personally use** A.I. technologies in a typical working week? Note: Among other things, A.I. technologies could include text generation using large language models (e.g. Microsoft Copilot), data or image processing using machine learning, and visual content creation."

Four choices are provided: "Not at all," "Up to 1 hour a week," "1 to 5 hours a week," and "More than 5 hours a week." The second AI use question asks:

> "Which of the following artificial intelligence (A.I.) technologies, if any, does your firm **currently use**? And which do you expect to make use of over **the next three years**?"

Respondents are directed to "Select all that apply" over two columns ("Currently using" and "Expect to use in the next 3 years") from "Text generation using Large Language Models," "Robotics," "Image processing using machine learning," "Data processing using machine learning," "Autonomous Vehicles," "Visual content creation," "Other A.I technology," and "We are not using /do not expect to use any A.I. technologies." Yotzov et al. (2026) find 72% of SBU respondents personally use AI (employment-weighted) and 78% of firms are using AI (employment-weighted). More relevant for our discussion is the 69% firm-level response reported in Meyer et al. (2025). We now look at potential reasons for differences in firm estimates across SBU and BTOS: timing, question content, respondent title, and sample characteristics.

Timing. Differences in timing can refer to both the reference period and the reference frame. In this paper, we focus on the pooled BTOS AI supplement results, but we can use the BTOS AI core results to use a more compatible reference period for comparison's sake. The SBU reports results collected over November 10-21, 2025, the BTOS AI core result for reference period November 3-16, 2025 is 17.3%. The

[45] Another firm survey also targets decision makers. The CFO Survey is conducted by the Federal Reserve Banks of Atlanta and Richmond in partnership with Duke University. As its name suggests, the survey targets financial officers and its questions about AI concern investments in AI. The monthly survey has about 500 responses. See The CFO Survey.

reference frames are "current" for SBU and "Between November 3-16" for BTOS. While these may seem similar at first glance, the BTOS reference frame is more precise in defining current usage. The SBU respondent could consider "current" over a much broader horizon (or over a narrower horizon); depending on how the SBU respondent interprets current, we would expect the AI use to be higher (broader interpretation) or lower (narrower interpretation). Moreover, the SBU could be capturing intermittent users or project-based users who may be missed by the BTOS time frame.

We can try to get a sense of the importance of reference frame by comparing the results for the BTOS core question (Question 23) with the BTOS supplement question about AI use in the last 6 months (Question 24). However, the two questions differ by more than the time frame dimension as Question 24 provides prompts for various business functions.[46] Keeping this important caveat in mind, we find that the AI use rate is 27.7% for Question 24 (see Table C.9).

Question wording. Building on this point about the importance of time frame and our earlier discussion about the core AI questions, the precise language used in the AI use questions can have an impact on the resulting estimates. There are many components to a question: the overall nature of the instrument and question intent, placement of the question in the instrument, question stem, examples provided (explicitly and through rollover functions on the electronic instrument), and responses provided. It is beyond the scope of this paper to address each of these in depth, but we provide some discussion about these here. In terms of the overall nature of the instrument, both the BTOS and the SBU AI collection ask qualitative questions that collect information through a series of checkboxes.[47]

In terms of question intent, as we noted in Section 2, the BTOS is attempting to capture formal AI integration, as opposed to incidental use, since more formal integration is more likely to drive long-term capital and labor adjustments. Other firm surveys, such as the SBU, are likely attempting to capture a broader spectrum of AI business use. Bick et al. (2026) focus on the pre-revision BTOS core question and compare it to results from the EU-ICT-Firm Survey which has a broader framing of its questions. Using the EU-ICT-Firm Survey, they find that for European firms, the share of firms using AI for any business purpose is about 5 times that of firms using AI for production. Using this broader framing brings the European worker and firm use closer together. They conclude that the pre-revision BTOS core framing was too narrow, and further, believe that it still too narrow post-revision (relative to the EU-ICT-Firm Survey).[48]

The placement of the question within the instrument could have a framing impact. The SBU firm question appears after the personal use question which has helped prime the respondent to think about AI use. This priming can help the SBU respondent think more closely about firm AI use leading to a more precise (and higher) response than otherwise. On the other hand, Bick et al. (2026, p. 16) point out that since this priming concerns the respondent's own use, it could "blur the distinction between

[46] Bick et al. (2026) make a similar point about the potentially confounding impact of time frame when using the first BTOS AI supplement to examine the difference between core and supplement questions (see Appendix B).

[47] As compared to more quantitative survey collections such as on the CFO Survey which asks for expenditures and investment in AI or the planned AI business expenditures question on the Census Bureau's Annual Integrated Economic Survey (AIES).

[48] Bick et al. (2026) compare results BTOS core and supplement questions from the first BTOS AI Supplement to try to get a sense of the importance of the narrow definition used in the core question. With the caveat the reference periods differ (see footnote above), they conclude that "providing more concrete examples than the standard question, the supplement appears to make AI adoption more salient to respondents" (Appendix B, p. VIII) as the rate is 8.8% as compared to the core rate of 5.0%.

personal and firm adoption" which would lead to a less precise (and inflated) estimate from the SBU. Several differences in the wording of the questions could lead to different answers.

Similarly, there are differences in the responses provided that could have an impact. SBU provides seven technologies while the BTOS core provides yes/no/do not know responses. In this sense, BTOS Question 24 is more similar to the SBU question in the sense that both provide detailed responses. However, the responses are very different: SBU provides technologies while BTOS provides business functions. The SBU question stem defines these as AI technologies, but not all of the SBU responses repeat this restriction. Three of the technologies provided in the responses do not necessarily use AI: "Robotics", "Autonomous vehicles," and "Visual content creation."[49] Depending upon whether the respondent uses the question stem prompt concerning AI to condition checking these boxes, the resulting estimate could overstate the importance of AI use. In designing questions and responses, there can be a tension between precision and visual clutter which we have also found when trying to design future iterations of the BTOS AI supplement. It would be useful to know how many respondents check one of the responses without an explicit mention of AI and also check the response that they are not using AI. In sum, the question wording is a promising avenue for collaborative research to understand the differences between surveys (especially SBU and BTOS).

Respondent type. An important characteristic of the survey concerns the intended respondent and the survey relationship with this respondent. As described above, the intended respondent to the SBU is a decision-maker who has been recruited for the panel through an initial telephone contact. The intended respondent to the BTOS is broader as it encompasses anyone with the appropriate knowledge and there is no recruiting mechanism. Thus, there is an implicit relationship with respondents for SBU but not for BTOS.[50]

We cannot address whether this implicit relationship has an impact on SBU respondents as compared to BTOS respondents. If the relationship does have an impact on respondents in terms of a greater probability of providing a perceived desirable response, it is interesting to note that what is considered "desirable" may have changed over time. Allen (2026) makes this point as well, in his discussion of "socially desirable responses." We can, however, address the issue of differences in respondent titles.

Yotzov et al. (2026, p. 2) argue that one potential reason for differences in adoption across the SBU and BTOS is that BTOS respondents are "typically non-executives, so may not be informed on AI use." Allen (2026) discusses this potential information asymmetry and attempts to get some sense of its importance by examining the prevalence of "do not know" responses in BTOS. As Allen (2026) notes "do not know" is generally about 10% and thus cannot fully account for the gap between BTOS and SBU.[51] We examine this concern using the fact that the BTOS collects the titles of the respondents. The November 2025 SBU sample has the following distribution of respondents: 51.1% CFO, 10.8% CEO, 17.7% senior management, and 20.6% other (Meyer, email to authors).[52] Yotzov et al. (2026) report

[49]These are similar to the responses provided in the 2018 Annual Business Survey (ABS). The 2018 ABS provides checkboxes for ten advanced technologies including "Robotics" and "Automated Guided Vehicles (AGV) or AGV Systems." McElheran et al. (2024) note that five of these ten technologies are "commonly associated with AI" (p. 381) including AGVs but not robotics.

[50]Neither pay their respondents, which Yotzov et al. (2026) note is an important consideration since payment can lead to falsification, see p.5.

[51]The response "Do not know" to the core current AI use question ranges from 10.4% to 11.2% during the AI supplement collection, more recently it is closer to 9%.

[52]Yotzov et al. (2026) report the statistics for SBU data averaged over January 2017 to January 2025 (Figure 2).

that CEOs have the highest reported self-use of AI. Those reporting not using AI at all are as follows: senior executives (30%), CFOs (27%), and CEOs (18%) (Yotzov et al., 2026, Figure A9). The authors view self-use as a proxy for firm use suggesting that we should expect to see differences in the firm use estimates based upon the title of the respondent. We look at this and other questions using BTOS data.

Appendix Table C.3 shows the firm-weighted shares of BTOS respondent types and the corresponding AI use rates. The most prevalent respondent type is top executives and owners, which includes titles like CEO, president, vice president, owner, partner, managing member, executive director. These ranks generally may have higher visibility into the macro-level impact of AI. They are usually the ones making strategic decisions to invest in AI, understanding the expected ROI, and driving broader workforce restructuring or hiring shifts. However, this group may also be the most susceptible to strategic signaling incentives, social desirability bias, or peer pressures and may overreport AI use, which may be mitigated if the survey respondent is not recruited. This group constitutes 50% of the respondents and core firm-level AI use rate (based on Question 23) within this group is 21%, somewhat higher than the average adoption rate of 17% across these groups.[53]

Operations and General Management includes titles like general manager, director, manager, business manager, which may have relatively high visibility into the practical, day-to-day applications of AI. These ranks may implement the AI tools and directly manage the workforce using them. Hence, they may be best for answering questions about how AI is changing specific tasks, changes in worker productivity, training bottlenecks, and localized impacts on team structures. This group constitutes 15% of the respondents and exhibits nearly average adoption rate (17%).

Finance and Accounting consists of titles like CFO, controller, accounting manager, accountant, CPA, and bookkeeper. These titles tend to be more specialized. Their view of AI may be concentrated on the financial costs of software/hardware, or specifically on AI applications within the accounting and finance verticals (e.g., automated bookkeeping, fraud detection). They may be best for answering questions about the capital expenditures associated with AI. Unless the firm is quite small, they may have limited visibility into how AI is impacting workflows in other departments like marketing or engineering. This group makes up 15% of the respondent population and reports a much lower AI use rate of 11%.

Similarly, Administrative Support group, which includes titles like office manager, administrative assistant, and secretary, accounts for 7% of the respondents and is associated with an AI use rate of 10%. There is, however, also a residual group ("Other"), amounting to about 15% of the respondents, which includes a variety of titles as well as individuals' names – these names may pertain to some top executives, owners, and managers, as well as others. This eclectic group reports a higher-than-average use rate of 20%.

Overall, while Top Executives and Owners group is associated with higher use rates (double the rates for finance and accounting and administrative support), these rates are not exceptionally large relative to the average. Confounding this assessment is the industry and firm size distribution of respondents. For instance, titles like CEO and CFO may be more prevalent in larger firms which have higher rates of AI use, and some respondent types may reflect industry specialization correlated with heavier AI use – such as an IT specialist or data scientist, professions which may be prevalent in Professional Services, for example. To address these concerns, Table C.4 reports the estimates from a linear probability model

[53]This does not exactly match the overall firm AI use rate as some respondent titles are missing.

that relates firm AI use indicator to respondent type categories with firm size class, industry, state and survey period fixed effects. The estimates indicate that, relative to the omitted group of top executives and owners, administrative support and finance and accounting groups have 10 pp lower use rates, while operations and general management and other groups exhibit 4 pp and 3 pp lower rates, respectively. In other words, there is a higher propensity for AI use reporting among the top executives and owners *even after controlling for observable confounders*. However, as discussed earlier, it is not clear whether this pattern represents this group's superior information on aspects of firm AI use or their propensity to potentially overreport due to social desirability or competitive pressures. Nevertheless, the rates in Table C.3 imply that even if all respondents were top executives or owners, the firm-weighted use rate would have been at most 21%.

Sample characteristics. Lastly, we turn to differences in sample characteristics. Here again, it is important to consider the underlying purpose of each survey. The SBU is intended to capture the responses of decision-makers. Given the recruitment required for this, it would not be surprising that the panel does not include many young businesses. The BTOS is intended to capture a nationally representative view of all private non-agricultural employer firms. Thus, we would expect one difference in sample characteristics would be firm size as measured by employment (and age, which we do not measure in this paper). In the November 2025 SBU sample, the median number of employees is 81 and the mean number of employees is 167 (Meyer, email to authors). In contrast, average firm size in the second BTOS supplement is 20 employees. Yotzov et al. (2026) compare the industry composition of the SBU to the Economic Census and find that the employment-weighted SBU is over-represented in Finance and Insurance (52), Manufacturing (31-33), and Professional and Business Services (54).[54]

To address the firm size issue, we calculate firm adoption rates conditional on firm size thresholds. Appendix Table C.5 shows that, as we move to higher size thresholds, firm-weighted adoption rates increase monotonically, reaching to 35% for 1000+ employee firms.[55]

When the same exercise is done for AI-heavy sectors (Table C.6), the adoption rates are uniformly much higher and peak at 57% for the largest firms. Table C.7 focuses only on Professional Services and Finance and Insurance. These two sectors have higher adoption rates and the largest firms in these sectors use AI at a rate of 63%. Finally, we tabulate adoption rates by size thresholds for AI-heavy sectors with the sample restricted to the respondent type "top executives and owners". In this case, adoption rates reach more than 70% for large size classes. Notably, the increase in AI use by firm size in AI-heavy sectors is more pronounced for "top executives and owners" relative to other respondents.

To summarize our findings from tables C.3-C.8, when comparing the BTOS to the SBU, the sample characteristics by industry and firm size are important and the issue of respondent title is more important in certain industries and firm sizes. Tables C.3 and C.4 also show that executives report more AI use. CEOs report more AI use, but recall that the SBU respondents are often CFOs. Table C.5 shows a monotonic increase in AI use by firm size. Since SBU sample over-represents AI-heavy sectors (NAICS

[54]See Figure A1 in Yotzov et al. (2026) which is for the SBU panel of more than 6,000 firms. However, this pattern also holds for the November 2025 SBU sample (Meyer, email to authors).

[55]Crane et al. (2025) attempt to broadly control for differences in firm size in their comparisons by focusing on employment-weighted results. Bick et al. (2026, Appendix B, p. VIII) attempt a similar exercise to our approach by restricting BTOS results by size groups. In their case, they restrict pre-revision BTOS results to those for firms with at least 10 employees and find that this does not materially change the estimates (because the firm-size adoption relationship is U-shaped during that period).

51, 52, 53, and 54), AI use rates climb to 49% firm-weighted and 52% employment weighted for firms with 200+ employees (Table C.6). Finally, when we pull together respondent title, firm size, and high AI sectors, the analogous use rates climb to 58% and 76% respectively (Table C.8). Recall that the SBU firm adoption rate is 69% and 78% employment-weighted. Overall, based on tables C.3-C.8, most of the differences between the BTOS and SBU adoption rates stems from SBU's focus on large firms in relatively AI intensive sectors. Respondent title also plays an important supporting role.

Comparison to RPS

We also highlight a worker-level survey, the Real Time Population Survey (RPS), in order to draw out some common themes across measures.[56] The RPS is designed to mirror the Current Population Survey (CPS) in terms of representation and concepts to allow for comparisons to CPS collections; for example, allowing them to compare AI adoption from RPS results to computer adoption from CPS results (Bick et al., 2025). Bick et al. (2026) adopt the RPS methodology for six European countries and report results from the US and international collections run May-June 2025 and January-February 2026.

For this international collection, the authors used an international commercial survey provider to administer the collection online from recruited respondents. Survey invitations were targeted to specific demographic groups designed to be representative of the country's "employed, at work" populations. The survey population is adults age 18-67 who were employed and at work the previous week with about 3,000 respondents per country for the January-February 2026 wave.

In addition to these considerations, the RPS also imposes some quality restrictions on the responses that they collect and use (see Bick et al. (2026) Appendix A.1 for a detailed discussion). These restrictions include dropping from the sample respondents who took the survey in less than 2 minutes and those who did not respond in the affirmative to the screener question: "We care about the quality of our survey data and hope to receive the most accurate measure of your opinions, so it is important to us that you thoughtfully provide your best answer to each question in the survey. Do you commit to providing your thoughtful and honest answers to the questions in this survey?" In addition, they include a captcha question and the survey provider "filtered out any responses that look suspicious" (Appendix A, p. I). It is not clear how many responses this filtered out for the US (nor whether it introduces any biases), but they note that overall 3-8% of responses per country were filtered out.

The RPS question specifically focuses on GenAI. It starts with a definition of AI that includes examples of existing AI tools (specifically, it lists "ChatGPT, Gemini, and Midjourney"). The question is: "Do you use Generative AI for your job? (No/Yes)." Given the focus on GenAI, this question is more comparable to BTOS question 31 which asks about use in the last six months and provides some examples of existing AI tools (specifically it lists "ChatGPT, Gemini, Copilot, Claude, Dall E, Midjourney and GitHub Copilot"). Notice that the RPS question has a potentially broader time frame (since it is not explicitly given). Using the January-February 2026 wave of the international RPS, they find that 43% of

[56] A more recent worker-level collection is through a Bureau of Labor Statistics (BLS) and Census Bureau partnership which added four questions on AI to the Household Trends and Outlook Pulse Survey (HTOPS). Respondents who worked for pay or profit in the past week are asked about their occupation (selecting from high-level standard occupation codes), use of AI in tasks for work (selecting from eleven options), frequency of AI use last week (selecting from three choices), and time saved through AI use (selecting from eight choices). The HTOPS collected this information over March 13-30 with plans to publish in spring 2026. In future work, we will also incorporate a discussion of HTOPS results.

workers use GenAI for their job (up from 39.4% in the May-June 2025 wave).

Our last comparison exercise attempts to bring all of this discussion together. Appendix Table C.9 takes a holistic approach and presents several AI use rate estimates implied by various survey questions. Depending on the type of AI use/measurement layer and the reference period, firm-weighted use rates range from 17.9% to 32.0%, and employment-weighted rates vary between 31.2% to 54.0%. Taking the RPS example from above, we can make a rough comparison using row 3 of Table C.9 which reports results of Question 31. Since we are comparing to a worker survey, employment-weighted is the more relevant comparison. In this case, BTOS reports GenAI use rate for worker tasks of 38.9% as compared to 43% from RPS. These rates demonstrate that while measurement layer, reference period, and weighting matter, nationwide AI use rate estimates based on the BTOS AI supplement rarely exceed 50%. When BTOS AI supplement rates exceed 50% it is for a sub-sample that is skewed towards larger firms or firms in certain industries (or a combination of these).

Summing up

In sum, it is important to keep in mind underlying survey differences in interpreting the AI use results from firm and worker surveys. Sectoral concentration plus size concentration in AI use implies caution/perspective when looking across surveys. The SBU is more about large firms in relatively AI-intensive sectors, the RPS focuses on worker use, and the BTOS is intended to be broad in scope (economy-wide) with a focus on material uses of AI. Allen (2026) makes a similar point about the importance of understanding these underlying differences in order to appreciate the complementary nature of the three perspectives on AI use provided by the three surveys.

10 Conclusion and future work

Using the second AI supplement in the BTOS, this paper provides new evidence on the microstructure of AI diffusion by analyzing multiple-layers of AI use within a firm: the extent of firm level and worker task use of AI, the degree of alignment between firm adoption and worker use, where AI is deployed across business functions and tasks, how firms report AI changing work through task substitution, augmentation, and creation, and how these mechanisms relate to labor and capital replacement behavior.

Some key findings from the analysis are as follows:

1. While growing, diffusion remains in the early stages across the general firm population. AI adoption has reached 18% of firms (32% employment-weighted), with worker-level use at 23% (41% emp-weighted).

2. Adoption remains heavily skewed toward large enterprises. Firms with 250+ employees report a 31% adoption rate, nearly double the 18% seen in micro-firms (1–10 employees).

3. Use is concentrated in knowledge-intensive sectors (Information, Professional Services, Finance, and Education) and multi-sector firms. In these sectors, employment-weighted adoption is projected to exceed 50% within the next six months.

4. Most firms are minimalist adopters, utilizing AI in only 1–3 business functions. While there is some specialization in marketing, strategy/R&D, and administrative functions, only 4% of functional users tend to deploy AI comprehensively across business functions.

5. Labor impacts are dominated by augmentation rather than substitution. Among firms shifting task structures, 66% engage exclusively in augmentation. While substitution is not common, its intensive margin (depth of replacement) has increased since 2024.

6. AI-driven capital substitution (16%) is over three times more common than AI-induced employment changes (5%). This stability aligns with an augmentation-heavy pattern where AI retools existing software and equipment rather than human roles.

7. Functional integration and operational investments are positively associated with sales increases, performance, and headcount declines. Conversely, worker-task breadth is positively associated with enhanced performance/sales, but not with displacing labor.

Overall, these results indicate a nascent stage of AI diffusion in business-function use, with adoption heavily skewed toward large firms and knowledge-intensive sectors. Although AI correlates with superior firm performance, its current impact is defined primarily by task augmentation rather than substitution, leaving firm employment largely unaffected. Both functional breadth and structural investment are positively associated with performance and sales growth and, to a lesser degree, with labor displacement. This suggests that proactive organizational adjustments are important for overcoming implementation frictions and traversing the productivity J-curve, though such actions also carry implications for headcount. Future research will focus on isolating the underlying mechanisms and establishing causal links between integration strategies and these firm-level outcomes.

The analysis highlights the importance of integrated, multi-dimensional measurement of AI adoption for understanding its economic impacts and policy relevance. This paper provides descriptive evidence using the timely and novel BTOS AI supplement, offering new insights into how different components of AI use relate to firm outcomes. In particular, we document meaningful correlations between various dimensions of AI adoption and measures of firm performance. A key next step is to link these rich survey-based measures to administrative microdata in order to examine realized outcomes such as employment, earnings, productivity, and firm dynamics. Such integration would also enable researchers to assess whether early patterns of AI adoption predict subsequent scaling, organizational change, and longer-run performance, while facilitating more rigorous causal analysis.

As was the case for the first BTOS AI Supplement, lessons learned from analyses of the second BTOS AI Supplement will help inform potential future collections. Capturing a quickly evolving technology is challenging and requires a flexible approach to collections (while balancing the desire to have consistent time series). We have benefitted greatly from the experiences of other surveys, especially the two highlighted in section 9, and hope to build on our strengths and those of others. For example, as we develop a potential third BTOS AI Supplement, we are exploring in our discussions with the SBU team, a collection that would attempt to capture the impact of AI-driven uncertainty on hiring. We look forward to seeing the directions that others take in their research using the newest BTOS AI Supplement results.

References

D. Acemoglu, G. Anderson, D. Beede, C. Buffington, E. Childress, E. Dinlersoz, L. Foster, N. Goldschlag, J. Haltiwanger, Z. Kroff, P. Restrepo, and N. Zolas. Technology, firms, and workers: Evidence from the 2019 annual business survey. Working Paper 30659, National Bureau of Economic Research, 2022.

D. Acemoglu, G. Anderson, D. Beede, C. Buffington, E. Childress, E. Dinlersoz, L. Foster, N. Goldschlag, J. Haltiwanger, Z. Kroff, P. Restrepo, and N. Zolas. Automation and the workforce: A firm-level view from the 2019 annual business survey. In Susanto Basu, Lucy Elridge, John Haltiwanger, and Erich Strassner, editors, *Technology, Productivity, and Economic Growth*. University of Chicago Press, 2025.

J. Allen. Monitoring AI adoption in the U.S. economy. *FEDS Notes*, April 2026. Washington: Board of Governors of the Federal Reserve System, April 03, 2026, https://doi.org/10.17016/2380-7172.4032.

S. Baslandze, Z. Edwards, J. Graham, T. McClure, B. Meyer, M. Sparks, S. Waddell, and D. Weitz. Artificial intelligence, productivity, and the workforce: Evidence from corporate executives. Working Paper 34984, National Bureau of Economic Research, 2026.

A. Bick, A. Blandin, and D. Deming. The rapid adoption of generative AI. Working Paper 2024-027B/027C/027F, Federal Reserve Bank of St. Louis, 2025. URL https://doi.org/10.20955/wp.2024.027.

A. Bick, A. Blandin, D. Deming, N. Fuchs-Schundeln, and J. Jessen. Mind the gap: AI adoption in europe and the US. Working Paper 34995, National Bureau of Economic Research, 2026.

K. Bonney, C. Breaux, C. Buffington, E. Dinlersoz, L. Foster, N. Goldschlag, J. Haltiwanger, Z. Kroff, and K. Savage. Tracking firm use of AI in real time: A snapshot from the business trends and outlook survey. Working Paper 26-25, Center for Economic Studies, U.S. Census Bureau, 2024a.

K. Bonney, C. Breaux, C. Buffington, E. Dinlersoz, L. Foster, N. Goldschlag, J. Haltiwanger, Z. Kroff, and K. Savage. The impact of AI on the workforce: Tasks versus jobs? *Economics Letters*, 244, 2024b.

E. Brynjolfsson, D. Rock, and C. Syverson. The productivity j-curve: How intangibles complement general purpose technologies. *American Economic Journal: Macroeconomics*, 13(1):333–372, 2021.

L. Crane, M. Green, and P. Soto. Measuring AI uptake in the workplace. *FEDS Notes*, February 2025. Board of Governors of the Federal Reserve System.

E. Dinlersoz, C. Dogan, and N. Zolas. Starting up AI. *Economics of Innovation and New Technology*, 2026. forthcoming.

L. A. Goodman. Exploratory latent structure analysis using both identifiable and unidentifiable models. *Biometrika*, 61(2):215–231, 1974.

A. Humlum and E. Vestergaard. Still waters, rapid currents: Early labor market transformation under generative AI. Working Paper 33777, National Bureau of Economic Research, 2026.

J. Magidson and J. K. Vermunt. Latent class models. In D. Kaplan, editor, *The SAGE Handbook of Quantitative Methodology for the Social Sciences*, pages 175–198. Sage Publications, Thousand Oaks, CA, 2004.

K. McElheran, F. Li, E. Brynjolfsson, Z. Kroff, E. Dinlersoz, L. Foster, and N. Zolas. AI adoption in america: Who, what, and where. *Journal of Economics & Management Strategy*, 33:375–415, 2024.

K. McElheran, M. J. Yang, Z. Kroff, and E. Brynjolfsson. The rise of industrial AI in america: Microfoundations of the productivity j-curve(s). Working Paper 25-27, Center for Economic Studies, U.S. Census Bureau, 2025.

B. Meyer, J. M. Barrero, N. Bloom, S. J. Davis, K. Foster, and E. Mihaylov. Survey of business uncertainty monthly report. Technical report, Federal Reserve Bank of Atlanta, 2025.

K. L. Nylund, T. Asparouhov, and B. O. Muthén. Deciding on the number of classes in latent class analysis and growth mixture modeling: A monte carlo simulation study. *Structural Equation Modeling: A Multidisciplinary Journal*, 14(4):535–569, 2007.

J. C. Short, G. T. Payne, and D. J. Ketchen Jr. Research on organizational configurations: Past accomplishments and future challenges. *Journal of Management*, 34(6):1053–1079, 2008.

J. K. Vermunt. Multilevel latent class models. *Sociological Methodology*, 33(1):213–239, 2003.

J. K. Vermunt. Latent class modeling with covariates: Two improved three-step approaches. *Political Analysis*, 18(4):450–469, 2017.

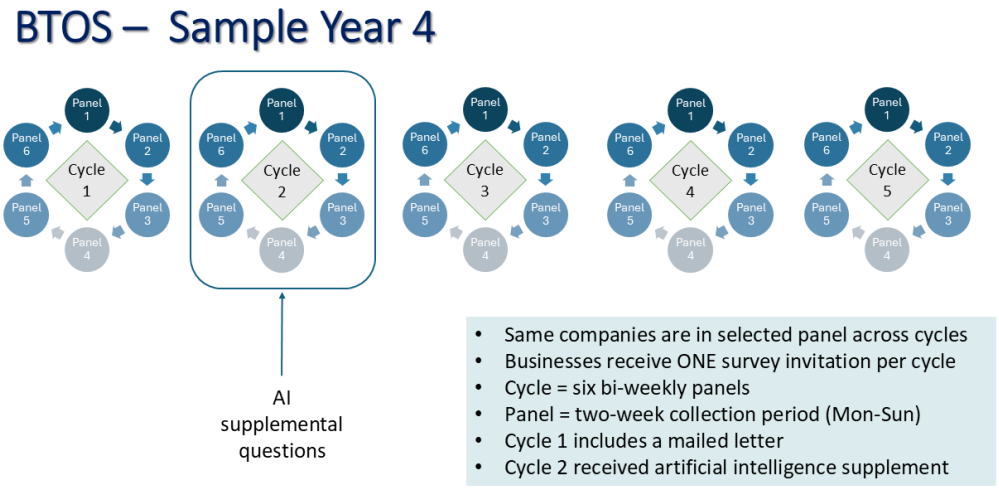
J. K. Vermunt and J. Magidson. Latent class analysis with sampling weights: A maximum-likelihood approach. *Sociological Methods & Research*, 36(1):87–111, 2007.

I. Yotzov, J. Barrero, N. Bloom, P. Bunn, S. Davis, K. Foster, A. Jalca, B. Meyer, P. Mizen, M. Navarette, P. Smietanka, G. Thwaiets, and B. Wang. Firm data on AI. Working Paper 34836, National Bureau of Economic Research, 2026.

N. Zolas, E. Brynjolfsson, K. McElheran, D. Beede, C. Buffington, N. Goldschlag, L. Foster, and E. Dinlersoz. Advanced technologies adoption and use by u.s. firms: Evidence from the annual business survey. Working Paper 28290, National Bureau of Economic Research, 2020.

Appendix

A BTOS collection and Cycle 2 questionnaire with AI supplement



Welcome to the Business Trends and Outlook Survey Content V4

Cycle 2 with AI Supplement Revised 07/23/2025

The U.S. Census Bureau is requesting your assistance to produce important data on current economic conditions, as well as the impact on our nation's economy when natural disasters or other emergencies occur.

Your response is especially important as it helps policy makers, government officials, and businesses like yours understand these effects. Please complete this quick survey; **it will take approximately 20 minutes.**

The data you provide will be aggregated to create a two-week snapshot of economic activity within the U.S. This survey is not intended to replace more comprehensive surveys that typically require the use of company records.

- Estimates are acceptable and encouraged.
- The survey will take approximately 20 minutes.
- This survey includes questions about this business's use of AI
- Please include all domestic operations of this consolidated business, including all locations, subsidiaries, and divisions.

1. FOR SINGLE UNITS ONLY Is the Employer Identification Number (EIN) used on this business's latest Internal Revenue Service Form 941, Employer's Federal Quarterly Tax Return?
   - Yes
   - No

If no, please provide EIN:

2. Which of the following best describes the current operational status of this business?

   - In business
   - Temporarily or seasonally inactive
   - Ceased operation
   - Sold or leased to another operator

3. Overall, how would you describe this business's current performance? Please include all domestic operations of this consolidated business, including all locations, subsidiaries, and divisions.

   - Excellent
   - Above average
   - Average
   - Below average
   - Poor

4. Between MMM DD – MMM DD, how did this business's operating revenues/sales/receipts change? Please compare this two-week period to the two weeks prior. Estimates are acceptable.

   - Increased
   - Decreased
   - No change

5. Between MMM DD – MMM DD, how did this business's number of paid employees change?

   - Increased
   - Decreased
   - No change

6. Between MMM DD – MMM DD, how did the total number of hours worked by this business's paid employees change?

   - Increased
   - Decreased
   - No change

7. Between MMM DD – MMM DD, how did the time it takes for this business to receive deliveries from suppliers change?

   - Increased
   - Decreased
   - No change
   - Not applicable

8. How would you describe this business's current inventories?

   - Larger than optimal
   - Optimal

- Smaller than optimal
- Not applicable

9. Between MMM DD – MMM DD, how did demand for this business's goods or services change?

   - Increased
   - Decreased
   - No change

10. Between MMM DD – MMM DD, how did the prices this business charges for its own goods or services change?

    - Increased
    - Decreased
    - No change

11. Between MMM DD – MMM DD, how did the prices this business pays for goods or services change?

    - Increased
    - Decreased
    - No change

12. In the last six months, in what ways did changes to interest rates negatively impact this business? Select all that apply.

    - Decreased profitability
    - Inability to service debt
    - Inability to refinance existing loans
    - Inability to invest in this business
    - Not applicable

13. In the last six months, did this business experience any monetary losses due to an extreme weather event (for example, hurricane, flood, drought, or heat wave)?

    - Yes
    - No

14. What type of extreme weather event caused the monetary losses? Select all that apply.

    - Wildfire
    - Earthquake
    - Hurricane
    - Tornado
    - Flood
    - Drought
    - High winds
    - Severe storm/hail
    - Snow/blizzard/ice

- Extreme temperature (heat or cold)
- Other

15. Six months from now, how do you think you will describe this business's performance?

    - Excellent
    - Above average
    - Average
    - Below average
    - Poor

16. Six months from now, how do you think this business's number of paid employees will have changed?

    - Increased
    - Decreased
    - No change

17. Six months from now, how do you think the total number of hours worked by this business's paid employees will have changed?

    - Increased
    - Decreased
    - No change

18. Six months from now, how do you think the time it takes for this business to receive deliveries from suppliers will have changed?

    - Increased
    - Decreased
    - No change
    - Not applicable

19. Six months from now, how do you think you will describe this business's inventories?

    - Larger than optimal
    - Optimal
    - Smaller than optimal
    - Not applicable

20. Six months from now, how do you think demand for this business's goods or services will have changed?

    - Increased
    - Decreased
    - No change

21. Six months from now, how do you think the prices this business charges for its own goods or services will have changed?

    - Increased

- Decreased
- No change

22. Six months from now, how do you think the prices this business pays for goods or services will have changed?

    - Increased
    - Decreased
    - No change

23. Between MMM DD – MMM DD, did this business use Artificial Intelligence (AI)[57] in any of its business functions? (Examples of AI: machine learning, natural language processing, virtual agents, voice recognition, etc.)

    - Yes
    - No
    - Do not know

24. In the last six months, did this business use Artificial Intelligence (AI) in any of the following business functions?

	Yes	**No**	**Don't know**
Production of goods (e.g., making or assembling products, construction)			
Provision of services, products, or merchandise (e.g., providing services, products, or merchandise to customers)			
Strategy and business development			
Finance and accounting			
Sales and marketing			
Customer service			
Research and development			
Information technology			
Human resources			
Public relations and communication			
Management and administration			
Sourcing, supply chains, and purchasing			
Quality management and control			
Distribution			
Legal and compliance			

Skip to 30 if the answer to 24 is No

[57] AI Definition: Computer systems and software that are able to perform tasks normally requiring human intelligence, such as decision-making, visual perception, speech recognition, and language processing. Types or applications of AI include machine learning, natural language processing, virtual agents, predictive analytics, machine vision, voice recognition, decision making systems, data analytics, image processing, etc.

25. In the last six months, did this business use Artificial Intelligence (AI) to do any of the following? Select all that apply.

    - Perform a task previously done by an employee
    - Supplement or enhance a task performed by an employee
    - Introduce a new task not previously done by an employee
    - None of the above

    Show 26 only if 25's response is perform a task previously done by an employee, otherwise skip to 27.

26. In the last six months, how many tasks previously done by employees were instead performed by Artificial Intelligence (AI)?

    - A small number
    - A moderate number
    - A large number

27. In the last six months, did this business use Artificial Intelligence (AI) to perform operations previously performed by existing software or equipment?

    - Yes
    - No
    - Do not know

28. In the last six months, how did the use of Artificial Intelligence (AI) affect this business's total employment?

    - Increased
    - Decreased
    - Did not change

29. In the last six months, to use Artificial Intelligence (AI), what changes did this business make? Select all that apply.

    - Trained current staff to use AI
    - Hired staff trained in AI
    - Purchased computing power or specialized equipment or software
    - Purchased cloud services or cloud storage
    - Changed data collection or data management practices
    - Developed new workflows
    - Used vendors or consulting services to install or integrate AI
    - Other (please describe:)
    - This business did not make any changes to use AI

30. In the last six months, did this business's employees use Artificial Intelligence (AI) to assist in any work-related tasks that support business functions? (e.g. software coding, writing emails, preparing visualizations, summarizing documents, gathering and processing information)

- Yes
- No
- Do not know

31. In the last six months, did this business's employees use Generative AI to assist in any work-related tasks? (Generative AI is a type of Artificial Intelligence that uses prompts or other inputs to create text, images, music, videos, or code. Some examples of Generative AI include ChatGPT, Gemini, Copilot, Claude, Dall E, MidJourney, and GitHub Copilot.)

    - Yes
    - No
    - Do not know

    Skip to 33 if No or Do not know is selected for 31

32. In the last six months, what work-related tasks did this business's employees use Generative AI to assist with? Select all that apply.

    - Information processing, paperwork, or filing
    - Writing or editing documents, emails, or communications
    - Interpreting, analyzing, translating, or summarizing documents
    - Customer support
    - Developing or researching new projects, processes, or products
    - Searching for information or technical help
    - Software coding or debugging
    - Data analysis or visualization
    - Tutoring, training, or learning
    - Other tasks (please describe:)

33. During the next six months, do you think this business will be using Artificial Intelligence (AI)2 in any of its business functions? (Examples of AI: machine learning, natural language processing, virtual agents, voice recognition, etc.)

    - Yes
    - No
    - Do not know

    Skip to 35, if no or don't know is selected for 33

34. During the next six months, do you think this business will be using Artificial Intelligence (AI) in any of the following business functions?

    If the answer to 34 is yes then skip 35

35. Why does this business not plan to use Artificial Intelligence (AI) during the next six months? Select all that apply.

    - Too expensive
    - AI is not a mature enough technology yet
    - Lack of knowledge on the capabilities of AI

	Yes	No	Don't know
Production of goods (e.g., making or assembling products, construction)			
Provision of services, products, or merchandise (e.g., providing services, products, or merchandise to customers)			
Strategy and business development			
Finance and accounting			
Sales and marketing			
Customer service			
Research and development			
Information technology			
Human resources			
Public relations and communication			
Management and administration			
Sourcing, supply chains, and purchasing			
Quality management and control			
Distribution			
Legal and compliance			

- Concerns about privacy/security
- Concerns about bias
- Lack of skilled workforce
- Lack of required data
- Laws and regulations prevent or restrict use of AI
- Previous or current use of AI did not meet expectations
- AI is not applicable to this business
- Other (please describe:)

36. Would you like to provide any information about your responses to these questions or this business? Remarks (500 characters)

37. For future Business Trends and Outlook Survey requests, please provide your preferred contact information:

    Contact name:

    Contact title:

    Email:

38. Please correct the following information for your company Doing Business As (DBA) name: Street:

    City/State:

    Suite/ Unit:

    Zip:

This collection has been approved by the Office of Management and Budget (OMB). The eight-digit OMB approval number is 0607-1022 and appears in the lower left corner of each reporting screen. Without this approval, we could not conduct this survey.

Authority and Confidentiality

Title 13, United States Code, Sections 131 and 182, authorizes the U.S. Census Bureau to conduct this collection and to request your voluntary assistance. The Census Bureau is required by Section 9 of the same law to keep your information confidential and use your responses only to produce statistics. The Census Bureau is not permitted to publicly release your responses in a way that could identify your business, organization, or institution. Per the Federal Cybersecurity Enhancement Act of 2015, your data are protected from cybersecurity risks through screening of the systems that transmit your data.

Burden Estimate Statement We estimate the Business Trends and Outlook Survey will take an average of 20 minutes to complete. Factors such as company size, complexity and activity will affect your actual time to complete the survey. This estimate includes the time to review instructions, search existing data sources, gather and maintain the data needed, and complete and review the survey. You may email comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to: Be sure to use "EID Survey Comments 0607-1022" as the subject.

B Construction of AI use and impact indices

This appendix provides the details of the construction of various indices used in the analysis. To ensure that all indices represent confirmed operational intensity, responses of "No" and "Don't know" in any survey question are treated as zeros (0), and "Yes" responses get a value of 1. This conservative approach anchors the indices on verified organizational activities and prevents the inflation of intensity scores due to respondent uncertainty.

For any given firm, the AI integration index, $S_{integration}$, is calculated as a neutral-weighted average of four components:

$$S_{integration} = \frac{1}{4}S_{firm} + \frac{1}{4}S_{worker} + \frac{1}{4}S_{invest} + \frac{1}{4}S_{impact} \tag{3}$$

The first three components measure three different types of AI use intensity. S_{firm} measures the breadth of use across 15 business functions, whereas S_{worker} focuses on the breadth of use across 10 worker-tasks. S_{invest} captures the breadth of organizational investment and changes to implement and use AI. Finally, S_{impact} is a composite measure of various impacts of AI use within the firm. Thus, while the first three components are "input" measures that capture where AI is used and what kind of changes and investments are made to enable use, the fourth component is more of an "outcome" measure that incorporates the firm-reported effects of AI use and investment on tasks, labor, and capital. All indices are designed to take values in the interval $[0, 1]$.

The first three components of $S_{integration}$ that measure aspects of use and implementation can be combined into an overall "use" index:

$$S_{use} = \frac{1}{3}S_{firm} + \frac{1}{3}S_{worker} + \frac{1}{3}S_{invest} \tag{4}$$

The four components (or sub-indices) of $S_{integration}$ are defined next.

B.1 Functional breadth index (S_{firm})

This index measures the horizontal breadth of AI adoption across 15 business functions identified in Question 24. Let $F_j \in \{0, 1\}$ be a binary indicator where $F_j = 1$ if the firm reports "Yes" for function j, and 0 otherwise. Functional breadth index is defined as:

$$S_{firm} = \frac{1}{15} \sum_{j=1}^{15} F_j \tag{5}$$

Note that $S_{firm} = 0$ corresponds to the case of "No" or "Don't Know" to all functions, and $S_{firm} = 1$ corresponds to the case of "Yes" to all functions.

B.2 Worker-task integration index (S_{worker})

Because Question 32 is contingent upon a positive response to Question 31, we define S_{worker} using a conditional approach.

Let $G \in \{0, 1\}$ represent the response to Question 31 (Generative AI use). Firms selecting "No" or "Do not know" for Question 31 receive a score of zero for G, correctly reflecting the lack of worker-level Generative AI tasks. Let $T_k \in \{0, 1\}$ represent the binary indicators for the 9 specific tasks listed in Question 32 plus the "Other" category. The worker integration index is then defined as:

$$S_{worker} = G \cdot \frac{1}{10} \sum_{k=1}^{10} T_k \tag{6}$$

B.3 Operational investment (S_{invest})

Operational investment is measured by the count of eight distinct organizational changes reported in Question 29. Let $c_m \in \{0, 1\}$ be the indicator for each change $m \in \{1, \ldots, 8\}$. Let Z be a binary indicator for the response "This business did not make any changes to use AI.".

The sub-index is calculated as:

$$S_{invest} = \begin{cases} 0 & \text{if } Z = 1 \\ \frac{1}{8} \sum_{m=1}^{8} c_m & \text{if } Z = 0 \end{cases} \tag{7}$$

This formulation ensures that firms explicitly reporting a lack of structural adjustment are anchored at zero.

B.4 Impact index (S_{impact})

This index measures the organizational commitment and the transformative nature of AI integration. It is composed of three sub-indices derived from Questions 25, 26, 27, and 28:

$$S_{impact} = \frac{1}{3} \left(T_{type} + A_{mag} + E_{imp} \right) \tag{8}$$

B.4.1 Task effect types (T_{type})

Based on Question 25, points are assigned as follows:

- 1 point: Introduce a new task not previously done by an employee
- 1 point: Perform a task previously done by an employee
- 1 point: Supplement or enhance a task performed by an employee
- 0 points: None of the above

The points are aggregated and normalized as follows:

$$T_{type} = \frac{1}{4} \sum_{l=1}^{4} e_k \tag{9}$$

where we give a point for each of the three task effect types that the firm undertakes, and zero for no task changes – without taking a stance on which task effect type is more impactful. Thus, firms that alter task structure get a positive weight, and the weight is higher the higher the number of different task effect types.

B.4.2 Automation magnitude (A_{mag})

The magnitude of task replacement is defined by both whether worker substitution occurs and the degree of worker task displacement, and whether capital substitution occurs (i.e. capital task replacement takes place in some form).

Let $R \in \{0, 1\}$ be an indicator from Question 25 representing whether AI performs tasks previously done by employees (i.e. task replacement/substitution). The task replacement score (T_{rep}) is based on Question 26 conditional on R:

$$T_{rep} = \begin{cases} \frac{\text{Score}_{Q26}}{3} & \text{if } R = 1 \\ 0 & \text{if } R = 0 \end{cases} \tag{10}$$

where

$$\text{Score}_{Q26} = \begin{cases} 1 & \text{if small number of tasks} \\ 2 & \text{if moderate number of tasks} \\ 3 & \text{if large number of tasks} \end{cases} \tag{11}$$

Thus, firms that engage in task replacement get a positive weight and this weight is higher the higher the intensity of task replacement.

The capital replacement score (K_{rep}) is a binary indicator from Question 27:

$$K_{rep} = \begin{cases} 1 & \text{if } Q27 = \text{“Yes”} \\ 0 & \text{if } Q27 \in \{\text{“No”}, \text{“Do not know”}\} \end{cases}$$

The final magnitude sub-index is:

$$A_{mag} = \frac{1}{2}(T_{rep} + K_{rep}) \tag{12}$$

$A_{mag} = 0$ if the firm engages in no worker task or capital replacement, and $A_{mag} = 1$ if the firm engages in both capital substitution and a high degree of task replacement.

B.4.3 Employment impact (E_{imp})

We define E_{imp} as a binary indicator of employment change due to AI use asked in Q28. Any reported increase or decrease in total employment attributed to AI is evidence of high-intensity integration:

$$E_{imp} = \begin{cases} 1 & \text{if Increased, Decreased} \\ 0 & \text{if Did Not Change} \end{cases} \tag{13}$$

B.5 Handling question non-response

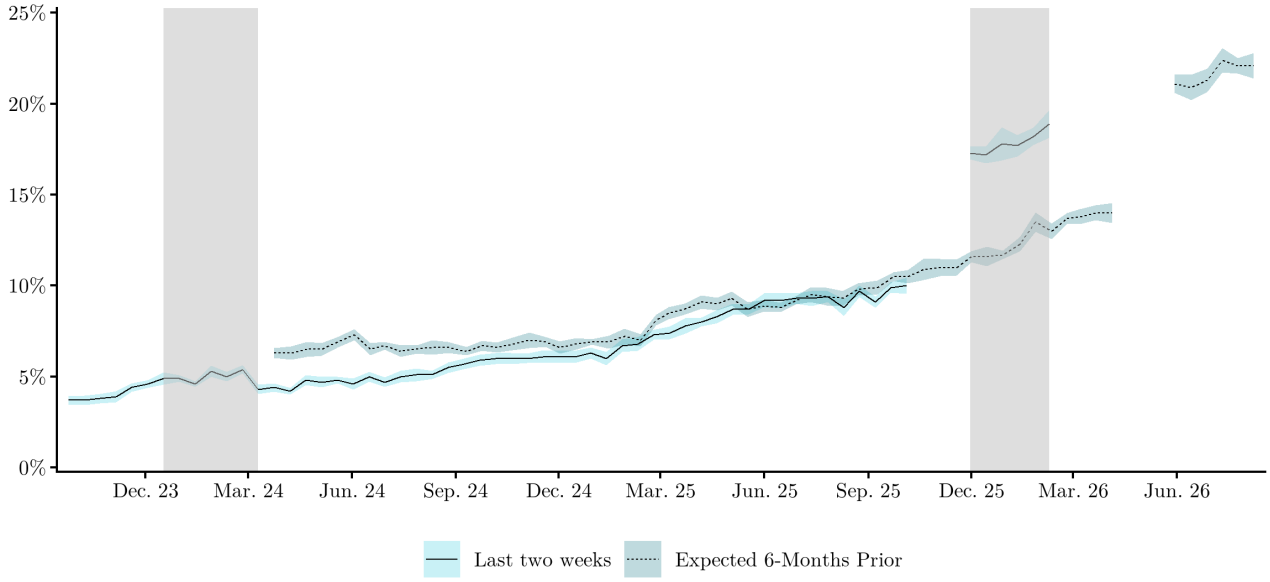
All "Don't know" responses within a question are assigned a value of zero (same as the value of "No" response), representing a lack of confirmed AI intensity. Similarly, for item non-response (completely unanswered questions), we assign a score of zero to the entire question. These two approaches are conservative and treat "Don't know" and item non-response as contributing zero value to index calculation.

B.6 Population estimates from indices

Finally, to calculate the population estimates based on any index, we weigh each firm's index with firm sampling weights or firm employment weights. These population estimates include the descriptive statistics of the indices and their correlations.

C Additional tables and figures

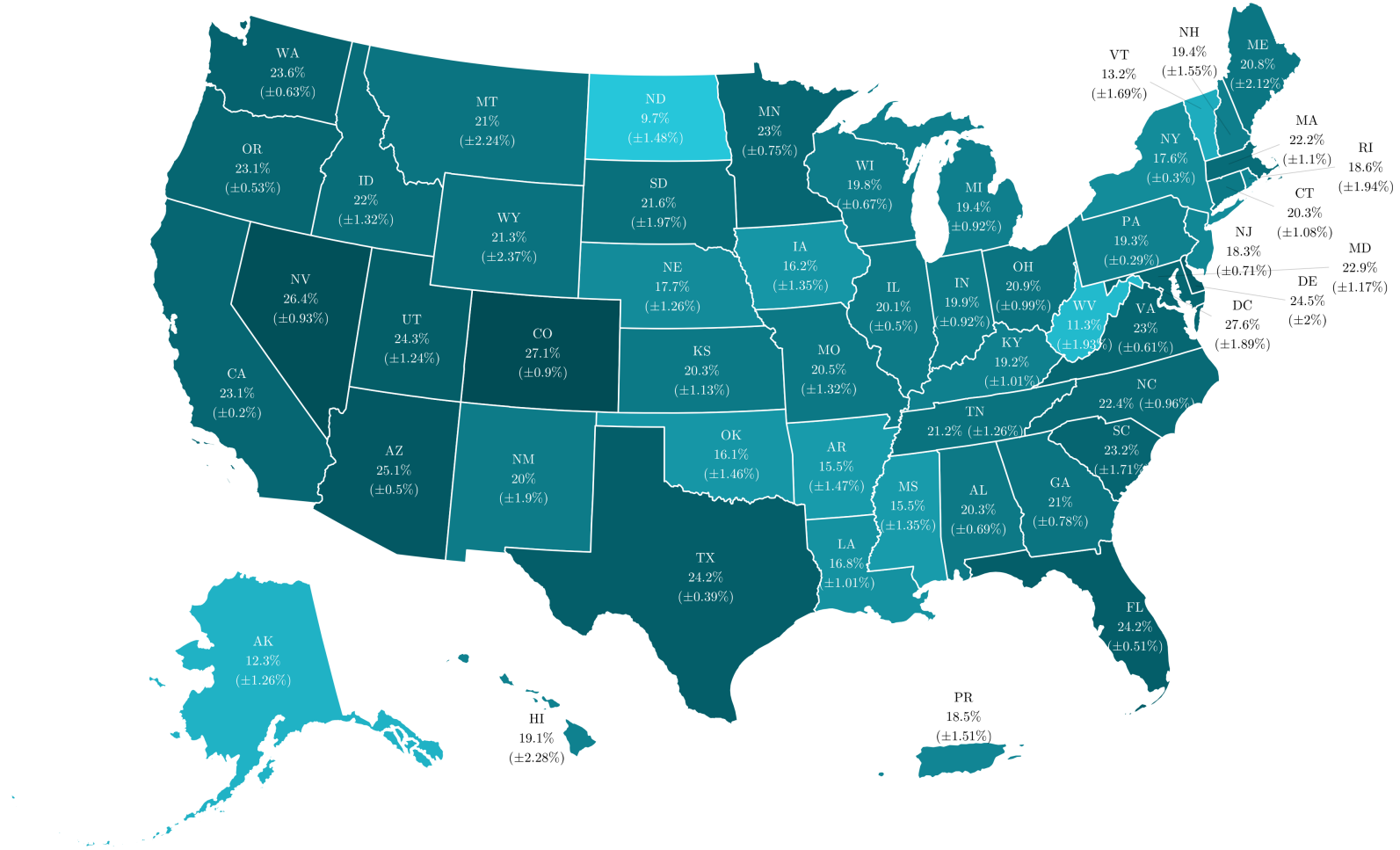
Figure C.1: Relationship between current (last two weeks) and future (next six months) AI use



Note: Shaded areas indicate 95% confidence intervals around BTOS estimates given BTOS response standard errors. Grey shaded areas represent AI supplement collection periods.

Figure C.2: AI use rates across states – next six months

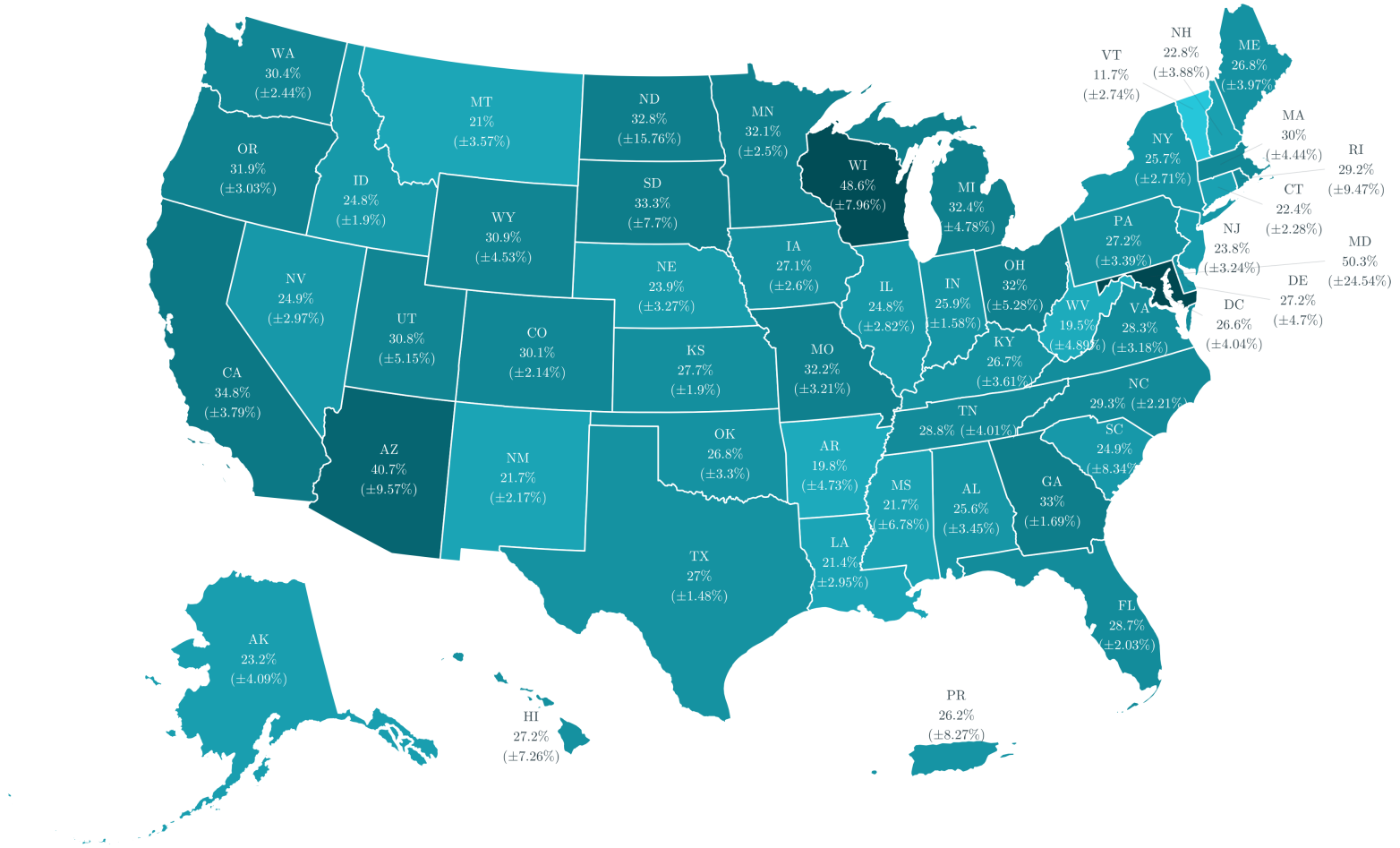
(a) Firm-weighted



Note: Values after estimates are ± standard errors.
Multi-State Firms: 38% (±1.3%)

Figure C.2: AI use rates across states – next six months (continued)

(b) Employment-weighted



Note: Values after estimates are ± standard errors.
Multi-State Firms: 58.7% (±5.55%)
(S) = Estimate suppressed due to quality concerns.

Figure C.3: Scree plot for LCA

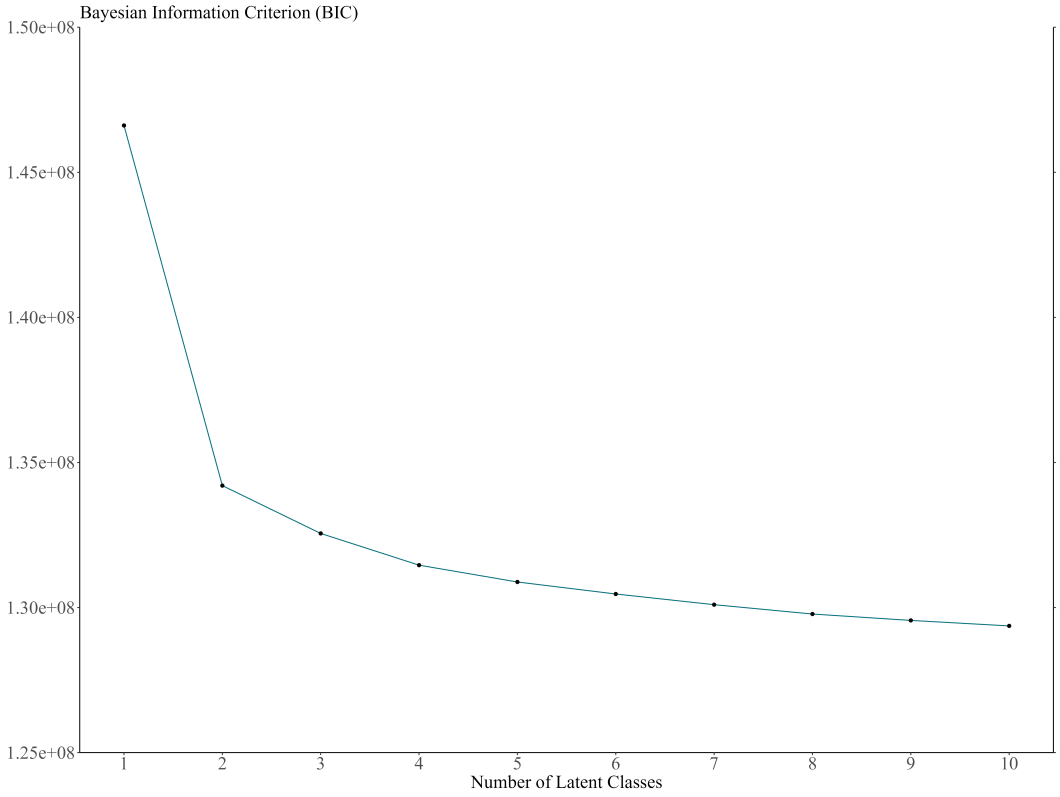


Table C.1: Detailed Shapley decomposition of regression explanatory power

	Performance above average		Sales increase	Employment decrease		
Predictor group	**Current**	**Future**	**Current**	**Current**	**Future**	**AI-driven**
Business functions						
Commercial, customer, and external relationships	14.8	12.1	**26.3**	**20.2**	13.4	7.9
Administration and governance	18.4	14.3	9.6	16.9	**25.2**	5.2
Production, fulfillment, and quality	1.5	2.2	2.4	11.7	13.2	13.5
Strategy, innovation, and technical	13.4	**26.0**	8.0	7.7	6.8	2.1
Subtotal: business functions	**48.1**	**54.6**	**46.3**	**56.5**	**58.6**	**28.7**
Organizational changes and investments						
Human capital and organizational change	10.5	7.8	15.5	9.3	9.0	**26.4**
No changes	1.3	3.6	11.3	6.6	3.1	7.4
Technical infrastructure, data, and implementation	7.8	8.5	4.4	4.1	13.8	16.4
Subtotal: changes and investments	**19.6**	**19.9**	**31.2**	**20.0**	**25.9**	**50.2**
Worker tasks						
General knowledge, documents, and communication	12.0	13.2	12.7	12.3	6.3	6.8
Specialized business, technical, and analytical	**20.2**	12.3	9.8	11.1	9.2	14.4
Subtotal: worker tasks	**32.2**	**25.5**	**22.5**	**23.4**	**15.5**	**21.2**
Within R^2	0.0072	0.0120	0.0061	0.0053	0.0047	0.0310

Notes: The underlying regressions include 34 indicators capturing AI use across 15 business functions, 9 organizational changes or investments, and 10 worker tasks, as well as the fixed effects described in the text. The indicators are organized into nine conceptually related predictor groups using a classification developed with the assistance of ChatGPT. Entries report each group's percentage contribution to the within R^2, as allocated by the Shapley decomposition. Subtotals aggregate contributions within the three broader dimensions. Boldface among the individual predictor groups indicates the largest contribution in each column. Subtotals may not sum to exactly 100 because of rounding. The final row reports the within R^2 allocated across the predictor groups.

Table C.2: Grouping of AI Survey Response Options

Group	Response options	No.	Description
Panel A. Business functions (Question 24)			
Production, fulfillment, and quality operations	Production of goods; Provision of services, products, or merchandise; Quality management and control; Distribution	4	Functions directly related to producing, delivering, and assuring the quality of the firm's output.
Commercial, customer, and external relationships	Sales and marketing; Customer service; Public relations and communication; Sourcing, supply chains, and purchasing	4	Functions centered on relationships with customers, suppliers, public audiences, and markets.
Strategy, innovation, and technical capability	Strategy and business development; Research and development; Information technology	3	Forward-looking and capability-building functions involving strategy, innovation, and technical systems.
Corporate administration, control, and governance	Finance and accounting; Human resources; Management and administration; Legal and compliance	4	Internal administrative, governance, compliance, and resource-management functions.
Panel B. Changes made to use AI (Question 29)			
Human capital and organizational process changes	Trained current staff to use AI; Hired staff trained in AI; Developed new workflows	3	Changes to worker capabilities and organizational processes associated with AI adoption.
Technical infrastructure, data, and implementation investments	Purchased computing power or specialized equipment or software; Purchased cloud services or cloud storage; Changed data collection or data management practices; Used vendors or consulting services to install or integrate AI	4	Investments in computing, software, cloud services, data infrastructure, and external implementation support.
No implementation changes	This business did not make any changes to use AI	1	AI use without reported complementary investments, retraining, workflow redesign, or other implementation changes.
Panel C. Employee Generative AI tasks (Question 32)			
General knowledge-work, document, and communication tasks	Information processing, paperwork, or filing; Writing or editing documents, emails, or communications; Interpreting, analyzing, translating, or summarizing documents; Searching for information or technical help; Tutoring, training, or learning	5	Broad, cross-functional knowledge-work tasks involving documents, communication, information retrieval, and learning.
Specialized business, technical, and analytical tasks	Customer support; Developing or researching new projects, processes, or products; Software coding or debugging; Data analysis or visualization	4	More specialized applications involving customer operations, innovation, software development, and quantitative analysis.

Notes: The table groups response options from Questions 24, 29, and 32 of the BTOS AI supplement based on similarity in economic and organizational content. "Other" response categories are omitted. For Question 29, "This business did not make any changes to use AI" is retained as a separate category.

Table C.3: The distribution of BTOS respondent types and AI use rate by respondent type

Respondent Type	**Proportion**	**AI Use Rate**
Top Executives & Owners	49.4%	20.8%
Other	14.9%	19.9%
Finance & Accounting	14.9%	10.6%
Operations & General Management	9.3%	17.3%
Administrative Support	7.1%	9.9%

Note: All statistics are firm-weighted.

Table C.4: OLS regression of firm AI use indicator on survey respondent types

Respondent type	**Coefficient estimate**
Administrative Support	-0.103***
	(0.004)
Finance & Accounting	-0.108***
	(0.003)
Operations & General Management	-0.043***
	(0.005)
Other	-0.031***
	(0.004)
N	106,600
R^2	0.0900
R^2 Within	0.0120

Note: The omitted category of respondents is Top Executives & Owners. Robust standard errors in parentheses. Regression is firm-weighted and includes fixed effects for firm size class, 3-digit NAICS industry, state, and survey period. Observation counts (N) are rounded in accordance with disclosure rules.
$^{+}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table C.5: Firm AI use rates conditional on size thresholds, all sectors

		Firm-weighted		**Employment-weighted**	
Firm Size Class (emp)	**Avg. Size**	**Current**	**Future**	**Current**	**Future**
10+	79	18.9%	24.6%	32.9%	42.9%
50+	322	24.3%	32.2%	36.6%	47.8%
100+	583	27.3%	36.1%	38.4%	50.1%
200+	1329	30.6%	40.0%	41.0%	53.3%
350+	1766	31.5%	40.8%	41.8%	54.4%
500+	2420	32.4%	41.0%	42.9%	55.5%
1000+	4627	35.1%	44.9%	45.3%	58.8%

Note: Current (future) refers to last two weeks (next six months).

Table C.6: Firm AI use rates conditional on size thresholds, high AI use sectors

		Firm-weighted		Employment-weighted	
Firm Size Class (emp)	**Avg. Size**	**Current**	**Future**	**Current**	**Future**
10+	65	35.4%	41.2%	45.0%	54.8%
50+	254	42.2%	49.7%	48.2%	59.3%
100+	419	44.8%	53.5%	49.5%	61.3%
200+	855	48.8%	57.3%	51.5%	64.2%
350+	1137	51.5%	62.9%	52.5%	66.5%
500+	1426	53.5%	64.1%	53.2%	67.3%
1000+	2403	57.0%	68.3%	54.2%	68.7%

Note: Current (future) refers to last two weeks (next six months). High AI use sectors are Information, Professional Services, Finance, Education and Real Estate.

Table C.7: Firm AI use rates conditional on size thresholds, Finance and Professional Services

		Firm-weighted		Employment-weighted	
Firm Size Class (Emp)	**Avg. Size**	**Current**	**Future**	**Current**	**Future**
10+	54	37.5%	42.6%	46.4%	54.5%
50+	215	44.6%	52.3%	50.3%	59.8%
100+	356	47.1%	56.5%	51.9%	62.0%
200+	762	53.6%	60.3%	55.0%	64.9%
350+	999	55.8%	65.6%	55.9%	67.1%
500+	1308	58.0%	64.9%	56.6%	67.0%
1000+	2223	62.6%	67.1%	57.8%	67.7%

Note: Current (future) refers to last two weeks (next six months).

Table C.8: Firm AI use rates conditional on firm size thresholds, high AI use sectors, respondents restricted to top executives & owners

		Firm-weighted		Employment-weighted	
Firm Size Class (Emp)	**Avg. Size**	**Current**	**Future**	**Current**	**Future**
10+	23	39.7%	45.1%	42.9%	49.4%
50+	109	44.7%	52.9%	50.1%	59.8%
100+	194	44.4%	60.2%	54.4%	66.7%
200+	636	57.6%	57.6%	75.8%	75.8%
350+	823	86.4%	86.4%	87.9%	87.9%
500+	(S)	(S)	(S)	(S)	(S)
1000+	(S)	(S)	(S)	(S)	(S)

Note: Current (future) refers to last two weeks (next six months). High AI use sectors are Information, Professional Services, Finance, Education and Real Estate. (S) = Estimate suppressed due to quality concerns.

Table C.9: AI use rates based on various measurement levels

			AI Use Rate	
Use type / Measurement layer	**Question(s)**	**Reference period**	**Firm-weighted**	**Employment-weighted**
Firm	23	Last 2 weeks	17.9%	31.2%
Worker task (AI)	30	Last 6 months	22.6%	40.6%
Worker task (GenAI)	31	Last 6 months	20.8%	38.9%
Firm or worker task (AI)	23, 30	Last 2 weeks and last 6 months	26.0%	44.7%
Firm or worker task (GenAI)	23, 31	Last 2 weeks and last 6 months	25.8%	46.4%
Worker task (AI) or worker task (GenAI)	30, 31	Last 6 months	26.3%	47.2%
Firm business function	24	Last 6 months	27.7%	37.1%
Firm business function or worker task (AI)	24, 30	Last 6 months	31.3%	48.8%
Firm business function or worker task (GenAI)	24, 31	Last 6 months	31.4%	51.0%
Firm or firm business function	23, 24	Last 2 weeks and last 6 months	28.0%	41.8%
Positive AI integration index ($S_{integration} > 0$)	24-29, 31, 32	Last 2 weeks and last 6 months	32.0%	54.0%